A computable prime-corrected Riemann–von Mangoldt clock: an exact defect identity, the prime-information threshold, and the location of the zeros of the Riemann zeta function in mean square

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Abstract

Let γ_k be the ordinates of the nontrivial zeros of the Riemann zeta function, counted with multiplicity, and let τ_k be the midpoint Riemann–von Mangoldt clock, $N_0(\tau_k) = k - \frac{1}{2}$. The clock is close to the zeros in height, but the ordinal displacement $s_k = k - \frac{1}{2} - N_0(\gamma_k)$ has mean square $\frac{1}{2\pi^2}\log\log T$ (Selberg, Korolev): the clock misplaces almost every zero by an unbounded number of positions. We prove, unconditionally: (A) an exact identity that turns sums of $\mathcal{G}(u_k + \frac{1}{2}) - \mathcal{G}(u_k - \frac{1}{2})$, $u_k = k - \frac{1}{2} - F(\gamma_k)$, over consecutive ordinates into dF-integrals of $g(N - F)$, where $\mathcal{G}' = g$, for every continuous g and every C^1 model F of the counting function N; (B) for the clock equation corrected by the Euler phase S_X of the primes $p \le X$, the mean square of the ordinal defect at the ordinates is $\frac{1}{2\pi^2}\log\frac{\log T}{\log X} + O(1 + \sqrt{\log\frac{\log T}{\log X}})$ uniformly for $2 \le X \le T^{1/2}$, so that it is bounded if and only if $\log X \asymp \log T$; (C) an exact finite-height accounting of the gain of the one-step clock $\tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$, $\mathcal{L}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$; (D) with $X = T^{1/\sqrt{\log\log T}}$ the one-step clock, computed from τ_k and the primes alone, has mean-square error $(\log\log\log T + O(\sqrt{\log\log\log T}))/\log^2 T$ in height, against $(2+o(1))\log\log T/\log^2 T$ for the clock, and the Newton-type step iterated $\log^* T + O(1)$ times, with cutoffs depending only on T through absolute constants that are not computed, locates the ordinates with a mean-square error of $O(1)$ mean spacings; the proof rests on an exact moment recursion, on Fujii's moments of $S(t+h) - S(t)$, and on a sampling inequality for maximal functions at points of bounded multiplicity; the sorted points form an increasing sequence with the same error; the ordinal defect has exponential tails, the corrected count with primes up to $T^{1/(C\sqrt{\lambda})}$ being off by more than λ positions on at most $2^{2-\sqrt{\lambda/C}}$ of the zeros, and the errors of the corrected clocks have all moments bounded in units of $\sqrt{H_m}$; (E) $\limsup_k |\tau_k - \gamma_k| = 2\pi \limsup_t |S(t)|/\log t$, and Backlund's method with the exponent profile generated from Bourgain's exponent pair by van der Corput's process gives $\limsup_k |\tau_k - \gamma_k| \le 0.6083074$, compared with $2\pi \cdot 0.10076 = 0.6331\ldots$ from the explicit bound of Bellotti and Wong; the new envelope is asymptotic and is not an effective finite-height estimate. No zero of the Riemann zeta function is used in the construction of the corrected clocks.

Keywords: Riemann zeta function, zeros of the Riemann zeta function, Riemann–von Mangoldt formula, argument of the Riemann zeta function, truncated Euler product, Newton-type iteration
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1. Introduction

1.1. The Riemann–von Mangoldt clock and the ordinates

Let $\zeta(s)$ be the Riemann zeta function, let $N(T)$ be the number of its nontrivial zeros $\rho = \beta + i\gamma$ with $0 < \gamma \le T$, counted with multiplicity, and list their ordinates as $\gamma_1 \le \gamma_2 \le \cdots$, with multiplicity. Away from ordinates the Riemann–von Mangoldt formula reads [27, §§9.3–9.4]

$$N(T) = N_0(T) + \frac{1}{\pi}\arg\zeta\Big(\frac{1}{2}+iT\Big) + O\Big(\frac{1}{T}\Big), \qquad N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}, \tag{1.1}$$

the argument of the Riemann zeta function being obtained by continuous variation along $2 \to 2+iT \to \frac{1}{2}+iT$ from $\arg\zeta(2) = 0$. The *midpoint Riemann–von Mangoldt clock* (RvM clock) is the sequence τ_k defined by

$$N_0(\tau_k) = k - \frac{1}{2}, \qquad \tau_k = \frac{2\pi\,(k-\frac{11}{8})}{W_0\Big((k-\frac{11}{8})/e\Big)} \qquad (k \ge 1), \tag{1.2}$$

where W_0 is the principal branch of the Lambert function [5]; substituting $w = W_0((k - 11/8)/e)$ directly in $we^w = (k-11/8)/e$ verifies the inversion. The other real branch at the negative argument for $k = 1$ gives a value below 2π and is excluded by the defining condition $\tau_k > 2\pi$. The clock is explicit and lies, in phase, half way between consecutive Gram points. This paper studies its relation to the ordinates γ_k: in which sense the clock describes them, and how much prime information has to be added to it. The relation is measured in units of zeros by the ordinal displacement and by its prime-corrected version, the *ordinal defect*,

$$s_k = k - \frac{1}{2} - N_0(\gamma_k), \qquad u_k(X) = s_k - S_X(\gamma_k), \qquad S_X(t) = -\frac{1}{\pi}\sum_{p\le X}\arg\Big(1-p^{-1/2-it}\Big). \tag{1.3}$$

$u_k(X)$ measures how well the true ordinate satisfies the corrected clock equation $N_0(t) + S_X(t) = k - \frac{1}{2}$; $|u_k(X)| < \frac{1}{2}$ means that the corrected count assigns to γ_k its correct index.

1.2. The approximation of the zeros by the clock and the primes

The approximation studied here starts from the explicit numbers τ_k and corrects them with the Euler phase of finitely many primes and prime powers, by one or several Newton-type steps for the corrected equation $N_0(t) + S_X(t) = k - \frac{1}{2}$ from $t = \tau_k$, with cutoffs that depend only on the height. Every corrected point is a closed expression in k, τ_k and the primes: the construction evaluates neither the Riemann zeta function nor any of its zeros. The ordinates γ_k enter only in the statements, to measure the error. Theorems A and C are exact identities; Theorems B and D are asymptotic statements in mean square over the zeros with $T < \gamma_k \le 2T$; Theorem E concerns the worst zero.

1.3. What is known and only used

The following facts are quoted with references in Section 3 and are not reproved.

- The displacement $\delta_k = \tau_k - \gamma_k$ is bounded, and $\limsup_k |\delta_k| \le 2\pi \cdot 0.10076 = 0.6331\ldots$, from the explicit bound of Bellotti and Wong for $N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}$ [1], which improves those of Trudgian [29] and of Hasanalizade, Shen and Wong [10]. For large T only, Trudgian proved $|S(T)| \le 0.1013\log T$ with Huxley's exponent [28, §5 of the preprint], which gives $\limsup_k |\delta_k| \le 0.6365$.

- The ordinal displacement s_k differs by less than 1 from minus the integer displacement of γ_k with respect to the Gram intervals, whose moments are those of a normal variable with variance $\frac{1}{2\pi^2}\log\log T$, by Selberg's formulae as proved by Korolev [18, 14]. Hence $\langle s_k^2\rangle \sim \frac{1}{2\pi^2}\log\log T$ and $\langle \delta_k^2\rangle \sim 2\log\log T/\log^2 T$ over $T < \gamma_k \le 2T$.
- In t-mean, $S(t) = \frac{1}{\pi}\arg\zeta(\frac{1}{2}+it)$ is asymptotically normal (Selberg [23], Tsang [31]), and Selberg's approximation of $S(t)$ by prime sums up to a small power of T holds in $L^{2k}(T,2T)$ [23, 24], in the form proved by Fujii [8, §3], together with Fujii's moments of $S(t+h)-S(t)$. These concern t-averages, not the ordinates.

1.4. Comparison with the cited antecedents

Table 1 compares statements about the same quantity wherever possible. A continuous t-average, a defect evaluated at a true ordinate, and the error of a point computed from the primes are different objects; a result about one requires a transfer argument before it applies to another. "Proved here" identifies the result established by the present proof, without asserting that no earlier equivalent formulation exists. The comparison concerns the cited antecedents, not an exhaustive priority survey.

Table 1: Cited antecedents, conclusions proved here, and the scope of the difference.

Object	Established antecedent	Present result and scope
Bare midpoint clock τ_k	The RvM main term and its inversion are classical. Selberg–Korolev displacement moments give $\langle(\tau_k-\gamma_k)^2\rangle_{\mathcal{I}_T} \sim 2\log\log T/\log^2 T$; see Remark 3.15 and [18, 14].	This is the baseline used here, not a contribution. The Lambert expression is not claimed as new; neither is the tendency of its mean-square height error to zero.
Prime approximation of $S(t)$	Selberg's approximation, the Fujii moment estimates, and the form recorded in [6, (3.6)] bound the residual of a short prime sum in continuous t-mean. Montgomery–Vaughan bounds its polynomial moments.	These are external inputs. They do not on their own state an error estimate for the specific output $\tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$; the sampling and transport estimates supply that step.
Exact conversion of moments and other test functions	The counting staircase and its jumps are classical. Moment and distribution formulae for Gram displacement are already available in [18, 14].	Theorems 4.1, 4.2 give the exact formula for any real C^1 model F, with both endpoint terms; no monotonicity of F is required. This is a general identity, not a claim to have discovered the displacement law.
Defect at the true ordinate, $u_k(X)$	Prime-sum approximation controls $S(t)-S_X(t)$ in t-mean. This is an input about the counting error, not an already constructed approximation to γ_k.	Theorems 5.1, 5.12 give $\langle u_k(X)^2\rangle_{\mathcal{I}_T} = (2\pi^2)^{-1}\log(1/c) + O(1+\sqrt{\log(1/c)})$, $c = \log X/\log T$, uniformly for $2 \le X \le T^{1/2}$, including an index-dependent cutoff. The difference is the quantified statement at the ordinates and its cutoff range.

Table 1 continued

Object	Established antecedent	Present result and scope
One explicit prime correction	The bare clock has the preceding $2\log\log T/\log^2 T$ baseline. Approximating the phase by primes is an established idea, not a new ingredient.	Theorem 7.3 controls the transport to τ_k. At the analytically specified cutoff, the height MSE is $(\log\log\log T + O(\sqrt{\log\log\log T}))/\log^2 T$. Its ratio to the bare-clock MSE tends to zero. Constants and onset are not evaluated.
Finite-height accounting	Expanding a squared error is elementary; it does not justify independent prime responses or a fitted gain.	Theorem 6.1 keeps the diagonal, cross term, transport, and exact nonnegative curvature term. It is an identity for a finite set, not a positive-gain guarantee at every height.
Further corrections and order	Newton-type correction and increasing rearrangement are standard operations. Rearrangement cannot increase a convex matching cost for two ordered real lists.	Theorem 8.11 gives uniform control through $\log^* T + O(1)$ prescribed corrections, with a multiplicity estimate. Corollary 8.15 gives a nondecreasing output with height MSE $O(\log^{-2} T)$. The construction and its transport control are the issue; the rate alone is not a priority claim. The cutoff constants are not yet numerical.
Tails	Moment inequalities and the conversion of moments to tail bounds are classical.	Theorem 8.3 gives the stated $2^{2-\sqrt{\lambda/C}}$ bound with its explicit admissible range of c; Corollary 8.13 treats the constructed points. These are asymptotic bounds with unspecified constants, not certified error probabilities for the available data.
Worst error of the bare clock	Bellotti–Wong [1, Thm. 1.1, Cor. 1.5] give the explicit coefficient 0.10076 for the logarithmic term of the counting and argument bounds, implying $\limsup \lvert\tau_k - \gamma_k\rvert \le 2\pi \cdot 0.10076$. Backlund's method and the exponent inputs are established tools.	Proposition 9.1 identifies the two limsups exactly. Theorem 9.2 evaluates the stated exponent-profile family, giving $2\pi C_\infty = 0.6083073\ldots$. This comparison improves that cited asymptotic coefficient; it is not a new explicit bound valid at a given height, nor a claim to optimize over all known profiles.
Accuracy at a specified height	Direct certified zero computations can locate individual zeros using evaluations of ζ; this is a different computational input from the prime-only clocks considered here.	No certified finite-height precision constant or individual-zero accuracy is obtained from the asymptotic O-terms here. No superiority at the available data heights is inferred from those terms.

Meaning of the mean-square scale.. The assertion $\langle \lvert x_k - \gamma_k\rvert^2\rangle \ll \log^{-2} T$ bounds the RMS error by a constant multiple of one mean spacing, since the local spacing is $1/\mathcal{L}(T) \sim 2\pi/\log T$. It does not assert an $o(1)$ fraction of a spacing, a numerical precision at a specified height, or a bound for each ordinate. Classical moment estimates already identify this natural scale: for example, [27, §9.25, (9.25.4)] bounds the third moment of consecutive gaps, and [6, (3.6)] supplies a fourth-moment prime approximation. A sublevel-integration construction is another route to this scale; its use of classical inputs does not establish priority for a particular formula. For approaching the zeros, the useful comparison is the error of the specified construction against the bare clock, together with its constants and effective range;

the latter remain open when they occur only in asymptotic O-terms.

1.5. What this paper adds

Let $\mathcal{I}_T = \{k : T < \gamma_k \le 2T\}$ and let $\langle\cdot\rangle_{\mathcal{I}_T}$ be the mean over $\mathcal{I}_T$. Theorems A–E are unconditional deductions from the external inputs stated in Section 3. Their precise statements are proved here; this description does not establish historical priority. In particular, the scale $O(\log^{-2} T)$ alone is not claimed as a new result. Theorem E applies Backlund's method in the form of [28, 29, 1] to exponents of [3, 12]. Table 1 separates the classical inputs from the conclusions for the specific corrected points.

A. Exact defect identity (Theorems 4.1 and 4.2). For every real C^1 model F of the counting function, every continuous g with primitive $\mathcal{G}$, and $u_k = k - \frac{1}{2} - F(\gamma_k)$, the sum of $\mathcal{G}(u_k + \frac{1}{2}) - \mathcal{G}(u_k - \frac{1}{2})$ over consecutive ordinates equals $\int g(N - F)\, dF$ up to two boundary terms. For $g(x) = x^{2j}$ this gives $\sum u_k^{2j}$ ($\frac{1}{12}$ per zero when $j = 1$); for $g(x) = e^{i\xi x}$ it identifies the characteristic function of the empirical law of u_k with a t-integral. No hypothesis on the zeros and no monotonicity of F is used.

B. How much prime information the clock needs (Theorems 5.1 and 5.12). Uniformly for $2 \le X \le T^{1/2}$, with $c = \log X/\log T$,

$$\frac{1}{\#\mathcal{I}_T} \sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{1}{2\pi^2} \log\frac{1}{c} + O\Big(1 + \sqrt{\log(1/c)}\Big), \tag{1.4}$$

and the same holds with the cutoff $X_k = \max(2, (\tau_k/2\pi)^c)$ varying with the index. A fixed set of primes leaves a mean-square defect $\sim (2\pi^2)^{-1} \log\log T$; primes up to T^c keep it bounded; boundedness holds if and only if $\liminf \log X/\log T > 0$ (Corollary 5.4). Thus merely letting the number of primes tend to infinity does not ensure a bounded ordinal defect: the logarithm of the cutoff must remain comparable with the logarithm of the height.

C. Exact accounting for the computable clock (Theorem 6.1). For the one-step clock $\tau_k^X = \tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$, the mean-square gain over any finite set of zeros equals a diagonal prime-response term minus a cross term, with no independence assumption; the residual of τ_k^X differs from $u_k(X)$ by an explicit transport term and by the exact nonnegative term $\chi_k = \frac{\gamma_k}{2\pi}(\frac{\tau_k}{\gamma_k} - 1 - \log\frac{\tau_k}{\gamma_k})$.

D. A computable clock that approaches the zeros (Theorems 7.3, 8.6 and 8.11). For $2 \le X \le T^{1/2}$ the residual $r_k(X) = \mathcal{L}(\tau_k)(\tau_k^X - \gamma_k)$ satisfies $\langle r_k(X)^2\rangle_{\mathcal{I}_T} = \frac{1}{2\pi^2}\log\frac{1}{c} + O((1 + \sqrt{\log(1/c)})(1 + c^2 \log\log T))$; the term $c^2 \log\log T$ is the transport between τ_k and γ_k (Lemma 7.2). With $X = T^{1/\sqrt{\log\log T}}$, or with the cutoff $\tau_k^{1/\sqrt{\log\log \tau_k}}$ read off the clock itself,

$$\frac{1}{\#\mathcal{I}_T} \sum_{k\in\mathcal{I}_T} \big(\tau_k^X - \gamma_k\big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}. \tag{1.5}$$

A second Newton-type step from τ_k^X, with cutoff $T^{1/\sqrt{\log\log\log T}}$, has mean-square error $(\log\log\log\log T + O(\sqrt{\log\log\log\log T}))/\log^2 T$ (Theorem 8.6), and the m-th step, with

the cutoffs (8.13) depending only on T, has error $(H_m + O(\sqrt{H_m}+1))/\log^2 T$ with absolute constants, where H_m is an iterated logarithm of T (Theorem 8.11); after $m^* = \log^* T + O(1)$ steps the error is $O(1)/\log^2 T$, that is, $O(1)$ mean spacings in mean square. The proof rests on an exact moment recursion, on a multiplicity bound for the corrected clock points of every level (Theorem 8.10, from Fujii's moments of $S(t+h)-S(t)$), and on a sampling inequality for maximal functions at points of bounded multiplicity. The second clock preserves the local order and spacing of the RvM clock outside a proportion $O(H_1^{-2})$ of the indices (Proposition 8.8). Sorting the points of a level never increases the error, so the final clock can be taken increasing (Corollary 8.15). The moments of every order of the errors are bounded in units of $\sqrt{H_j}$, which gives tails $\ll_q \lambda^{-2q}$ for the j-th clock (Corollary 8.13); and for the defect at the ordinates, with primes up to T^c and $\exp(-(\lambda/C)^{3/2}) \le c \le 1/(C\sqrt{\lambda})$, the corrected count is off by more than λ positions on at most $2^{2-\sqrt{\lambda/C}}$ of the zeros (Theorem 8.3), where Chebyshev's inequality would give $\lambda^{-2}\log\lambda$.

E. The pointwise envelope (Proposition 9.1 and Theorem 9.2). $\limsup_k |\tau_k - \gamma_k| = 2\pi \limsup_{t\to\infty} |S(t)|/\log t$, also for the corrected clocks with $X \le (\log T)^{2-\varepsilon}$; and for every exponent profile $\mathfrak{e}$ of the Riemann zeta function on $[\frac12, 1]$,

$$\limsup_{k\to\infty} |\tau_k - \gamma_k| \le \frac{1}{\log 2}\int_0^1 \frac{\mathfrak{e}(\sigma)\,d\sigma}{\sqrt{\sigma(2-\sigma)}}, \tag{1.6}$$

$\mathfrak{e}$ being extended to $[0, \frac12)$ by the functional equation. The profiles $\mathfrak{e}_J$ generated from Bourgain's exponent pair by J applications of van der Corput's $\mathscr{A}$-process give a decreasing sequence of bounds with limit $C_\infty = 0.0968151142790\ldots$, so $\limsup_k |\tau_k - \gamma_k| \le 2\pi C_\infty = 0.6083073\ldots$ and $\limsup_t |S(t)|/\log t \le 0.09681512$, compared with 0.10076 in [1] and 0.1013 in [28] (Remark 9.3). This is an asymptotic bound and does not replace the explicit bounds of [1], valid for all $T \ge e$.

Main finding. The Euler phase of the primes up to a power of the height turns the clock from an approximation in height into an approximation in position. For the bare clock the error in units of zeros, $s_k = \mathcal{L}(\xi_k)\delta_k$, has mean square $\sim \frac{1}{2\pi^2}\log\log T$. With $X = T^c$ the error in units of zeros of the corrected clock equation has bounded mean square (Theorem B), so outside a proportion $\ll (1+\log(1/c))/A^2$ of the zeros it is at most A; and the computable clock, which uses only τ_k and the primes, keeps the gain up to the transport between τ_k and γ_k (Theorem D). With $X = T^{1/\sqrt{\log\log T}}$,

$$\frac{\langle(\tau_k^X - \gamma_k)^2\rangle_{\mathcal{I}_T}}{\langle(\tau_k - \gamma_k)^2\rangle_{\mathcal{I}_T}} \sim \frac{\log\log\log T}{2\log\log T} \longrightarrow 0, \tag{1.7}$$

and in units of zeros the mean square drops from $\frac{1}{2\pi^2}\log\log T$ to $\frac{1}{4\pi^2}\log\log\log T$; iterating the Newton-type step brings it down to $O(1)$. The gain is a statement in mean square; for the worst zero, Proposition 9.1 shows that no cutoff $X \le (\log T)^{2-\varepsilon}$ changes $\limsup |\tau_k - \gamma_k|$, and the bound (1.6) comes from the exponents of the Riemann zeta function, not from the primes.

1.6. Degree of proof

Theorem, *Proposition*, *Lemma* and *Corollary* are proved here unconditionally, from the external results of Section 3; *Question* names an open problem. No statement of this paper rests on a computation with the zeros. The only numbers that are not closed expressions are the values of the integral (9.1) for piecewise linear profiles with rational vertices, computed in closed form by the two programs named in the Data availability statement; the constants in the O-terms of Theorems B–D are absolute but not computed.

Table 2: Main results: degree of proof and role.

result	status	role
Theorems 4.1, 4.2 (A)	unconditional	statistics of the errors at the zeros as t-integrals
Theorems 5.1, 5.12 (B)	unconditional	bounded error in units of zeros iff $\log X \asymp \log T$
Theorem 6.1 (C)	unconditional, exact	finite-height balance of the gain of one correction
Theorems 7.3, 8.6, 8.11 (D)	unconditional	zero-free corrected points with mean-square error $O(1)$ mean spacings
Theorem 8.3, Corollaries 8.13, 8.15	unconditional	tails of the defect and of the corrected clocks; an increasing computable clock
Proposition 9.1, Theorem 9.2 (E)	unconditional, asymptotic	the worst error of the clock is the extreme size of $S(t)$
Questions 10.1–10.3	open	what remains to be proved

1.7. Organization

Section 2 lists the notation and Section 3 the external results. Sections 4–7 prove Theorems A–D, Section 8 iterates the Newton-type step uniformly in the number of steps, and Section 9 proves Theorem E. Section 10 lists what is open.

2. Notation

2.1. List of symbols

The notation below fixes the meaning of the recurring symbols. Coefficients and dummy indices explicitly introduced inside a proof are local. The letter μ is not used (it is reserved for the Möbius function), π is only the number $3.14159\ldots$, and prime counts are written $\#\{p \le y\}$. Dummy variables of integration (t, y) are local.

symbol	meaning
$\zeta(s)$	the Riemann zeta function
$\rho_k = \beta_k + i\gamma_k$	nontrivial zeros of the Riemann zeta function, $0 < \gamma_1 \le \gamma_2 \le \cdots$ with multiplicity

symbol	meaning
$\nu(\gamma)$	number of zeros of the Riemann zeta function with ordinate γ (multiplicities included)
$N(t)$, $N(t^{\pm})$	number of zeros with $0 < \gamma \le t$; one-sided limits
ϑ, Θ_0, ϖ	Riemann–Siegel theta function; $\Theta_0(t) = \frac{t}{2}\log\frac{t}{2\pi e} - \frac{\pi}{8}$; $\varpi = \vartheta - \Theta_0$
g_m	Gram points, $\vartheta(g_m) = m\pi$
N_0, $\mathcal{L}$	$N_0 = \Theta_0/\pi + 1$; $\mathcal{L} = N_0'$, $\mathcal{L}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$
S, S^*, G	$S(t) = \frac{1}{\pi}\arg\zeta(\frac{1}{2}+it)$; midpoint value at ordinates; $G = N - N_0 = S + \varpi/\pi$
$\mathcal{I}_T$, $\langle\cdot\rangle_I$	$\{k : T < \gamma_k \le 2T\}$; mean over k in a finite index set I
τ_k, τ_x, W_0	RvM clock (2.5); the same for real $x \ge 1$; principal Lambert function
s_k, δ_k, ξ_k	$s_k = k - \frac{1}{2} - N_0(\gamma_k)$; $\delta_k = \tau_k - \gamma_k$; mean-value point in $s_k = \mathcal{L}(\xi_k)\delta_k$
Δ_k	integer displacement of γ_k with respect to the Gram intervals (Theorem 3.4)
$\mathfrak{d}_k$	$s_k + \Delta_k$, $\lvert\mathfrak{d}_k\rvert < 1$ (Remark 3.15)
$\mathfrak{s}_T$	$(\frac{1}{2\pi^2}\log\log T)^{1/2}$ (Remark 3.15)
Λ, a_n	von Mangoldt function; $a_n = -\Lambda(n)/(\pi\sqrt{n}\log n)$
A, A_X	set of prime powers with $\sum_{n\in A}\lvert a_n\rvert < \infty$; $A_X = \{p^m : p \le X\}$
S_A, S_X	$\sum_{n\in A} a_n \sin(t\log n)$; S_{A_X}
v_A, v_X	$\frac{1}{2}\sum_{n\in A} a_n^2$; v_{A_X}
σ_X, M_X	bounds for $\lvert S_X\rvert$ and $\lvert S_X'\rvert$, (2.11)
F, F_X	a real C^1 model of N; $F_X = N_0 + S_X$
u_k, $u_k(X)$	$k - \frac{1}{2} - F(\gamma_k)$; $k - \frac{1}{2} - F_X(\gamma_k)$
Ψ_j, κ_j	sawtooth polynomial (4.2); $\kappa_j = \frac{(2j)!}{j!(2\pi)^{2j}}$
X, c, C, T_0	prime cutoff; $c = \log X/\log T$; absolute constants
Y, Y_0, S_Y^{pr}	polynomial length; $Y_0 = T^{1/160}$; $S_Y^{\mathrm{pr}}(t) = -\frac{1}{\pi}\sum_{p<Y} p^{-1/2}\sin(t\log p)$
$\mathcal{J}$, V, $\mathcal{W}$, $\mathcal{W}_1$	primes between X and Y_0; $V = \sum_{p\in\mathcal{J}} p^{-1}$; $\mathcal{W} = S_{Y_0}^{\mathrm{pr}} - S_X$; its part over $\mathcal{J}$ (5.4)
E_0, E_X, $\mathcal{E}_2$	$S - S_{Y_0}^{\mathrm{pr}}$; $S - S_X$; $\int_T^{2T} E_X^2\,dt$
z_n, $\mathcal{D}$, $\mathcal{D}_1$, $\mathcal{D}_2$, η_n, η_n'	coefficients and Dirichlet polynomials in Lemmas 5.9–5.10
τ_k^X, δ_k^X, $r_k(X)$, χ_k	one-step clock; $\tau_k^X - \gamma_k$; $\mathcal{L}(\tau_k)\delta_k^X$; second-order error (6.3)
$\mathcal{R}_n^{\gamma}(I)$, $\mathcal{R}_n^{\tau}(I)$	prime-power responses at the zeros and at the clock points (6.1)
$\mathrm{Cr}_A(I)$, $D_X(I)$	cross term (6.2); diagonal term (6.5)
λ	real multiplier of an Euler phase
ς_n, $\mathbf{Q}$, $\mathbf{b}$, $\boldsymbol{\lambda}$, $\Upsilon_{\boldsymbol{\lambda}}$	$\varsigma_n(k) = \sin(\tau_k\log n)$; Gram matrix; projection vector; coefficients; trigonometric correction (Remark 6.3)
t_k^X	root of $F_X(t) = k - \frac{1}{2}$ (Remark 5.7)
ℓ_n	$\log n$
ψ, t_1, t_2	test function and endpoints in Lemma 4.7
y_0, w_T	fixed threshold; threshold tending to infinity (Corollary 5.4)
$\mathfrak{g}$, $\mathfrak{G}$, ω	continuous test function and a primitive (Theorem 4.2); real frequency variable
$\mathfrak{t}_k(X)$, y_k	transport term $S_X(\gamma_k) - S_X(\tau_k)$; one-step correction $S_X(\tau_k)/\mathcal{L}(\tau_k)$

symbol	meaning
φ, $\mathfrak{h}$, $\mathcal{M}$, $\mathfrak{b}_T$, $\mathcal{U}_k$	cutoff function; $\varphi S'_X$; centred Hardy–Littlewood maximal operator; $2\pi/\log(3T)$; sampling intervals (Lemma 7.2)
X_k, $\tilde{u}_k$, $X_\pm$, $V_\pm$	variable cutoff; its defect; extreme cutoffs in a window; $\sum_{X_-<p\le X_+} p^{-1}$ (Theorem 5.12)
$\mathcal{Q}$, $\mathfrak{w}$	maximal partial prime sum; smooth window (Lemma 5.11)
$\mathcal{I}_T^\tau$, τ_k^{loc}	$\{k : T \le \tau_k < 2T\}$; one-step clock with cutoff $\tau_k^{1/\sqrt{\log\log\tau_k}}$ (Corollary 7.5)
$\mathfrak{a}$, BW_S	$\limsup_{t\to\infty} \lvert S(t)\rvert/\log t$; Bellotti–Wong bound for $\lvert S(t)\rvert$ (Theorem 3.2)
$\mathfrak{e}$, (k,l)	exponent profile of the Riemann zeta function; exponent pair (Section 9)
$\sigma_\circ$, ϱ, $\varkappa$, $\sigma^\sharp$, $\delta^\sharp$	centre and radius of the Jensen disc; power in Backlund's function; parameters of Backlund's trick (Theorem 3.14)
$x_k^{(j)}$, $e_{j,k}$, $\chi_{j,k}$, $\lVert\cdot\rVert_{r,I}$	iterated Newton points (8.1); their errors; curvature terms; discrete norms (Section 8)
H_j, c_j, y_k, $\tilde{n}_j$	$H_0 = \log\log T$, $H_j = 2\log(1/c_j)$ and the cutoff law (8.13); $y_k = S_{X_1}(\tau_k)/\mathcal{L}(\tau_k)$; multiplicity functions (Lemma 8.4, Theorem 8.10)
$A_j(r)$, m^*, $C_\star$, H^*, $B_\star$, C_5, C_6	normalized moments of the errors, number of steps and constants of Theorem 8.11
$n_{\mathsf{R}}(u)$, C_1, C_2, C_3	number of ordinates of the window within $\mathsf{R}\mathfrak{b}_T$ of u; constants of Lemma 8.9 and Theorem 8.10
M, P, F, s, W, y	local names in proofs: $\#\mathcal{I}_T$; number of close pairs; g^{4r} (Lemma 8.5); window and scale (Lemma 5.11)
$\mathfrak{k}$, $\mathfrak{k}_j$	multiplicity constants: least $\mathfrak{k} \ge 1$ with $\int \tilde{n}^2 \le \mathfrak{k}\,\mathfrak{b}_T \#\mathcal{I}_T$ (Lemma 8.5, Theorem 8.11)
$y_k^{(j)}$, $\mathfrak{r}_{j,k}$, $d_{j,k}$	$\tau_k - x_k^{(j)}$ and the remainder of Lemma 8.7; gaps $x_{k+1}^{(j)} - x_k^{(j)}$ (Proposition 8.8)
g_j	$\mathcal{M}(\varphi_j S'_{X_j})$, maximal function of the derivative of the j-th Euler phase (Proposition 8.1)
$\mathcal{A}_p$	norm of the maximal operator on L^p (Theorem 3.10)
$\mathscr{A}$	van der Corput's $\mathscr{A}$-process on exponent pairs (Theorem 3.12)
$\varepsilon_\circ$, q, C_{b}, C_{d}	parameter of the Jensen disc (proof of Theorem 9.2); fixed exponent $\log n = \mathsf{q}\log T$ (Remark 6.4); constants in the proof of Lemma 8.2
$\mathfrak{f}_\varkappa$, $\mathcal{Z}_\varkappa$, $F_\circ$	Backlund's auxiliary function, its zero sum and the majorant (Theorem 3.14)

2.2. Counting functions

Let $\vartheta(t) = \Im \log\Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2}\log\pi$ for $t > 0$. For $t > 0$ not an ordinate [27, Thms. 9.3–9.4],

$$N(t) = \frac{1}{\pi}\vartheta(t) + 1 + S(t), \qquad S(t) = \frac{1}{\pi}\arg\zeta\Big(\tfrac{1}{2} + it\Big) = O(\log t), \tag{2.1}$$

with $\arg\zeta$ defined by continuous variation from 2 to $2 + it$ and then horizontally to $\frac{1}{2} + it$. At an ordinate we keep the right-continuous N and define S by (2.1); $S^*(\gamma) = \frac{1}{2}(S(\gamma^-) + S(\gamma^+))$.

Put

$$\Theta_0(t) = \frac{t}{2}\log\frac{t}{2\pi e} - \frac{\pi}{8}, \qquad N_0(t) = \frac{1}{\pi}\Theta_0(t) + 1, \qquad \varpi(t) = \vartheta(t) - \Theta_0(t). \tag{2.2}$$

By Stirling's formula $\varpi(t) = \frac{1}{48t} + O(t^{-3})$ and $\varpi'(t) = O(t^{-2})$; in particular $|\varpi(t)|/\pi < \frac{1}{100}$ for $t \geq 14$. For all $t > 0$,

$$N(t) = N_0(t) + S(t) + \frac{\varpi(t)}{\pi}, \qquad \mathcal{L}(t) = N_0'(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}. \tag{2.3}$$

Since $N(2T) - N(T) = N_0(2T) - N_0(T) + O(\log T)$,

$$\#\mathcal{I}_T = T\mathcal{L}(T) + O(T). \tag{2.4}$$

2.3. The RvM clock and the displacements

N_0 is C^∞ and strictly increasing on $(2\pi, \infty)$, with $N_0(2\pi) = -\frac{1}{8}$ and $N_0(t) \to \infty$. Hence for every real $x \geq 1$ the equation $N_0(t) = x - \frac{1}{2}$ has exactly one solution $\tau_x > 2\pi$. Writing $t = 2\pi y$, the equation is $y\log(y/e) = x - \frac{11}{8} \geq -\frac{3}{8}$; its root $y > 1$ is $y = (x - \frac{11}{8})/W_0((x - \frac{11}{8})/e)$, with $y = e$ if $x = \frac{11}{8}$ [5]. Thus

$$\tau_x = \frac{2\pi\,(x - \frac{11}{8})}{W_0\Big((x - \frac{11}{8})/e\Big)}, \qquad \frac{d\tau_x}{dx} = \frac{1}{\mathcal{L}(\tau_x)} = \frac{2\pi}{\log(\tau_x/2\pi)}, \tag{2.5}$$

the derivative by implicit differentiation. Since $\Theta_0(\tau_k) = \pi(k - \frac{3}{2})$ and $\varpi = O(1/t)$, τ_k lies in phase half way between g_{k-2} and g_{k-1}, up to $O(\tau_k^{-1}(\log\tau_k)^{-1})$.

Definition 2.1. For $k \geq 1$, $s_k = k - \frac{1}{2} - N_0(\gamma_k)$ and $\delta_k = \tau_k - \gamma_k$.

Since $\gamma_1 > 14 > 2\pi$ and $\tau_k > 2\pi$, the mean value theorem applied to $s_k = N_0(\tau_k) - N_0(\gamma_k)$ gives

$$s_k = \mathcal{L}(\xi_k)\,\delta_k \qquad \text{for some } \xi_k \text{ between } \gamma_k \text{ and } \tau_k. \tag{2.6}$$

Lemma 2.2. *For every $k \geq 1$,*

$$N(\gamma_k^-) - N_0(\gamma_k) + \frac{1}{2} \leq s_k \leq N(\gamma_k^+) - N_0(\gamma_k) - \frac{1}{2}, \tag{2.7}$$

and

$$\left|s_k - \frac{\varpi(\gamma_k)}{\pi}\right| \leq \max\Big(|S(\gamma_k^-)|, |S(\gamma_k^+)|\Big) - \frac{1}{2}. \tag{2.8}$$

Proof. The indices j with $\gamma_j = \gamma_k$ are exactly $N(\gamma_k^-) + 1, \ldots, N(\gamma_k^+)$, so $N(\gamma_k^-) + 1 \leq k \leq N(\gamma_k^+)$; subtract $\frac{1}{2} + N_0(\gamma_k)$. By (2.3) at $\gamma_k^\pm$ and continuity of N_0 and ϖ, (2.7) reads $S^- + \frac{1}{2} \leq x \leq S^+ - \frac{1}{2}$ with $x = s_k - \varpi(\gamma_k)/\pi$, $S^- = S(\gamma_k^-)$, $S^+ = S(\gamma_k^+)$, and $S^+ - S^- = \nu(\gamma_k) \geq 1$. Put $\mathsf{m} = \max(|S^-|, |S^+|)$. Then $x \leq S^+ - \frac{1}{2} \leq \mathsf{m} - \frac{1}{2}$ and $x \geq S^- + \frac{1}{2} \geq -\mathsf{m} + \frac{1}{2}$, which is (2.8). □

2.4. Euler phases and the corrected clock

For $n \geq 2$ let $a_n = -\Lambda(n)/(\pi\sqrt{n}\log n)$. For a set A of prime powers with $\sum_{n\in A}|a_n| < \infty$,

$$S_A(t) = \sum_{n\in A} a_n \sin(t\log n), \qquad v_A = \frac{1}{2}\sum_{n\in A} a_n^2; \tag{2.9}$$

the series converges absolutely and uniformly in t, so S_A is continuous and bounded. For $X \geq 2$, $\sum_{n\in A_X}|a_n| \leq \frac{1}{\pi}\sum_{p\leq X}\sum_{m\geq 1}\frac{1}{mp^{m/2}} < \infty$ and, since $\arg(1-z) = -\Im\sum_{m\geq 1} z^m/m$ for $|z| < 1$ with the principal argument,

$$S_X(t) = -\frac{1}{\pi}\sum_{p\leq X}\arg\bigl(1-p^{-1/2-it}\bigr) = -\frac{1}{\pi}\sum_{p\leq X}\sum_{m\geq 1}\frac{\sin(mt\log p)}{m\,p^{m/2}} = S_{A_X}(t). \tag{2.10}$$

Each factor has positive real part: $1 - p^{-1/2}\cos(t\log p) > 0$. Its argument is therefore the single continuous value $\arctan(p^{-1/2}\sin(t\log p)/(1-p^{-1/2}\cos(t\log p)))$ in $(-\pi/2, \pi/2)$. The phase in (2.10) is the sum of these values; reducing the argument of the entire product modulo 2π would change the function. Thus no branch is selected from the zeros or by fitting. The series of derivatives converges uniformly, because $\sum_{p\leq X}\log p\sum_{m\geq 1}p^{-m/2} < \infty$; hence $S_X \in C^\infty(\mathbb{R})$, its derivatives may be taken termwise, and

$$|S_X(t)| \leq \sigma_X := \frac{1}{\pi}\sum_{p\leq X}\sum_{m\geq 1}\frac{1}{m\,p^{m/2}}, \qquad |S_X'(t)| \leq M_X := \frac{1}{\pi}\sum_{p\leq X}\frac{\log p}{\sqrt{p}-1}, \tag{2.11}$$

with $\sigma_X \ll \sqrt{X}/\log X$ and $M_X \ll \sqrt{X}$ by the prime number theorem.

Definition 2.3. For $X \geq 2$ let $F_X = N_0 + S_X$ and $u_k(X) = k - \frac{1}{2} - F_X(\gamma_k) = s_k - S_X(\gamma_k)$. The one-step clock and its residual are

$$\tau_k^X = \tau_k - \frac{S_X(\tau_k)}{\mathcal{L}(\tau_k)}, \qquad \delta_k^X = \tau_k^X - \gamma_k, \qquad r_k(X) = \mathcal{L}(\tau_k)\,\delta_k^X. \tag{2.12}$$

Since $F_X(\tau_k) - (k-\frac{1}{2}) = S_X(\tau_k)$, τ_k^X is the first step of $t \mapsto t - (F_X(t) - k + \frac{1}{2})/\mathcal{L}(t)$ from $t = \tau_k$. At a simple ordinate, (2.3) gives $N_0(\gamma_k) + S^*(\gamma_k) + \varpi(\gamma_k)/\pi = k - \frac{1}{2}$, so $u_k(X) = S^*(\gamma_k) - S_X(\gamma_k) + \varpi(\gamma_k)/\pi$ is the error of the prime approximation of the argument of the Riemann zeta function at the zero.

3. External results

The results of this section are known; they are used, not reproved.

Theorem 3.1 ([27, Thms. 9.4, 9.9 and §9.12], [20])**.** *$S(t) = O(\log t)$; $\gamma_{k+1} - \gamma_k$ is bounded; and $S_1(t) = \int_0^t S(y)\,dy = O(\log t)$.*

Theorem 3.2 (Bellotti–Wong [1, Thm. 1.1, Cor. 1.5 and (1.2)])**.** *For $T \geq e$, $\bigl|N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}\bigr| \leq \mathrm{BW}(T)$ and $|S(T)| \leq \mathrm{BW}_S(T)$, where*

$$\mathrm{BW}_S(t) = 0.10076\log t + \min\{0.24460\log\log t + 7.20844,\ 1.68845\log\log t + 1.50956\}. \tag{3.1}$$

Moreover $|S(T)| \leq 2.5167$ for $0 \leq T \leq 30\,610\,046\,000$.

Earlier explicit versions are due to Trudgian [29] and to Hasanalizade, Shen and Wong [10]. For T sufficiently large, Trudgian [28, §5 of the preprint] proved $|S(T)| \le 0.1013 \log T$ with Huxley's exponent $\frac{32}{205}$ for $\zeta(\frac12 + it)$. The last bound rests on Platt's rigorous isolation of the zeros up to that height [1, (1.2)].

Theorem 3.3 (Selberg [23]; see [8, p. 225]). *For every integer $j \ge 1$, $\int_T^{2T} S(t)^{2j}\,dt = \kappa_j\, T(\log\log T)^j + O_j\Big(T(\log\log T)^{j-1/2}\Big)$.*

Theorem 3.4 (Selberg; Korolev [18], [14, Thm. 4.2]). *Let Δ_k be the integer such that $g_{k+\Delta_k-2} < \gamma_k \le g_{k+\Delta_k-1}$ (in [14] the Gram points are $t_m = g_{m-1}$ and $t_{m-1} < \gamma_k \le t_m$ with $m = k + \Delta_k$), and let $0 < \varepsilon_0 < 10^{-3}$. For every fixed $j \ge 1$, as $n_0 \to \infty$, uniformly for $n_0^{27/82+0.9\varepsilon_0} \le h_0 \le n_0^{27/82+\varepsilon_0}$,*

$$\sum_{n_0<k\le n_0+h_0} \Delta_k^{2j} = \kappa_j h_0 (\log\log n_0)^j \Big(1 + o(1)\Big), \qquad \Big|\sum_{n_0<k\le n_0+h_0} \Delta_k^{2j-1}\Big| \ll_j h_0 (\log\log n_0)^{j-1}. \tag{3.2}$$

Theorem 3.5 (Selberg's approximation in the form of Fujii [8, §3, pp. 232–233]). *For every integer $k \ge 1$, $\int_T^{2T} \Big(S(t) - S^{\mathrm{pr}}_{Y_k}(t)\Big)^{2k} dt \le (Ck^2)^{2k}\, T$ with $Y_k = T^{1/(80k)}$ and an absolute constant C. In particular $\int_T^{2T} (S - S^{\mathrm{pr}}_{Y_0})^4 \ll T$ with $Y_0 = Y_2 = T^{1/160}$.*

In [8, §3] Selberg's difference between $\pi^{-1}\arg L(\frac12 + it, \chi)$ and a prime sum is written as five explicit terms, each bounded on p. 233 in $L^{2k}(T, T+H)$ by $\ll k^2 H^{1/2k}$ under the hypotheses of Fujii's Main Theorem, which restrict q, H and h but not k. For the principal character (so that the L-function is the Riemann zeta function), $H = T$ and Fujii's parameter $a_2 = \frac14$, the prime sum runs over $p < x^3$ with $x = T^{a_2/60k}$, that is $p < T^{1/(80k)} = Y_k$; the five terms are bounded by $k^{1/2}$, k^2, k^2, $k^{1/2}$ and $k^{3/2}$ times $H^{1/2k}$, with absolute implied constants, whence the constant $(Ck^2)^{2k}$; and $\frac{1}{\pi}\Im p^{-1/2-it} = -\frac{1}{\pi} p^{-1/2}\sin(t\log p)$. For $k = 2$ this is the form used in Section 5; $k = 4$, with $Y_4 = T^{1/320}$, and the general k are used in Section 8. The uniformity in k of Selberg's approximation is stated with explicit constants in other places: Korolev [18, Lemma 4] proves $\sum_{N<n\le N+M}(S(t_n+0) + V_y(t_n+0))^{2k} \le (Ae^{-4}k)^{2k} M$ at the Gram points t_n, with $V_y(t) = \frac{1}{\pi}\sum_{p<y} p^{-1/2}\sin(t\log p)$, $y = x^{1/(4k)}$, $x = t_N^{0.1\varepsilon}$, uniformly for $1 \le k \le \frac{1}{192}\log x$, and [18, Thm. 2] bounds the $2k$-th moments of $S(t_n+0) - S(t_{n-1}+0)$ by $2Mk(4k\sqrt{B})^{2k}$ in the same range; a t-mean version over short intervals is [15, Lemma 3.13], as quoted in [16, Lemma 12]; for $\log|\zeta(\frac12 + it)|$ the bound $\ll A^k k^{4k}$, of the same shape $(Ck^2)^{2k}$, is [17, Lemma 1], after Tsang [31, p. 60].

Theorem 3.6 (Fujii [8, Main Theorem, p. 225]). *For every integer $k \ge 1$ there is C_k such that, for $T \ge T_0$ and $0 < h \le 1$,*

$$\int_T^{2T} \Big(S(t+h) - S(t)\Big)^{2k} dt \le C_k\, T \Big(\log(2 + h\log T)\Big)^k.$$

Part II of Fujii's paper [9] carries corrections to Part I; the two statements quoted here are those of Part I, pp. 225 and 232–233. Fujii's statement is an asymptotic formula with main term $c_k H(2\log(2 + h\log T))^k$ and relative error $O((Ak)^{3k}(\log(2 + h\log T))^{-1/2})$ for $T^{1/2+a_2} \le H \le T$; the upper bound with $H = T$ follows for all $0 < h \le 1$. Only $k = 1, 2$ are used (Lemma 8.9).

Theorem 3.7 (Montgomery–Vaughan [22, Cor. 3]). *For complex z_n, $\int_T^{2T}\big|\sum_{n\le Y} z_n n^{-it}\big|^2 dt = \sum_{n\le Y}|z_n|^2\big(T+O(n)\big)$ with an absolute implied constant.*

(Replace z_n by $z_n n^{-iT}$ to pass from $[0,T]$ to $[T,2T]$.)

Theorem 3.8 (Mertens [21]; see [26, Ch. I.1, Thms. 1.8 and 1.10]). *For $y\ge 2$, $\sum_{p\le y}\frac{\log p}{p} = \log y + O(1)$ and $\sum_{p\le y}\frac{1}{p} = \log\log y + O(1)$.*

Theorem 3.9 (Selberg [23]; Tsang [31]; see [6, (1.1)]). *For fixed real $y_1<y_2$, $\frac{1}{T}\operatorname{meas}\{t\in[T,2T] : y_1 < S(t)/\mathfrak{s}_T \le y_2\} \to \frac{1}{\sqrt{2\pi}}\int_{y_1}^{y_2} e^{-y^2/2}dy$, where $\mathfrak{s}_T = (\frac{1}{2\pi^2}\log\log T)^{1/2}$.*

In [6] the statement is that $\log\zeta(\frac12+it)/\sqrt{\log\log T}$ tends to a standard complex Gaussian for t uniform in $[T,2T]$; its imaginary part is $\pi S(t)/\sqrt{\log\log T}$, of variance $\frac12$.

Theorem 3.10 (Hardy–Littlewood [25, Thm. I.1], as quoted in [13, (1.2)]; Kinnunen [13, Thm. 1.4]). *Let $\mathcal{M}f(x) = \sup_{y>0}\frac{1}{2y}\int_{x-y}^{x+y}|f|$ be the centred maximal function of $f\in L^1_{\mathrm{loc}}(\mathbb{R})$. For $1<p<\infty$ and $f\in L^p(\mathbb{R})$, $\|\mathcal{M}f\|_p \le \mathcal{A}_p\|f\|_p$, where $\mathcal{A}_p$ is bounded for $p\ge 2$. If f and f' belong to $L^p(\mathbb{R})$, $1<p<\infty$, then $\mathcal{M}f$ has a locally absolutely continuous representative, $(\mathcal{M}f)'\in L^p(\mathbb{R})$, and $|(\mathcal{M}f)'|\le \mathcal{M}(f')$ almost everywhere.*

Theorem 3.11 (Carleson [4], Hunt [11]; on $\mathbb{R}$ see [19]). *For $f\in L^2(\mathbb{R})\cap L^4(\mathbb{R})$ with Fourier transform $\mathcal{F}f(\omega)=\int_{\mathbb{R}} f(t)e^{-i\omega t}dt$, let $\mathcal{C}f(x) = \sup_{\omega_1\in\mathbb{R}}\big|\frac{1}{2\pi}\int_{-\infty}^{\omega_1}\mathcal{F}f(\omega)e^{i\omega x}d\omega\big|$. Then $\|\mathcal{C}f\|_4 \ll \|f\|_4$.*

In [19] the statement is that $f\mapsto \sup_{\xi\in\mathbb{R}}|\mathcal{H}(e^{i\xi\cdot}f)|$, $\mathcal{H}$ the Hilbert transform, is bounded on $L^p(\mathbb{R})$ for $1<p<\infty$; since the partial Fourier integral up to ω_1 is $e^{i\omega_1 x}\cdot\frac12(\mathrm{Id}\pm i\mathcal{H})(e^{-i\omega_1\cdot}f)(x)$, the operator $\mathcal{C}f$ is at most $\frac12(|f|+\sup_\xi|\mathcal{H}(e^{i\xi\cdot}f)|)$.

Theorem 3.12 ([12, Lemma 2.8 and (7.57)]; Bourgain [3, §5, Thm. 6]). *(a) If (k,l) is an exponent pair, so is $\mathscr{A}(\mathsf{k},\mathsf{l}) = \big(\frac{\mathsf{k}}{2\mathsf{k}+2}, \frac12+\frac{\mathsf{l}}{2\mathsf{k}+2}\big)$. (b) If (k,l) is an exponent pair with $\mathsf{l}-\mathsf{k}\ge\frac12$, then $\zeta(\mathsf{l}-\mathsf{k}+it)\ll|t|^{\mathsf{k}}\log|t|$ for $|t|\ge 2$. (c) For every $\varepsilon>0$, $(\frac{13}{84}+\varepsilon, \frac{55}{84}+\varepsilon)$ is an exponent pair.*

For (b): [12, (7.57)] gives $\zeta(\sigma+it)\ll|t|^{(\mathsf{k}+\mathsf{l}-\sigma)/2}\log|t|$ for $\frac12\le\sigma\le\mathsf{l}-\mathsf{k}$; take $\sigma=\mathsf{l}-\mathsf{k}$.

Theorem 3.13 (Phragmén–Lindelöf principle in a strip [1, Lemma 2.5], [29, Lemma 3]). *Let $\sigma_1<\sigma_2$ and $e_1\ge e_2\ge 0$. If $|\zeta(\sigma_i+it)|\le C_i|t|^{e_i}\log|t|$ for $|t|\ge 2$ and $i=1,2$, then $|\zeta(\sigma+it)|\ll|t|^{e(\sigma)}\log|t|$ uniformly for $\sigma_1\le\sigma\le\sigma_2$ and $|t|\ge 2$, where e is the linear function with $e(\sigma_i)=e_i$.*

(Apply [1, Lemma 2.5] to the entire function $(s-1)\zeta(s)$ of order 1, with exponents $1+e_i$ and one power of the logarithm on the two lines.)

Theorem 3.14 (Backlund's method [1, Lemmas 2.1, 2.3 and 2.4], after [29, 10]). *Let T be large, $\varkappa\in\mathbb{N}$, $\mathfrak{f}_\varkappa(s) = \frac12\big[((s+iT-1)\zeta(s+iT))^\varkappa + ((s-iT-1)\zeta(s-iT))^\varkappa\big]$, and for a disc $D(\sigma_\circ,\varrho)$ let $\mathcal{Z}_\varkappa(\sigma_\circ,\varrho) = \frac{1}{\varkappa}\sum_z \log\frac{\varrho}{|z-\sigma_\circ|}$ over the zeros z of $\mathfrak{f}_\varkappa$ in $D(\sigma_\circ,\varrho)$. Let $1<\sigma_\circ<\varrho$, $\sigma^\sharp = \sigma_\circ + (\sigma_\circ-\frac12)^2/\varrho$, $\delta^\sharp = 2\sigma_\circ - \sigma^\sharp - \frac12$, and assume $0<\delta^\sharp<\frac12$.*

(a) If $F_\circ:[-\pi,\pi]\to\mathbb{R}$ is even and $F_\circ(\theta)\ge\frac{1}{\varkappa}\log|\mathfrak{f}_\varkappa(\sigma_\circ+\varrho e^{i\theta})|$ for all $\varkappa$ and θ, then there are $\varkappa_m\to\infty$ with $\limsup_m\mathcal{Z}_{\varkappa_m}(\sigma_\circ,\varrho)\le\log\frac{\zeta(\sigma_\circ)}{\sqrt{(\sigma_\circ-1)^2+T^2}\,\zeta(2\sigma_\circ)}+\frac{1}{\pi}\int_0^\pi F_\circ(\theta)\,d\theta$.

(b) For every $\varkappa$, $\Big|\Delta^{1/2}_{\sigma=\sigma^\sharp}\arg\Big((\sigma+iT-1)\zeta(\sigma+iT)\Big)\Big|\le\frac{\pi\mathcal{Z}_\varkappa(\sigma_\circ,\varrho)}{2\log(\varrho/(\sigma_\circ-\frac12))}+\frac{\pi}{100}+\frac{3\pi}{2\varkappa}+\frac{\pi}{4}$.

In (b) the term $\frac{\pi}{100}$ bounds $\frac12E(T,\delta^\sharp)$ of [1, Lemma 2.3], by the monotonicity in $\delta^\sharp$ and the bound for $E(T,\frac12)$ in [1, Lemma 2.4]; (a) is [1, Lemma 2.1], which also requires $\sigma_\circ-\varrho<\frac12<1<\sigma_\circ<\sigma^\sharp<\sigma_\circ+\varrho$, a consequence of $1<\sigma_\circ<\varrho$.

Remark 3.15 (Known consequences for the clock). (a) *Moments and law of s_k.* Since $\vartheta(\gamma_k)/\pi\in(k+\Delta_k-2,k+\Delta_k-1]$ and $\vartheta/\pi=N_0-1+\varpi/\pi$, one has $s_k+\Delta_k=k+\Delta_k-\frac12-N_0(\gamma_k)\in[-\frac12+\frac{\varpi(\gamma_k)}{\pi},\frac12+\frac{\varpi(\gamma_k)}{\pi})$, so $s_k=-\Delta_k+\mathfrak{d}_k$ with $|\mathfrak{d}_k|<1$. The index set $\mathcal{I}_T$ is the integer interval $(N(T),N(2T)]$, of length $\asymp T\log T$. Cover it by consecutive windows $(n_0,n_0+h_0]$ with $h_0=\lfloor n_0^{27/82+0.95\varepsilon_0}\rfloor$, the last one shifted to end at $N(2T)$, so that it overlaps the previous one in at most $h_0=O(T^{-1/2}\#\mathcal{I}_T)$ indices; in each window $\log\log n_0=\log\log T+o(1)$. Summing (3.2) over the windows, and bounding the doubly counted indices by one window sum (for odd powers, by Hölder's inequality and the even moments), gives $\langle\Delta_k^{2j}\rangle_{\mathcal{I}_T}=\kappa_j(\log\log T)^j(1+o(1))$ and $\langle\Delta_k^{2j-1}\rangle_{\mathcal{I}_T}=O_j((\log\log T)^{j-1})$. Expanding $s_k^r=(-\Delta_k+\mathfrak{d}_k)^r$ and bounding $\langle|\Delta_k|^{r-i}\rangle_{\mathcal{I}_T}\le\langle\Delta_k^{2\lceil r/2\rceil}\rangle_{\mathcal{I}_T}^{(r-i)/2\lceil r/2\rceil}\ll(\log\log T)^{(r-i)/2}$ for $1\le i\le r$, one obtains for every fixed $j\ge1$

$$\langle s_k^{2j}\rangle_{\mathcal{I}_T}=\kappa_j(\log\log T)^j\Big(1+o(1)\Big),\qquad\langle s_k^{2j-1}\rangle_{\mathcal{I}_T}=O_j\Big((\log\log T)^{j-1}\Big).\tag{3.3}$$

With $\mathfrak{s}_T^2=\frac{\log\log T}{2\pi^2}$, so that $\kappa_j(\log\log T)^j=\frac{(2j)!}{j!2^j}\mathfrak{s}_T^{2j}$, the normalized moments $\langle(s_k/\mathfrak{s}_T)^r\rangle_{\mathcal{I}_T}$ therefore tend to the moments of a standard normal variable Z for every $r\ge1$: the even ones by the first relation, the odd ones to 0 because $(\log\log T)^{j-1}/\mathfrak{s}_T^{2j-1}\ll_j(\log\log T)^{-1/2}$. The even moments alone determine only the law of $|s_k|$; it is the odd-moment bound in (3.2) that gives the sign. The normal law is determined by its moments, so the method of moments [2, Thm. 30.2] shows that the empirical law of $s_k/\mathfrak{s}_T$ on $\mathcal{I}_T$ tends to that of Z. Only consequences for $|s_k|$ are used in this paper: $\langle s_k^2\rangle_{\mathcal{I}_T}\sim\frac{1}{2\pi^2}\log\log T$, $\limsup_k|s_k|=\infty$, and $\#\{k\in\mathcal{I}_T:|s_k|\le y_0\}=o(\#\mathcal{I}_T)$ for every fixed $y_0>0$; for these the even moments suffice.

(b) *Bound for δ_k.* Theorem 3.2, applied as $T\to\gamma_k^\pm$ (both sides have one-sided limits), gives $|N(\gamma_k^\pm)-N_0(\gamma_k)+\frac78|\le\mathrm{BW}(\gamma_k)$, and (2.7) gives $-\mathrm{BW}(\gamma_k)-\frac38\le s_k\le\mathrm{BW}(\gamma_k)-\frac{11}{8}$. As $\mathcal{L}$ is positive and increasing on $(2\pi,\infty)$ and $s_k=\int_{\gamma_k}^{\tau_k}\mathcal{L}$,

$$-\frac{\mathrm{BW}(\gamma_k)+3/8}{\mathcal{L}(\tau_k)}\le\delta_k\le\frac{\mathrm{BW}(\gamma_k)-11/8}{\mathcal{L}(\gamma_k)}\qquad(k\ge1).\tag{3.4}$$

Hence $\delta_k=O(1)$, $\mathcal{L}(\tau_k)/\mathcal{L}(\gamma_k)\to1$, and $\limsup_k|\delta_k|\le2\pi\cdot0.10076=0.6331\ldots$, an immediate consequence of [1].

4. Theorem A: an exact defect identity at the zeros

Theorem 4.1. *Let $1\le k_1<k_2$ be indices, F a real C^1 function on an open interval containing $[\gamma_{k_1},\gamma_{k_2}]$, and $u_k=k-\frac12-F(\gamma_k)$ for $k_1\le k\le k_2$. For every integer $j\ge1$,*

$$\int_{\gamma_{k_1}}^{\gamma_{k_2}}\Big(N(t)-F(t)\Big)^{2j}F'(t)\,dt=\sum_{k=k_1+1}^{k_2-1}\Psi_j(u_k)+\frac{(\frac12+u_{k_1})^{2j+1}+(\frac12-u_{k_2})^{2j+1}}{2j+1},\tag{4.1}$$

where

$$\Psi_j(u) = \frac{(\frac{1}{2}+u)^{2j+1} + (\frac{1}{2}-u)^{2j+1}}{2j+1} = \sum_{l=0}^{j} \binom{2j+1}{2l} \frac{4^{l-j}}{2j+1}\, u^{2l} = u^{2j} + \sum_{l<j} \binom{2j+1}{2l} \frac{4^{l-j}}{2j+1} u^{2l}. \tag{4.2}$$

In particular, since $\Psi_1(u) = u^2 + \frac{1}{12}$,

$$\sum_{k=k_1+1}^{k_2-1} u_k^2 = \int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(N(t) - F(t)\Big)^2 F'(t)\, dt - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac{1}{2}+u_{k_1})^3 + (\frac{1}{2}-u_{k_2})^3}{3}. \tag{4.3}$$

Proof. Existence. On $[\gamma_{k_1}, \gamma_{k_2}]$ the function N is a step function with finitely many jumps (Theorem 3.1) and F, F' are continuous; the integrand is bounded and piecewise continuous, hence Riemann integrable.

One gap. Fix k with $k_1 \le k < k_2$. If $\gamma_k < \gamma_{k+1}$, then $N(t) = k$ for $t \in (\gamma_k, \gamma_{k+1})$, because exactly k ordinates counted with multiplicity are $\le t$. The function $t \mapsto -(k - F(t))^{2j+1}/(2j+1)$ is C^1 with derivative $(k - F(t))^{2j} F'(t)$ (chain rule; no monotonicity of F is needed), so

$$\begin{aligned} \int_{\gamma_k}^{\gamma_{k+1}} \Big(k - F(t)\Big)^{2j} F'(t)\, dt &= \frac{(k - F(\gamma_k))^{2j+1} - (k - F(\gamma_{k+1}))^{2j+1}}{2j+1} \\ &= \frac{(\frac{1}{2}+u_k)^{2j+1} + (\frac{1}{2}-u_{k+1})^{2j+1}}{2j+1}, \end{aligned} \tag{4.4}$$

because $k - F(\gamma_k) = \frac{1}{2} + u_k$ and $k - F(\gamma_{k+1}) = -\frac{1}{2} + u_{k+1}$. If $\gamma_k = \gamma_{k+1}$, the left side of (4.4) is 0; then $u_{k+1} = u_k + 1$, and the right side equals $((\frac{1}{2}+u_k)^{2j+1} + (-\frac{1}{2}-u_k)^{2j+1})/(2j+1) = 0$. So (4.4) holds for every k, and the values of N at the finitely many ordinates do not affect the integrals.

Summation and boundary terms. Summing (4.4) over $k_1 \le k < k_2$ (a finite sum), each k with $k_1 < k < k_2$ contributes $(\frac{1}{2}+u_k)^{2j+1}$ from the gap starting at γ_k and $(\frac{1}{2}-u_k)^{2j+1}$ from the gap ending at γ_k, that is $\Psi_j(u_k)$; the unmatched terms are $(\frac{1}{2}+u_{k_1})^{2j+1}$ and $(\frac{1}{2}-u_{k_2})^{2j+1}$. In the binomial expansion of $(\frac{1}{2}+u)^{2j+1} + (\frac{1}{2}-u)^{2j+1}$ the odd powers of u cancel and u^{2l} has coefficient $2\binom{2j+1}{2l} 2^{2l-2j-1}$, which gives (4.2); the leading coefficient is $\binom{2j+1}{2j}/(2j+1) = 1$. □

Theorem 4.2. *Let $k_1 < k_2$, F and u_k be as in Theorem 4.1, and let $\mathfrak{g} : \mathbb{R} \to \mathbb{C}$ be continuous with a primitive $\mathfrak{G}$. Then*

$$\int_{\gamma_{k_1}}^{\gamma_{k_2}} \mathfrak{g}\Big(N(t) - F(t)\Big) F'(t)\, dt = \sum_{k=k_1+1}^{k_2-1} \Big[\mathfrak{G}\Big(u_k + \tfrac{1}{2}\Big) - \mathfrak{G}\Big(u_k - \tfrac{1}{2}\Big)\Big] + \mathfrak{G}\Big(u_{k_1} + \tfrac{1}{2}\Big) - \mathfrak{G}\Big(u_{k_2} - \tfrac{1}{2}\Big). \tag{4.5}$$

Proof. The integrand is bounded and piecewise continuous, as in the proof of Theorem 4.1. If $\gamma_k < \gamma_{k+1}$, then $N = k$ on (γ_k, γ_{k+1}) and $t \mapsto -\mathfrak{G}(k - F(t))$ is C^1 with derivative $\mathfrak{g}(k - F(t))F'(t)$, so the integral over $[\gamma_k, \gamma_{k+1}]$ is $\mathfrak{G}(k - F(\gamma_k)) - \mathfrak{G}(k - F(\gamma_{k+1})) = \mathfrak{G}(u_k + \frac{1}{2}) - \mathfrak{G}(u_{k+1} - \frac{1}{2})$. If $\gamma_k = \gamma_{k+1}$, the integral is 0 and $u_{k+1} = u_k + 1$, so the same formula holds. Summing over $k_1 \le k < k_2$, each k with $k_1 < k < k_2$ contributes $\mathfrak{G}(u_k + \frac{1}{2})$ from the gap that starts at γ_k and $-\mathfrak{G}(u_k - \frac{1}{2})$ from the gap that ends there. □

For $\mathfrak{g}(x) = x^{2j}$ this is (4.1), since $\mathfrak{G}(u + \frac{1}{2}) - \mathfrak{G}(u - \frac{1}{2}) = \Psi_j(u)$.

Corollary 4.3. *For real $\omega \neq 0$,*

$$\frac{2\sin(\omega/2)}{\omega}\sum_{k=k_1+1}^{k_2-1} e^{i\omega u_k} = \int_{\gamma_{k_1}}^{\gamma_{k_2}} e^{i\omega(N(t)-F(t))}F'(t)\,dt - \frac{e^{i\omega(u_{k_1}+1/2)} - e^{i\omega(u_{k_2}-1/2)}}{i\omega}. \tag{4.6}$$

Proof. Theorem 4.2 with $\mathfrak{g}(x) = e^{i\omega x}$ and $\mathfrak{G}(x) = e^{i\omega x}/(i\omega)$, for which $\mathfrak{G}(u+\frac{1}{2}) - \mathfrak{G}(u-\frac{1}{2}) = e^{i\omega u}\cdot 2\sin(\omega/2)/\omega$. □

Corollary 4.4 (Signed law of s_k from the t-law of S)**.** *For fixed real ξ, $\langle e^{i\xi s_k/\mathfrak{s}_T}\rangle_{\mathcal{I}_T} \to e^{-\xi^2/2}$. Hence the empirical law of $s_k/\mathfrak{s}_T$ on $\mathcal{I}_T$ tends to the standard normal law.*

Proof. Let $\xi \neq 0$, $\omega = \xi/\mathfrak{s}_T$, $k_1 = N(T)$ and $k_2 = N(2T)+1$, so that $\{k_1+1,\dots,k_2-1\} = \mathcal{I}_T$. Apply (4.6) with $F = N_0$, so $u_k = s_k$ and $N - F = G$. By Theorem 3.1 the intervals $[\gamma_{k_1}, T]$ and $[2T, \gamma_{k_2}]$ have bounded length, so the integral equals $\int_T^{2T} e^{i\omega G}\mathcal{L}\,dt + O(\log T)$; the boundary term is $O(1/|\omega|) = O(\mathfrak{s}_T)$; and $2\sin(\omega/2)/\omega = 1 + O(\omega^2)$. Divide by $\#\mathcal{I}_T = \int_T^{2T}\mathcal{L}\,dt + O(\log T)$. On $[T, 2T]$, $|e^{i\omega G} - e^{i\omega S}| \le |\omega||\varpi|/\pi \ll (\mathfrak{s}_T T)^{-1}$ and $\mathcal{L}(t) = \mathcal{L}(T)(1 + O(1/\log T))$, whence

$$\langle e^{i\omega s_k}\rangle_{\mathcal{I}_T} = \frac{1}{T}\int_T^{2T} e^{i\xi S(t)/\mathfrak{s}_T}dt + O\Big(\omega^2 + \frac{1}{\log T}\Big).$$

By Theorem 3.9 and the definition of convergence in distribution, applied to the bounded continuous function $y \mapsto e^{i\xi y}$, the integral tends to $e^{-\xi^2/2}$, and $\omega \to 0$. The last statement is Lévy's continuity theorem [2, Thm. 26.3]. □

Remark 4.5. $\Psi_j(u) = \int_{u-1/2}^{u+1/2} x^{2j}dx$ is the mean of x^{2j} over the unit interval centred at u; in particular $\Psi_1(u) = u^2 + \frac{1}{12}$ and $\frac{1}{12} = \int_{-1/2}^{1/2} x^2 dx$. The constant comes from the unit jump of $N - F$ at each zero, through the primitive in Theorem 4.2; when F is not monotone, $N - F$ need not run through an interval of length one between consecutive zeros, and the identities remain exact with the integral counted with sign. With $F = N_0$, Theorem 3.3 and induction on j through the lower-degree terms of Ψ_j recover the even moments in (3.3) without Gram intervals; Corollary 4.4 gives the sign without Korolev's odd moments. The main purpose of Theorem 4.1 is Theorem 5.1, where $F = F_X$ need not be monotone and no Gram-interval structure is available.

Corollary 4.6. *For $X \ge 2$ and $1 \le k_1 < k_2$,*

$$\begin{aligned}\sum_{k=k_1+1}^{k_2-1} u_k(X)^2 = {} & \int_{\gamma_{k_1}}^{\gamma_{k_2}}\Big(S - S_X + \frac{\varpi}{\pi}\Big)^2\big(\mathcal{L} + S_X'\big)dt \\ & - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac{1}{2} + u_{k_1}(X))^3 + (\frac{1}{2} - u_{k_2}(X))^3}{3}.\end{aligned} \tag{4.7}$$

Proof. F_X is C^∞ on $(0,\infty)$ (Section 2.4), $F_X' = \mathcal{L} + S_X'$, and $N - F_X = S - S_X + \varpi/\pi$ by (2.3); apply Theorem 4.1 with $j = 1$. □

Lemma 4.7 (First-moment identity)**.** *Let $2\pi < t_1 < t_2$ be points which are not ordinates and $\psi \in C^1[t_1, t_2]$, real or complex. Then*

$$\sum_{t_1 < \gamma_k \le t_2} s_k\psi(\gamma_k) = \int_{t_1}^{t_2} G(t)\psi(t)\mathcal{L}(t)\,dt + \frac{1}{2}\Big[\psi\,G^2\Big]_{t_1}^{t_2} - \frac{1}{2}\int_{t_1}^{t_2} G(t)^2\psi'(t)\,dt. \tag{4.8}$$

Proof. On $[t_1,t_2]$ there are finitely many ordinates, none at t_1 or t_2. The function $G=N-N_0$ is right-continuous and of bounded variation, C^∞ with $G'=-\mathcal{L}$ between ordinates, and it jumps by $\nu(\gamma)$ at an ordinate γ. Let $G^*(\gamma)=\frac12(G(\gamma^-)+G(\gamma^+))$. The indices with $\gamma_k=\gamma$ are $N(\gamma^-)+1,\dots,N(\gamma^-)+\nu(\gamma)$, so $\sum_{\gamma_k=\gamma}(k-\frac12)=\nu(\gamma)N(\gamma^-)+\frac{\nu(\gamma)^2}{2}=\nu(\gamma)\frac{N(\gamma^-)+N(\gamma^+)}{2}$ and, N_0 being continuous, $\sum_{\gamma_k=\gamma}s_k=\nu(\gamma)G^*(\gamma)$. The left side of (4.8) therefore equals the finite sum $\sum_{t_1<\gamma<t_2}\nu(\gamma)G^*(\gamma)\psi(\gamma)$ over distinct ordinates. As Lebesgue–Stieltjes measures on $(t_1,t_2]$,

$$d(G^2)=-2G\mathcal{L}\,dt+\sum_{t_1<\gamma<t_2}2G^*(\gamma)\nu(\gamma)\,\delta_\gamma, \tag{4.9}$$

because $(G^2)'=2GG'=-2G\mathcal{L}$ between ordinates and $G(\gamma^+)^2-G(\gamma^-)^2=(G(\gamma^+)+G(\gamma^-))\nu(\gamma)=2G^*(\gamma)\nu(\gamma)$. Integrating the bounded Borel function ψ against (4.9) gives $\sum_{t_1<\gamma<t_2}\nu(\gamma)G^*(\gamma)\psi(\gamma)=\frac12\int_{(t_1,t_2]}\psi\,d(G^2)+\int_{t_1}^{t_2}G\psi\mathcal{L}\,dt$. Since ψ is C^1 and G^2 is right-continuous of bounded variation and continuous at t_1 and t_2, integration by parts for Lebesgue–Stieltjes integrals [7, §3.5] gives $\int_{(t_1,t_2]}\psi\,d(G^2)=[\psi G^2]_{t_1}^{t_2}-\int_{t_1}^{t_2}G^2\psi'\,dt$. All integrands are bounded and measurable, so all integrals exist. □

5. Theorem B: how much prime information the clock needs

Theorem 5.1. *There are absolute constants T_0 and C such that for $T\ge T_0$ and every X with $2\le X\le T^{1/2}$, writing $c=\log X/\log T$,*

$$\Big|\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}u_k(X)^2-\frac{1}{2\pi^2}\log\frac1c\Big|\le C\Big(1+\sqrt{\log(1/c)}\Big). \tag{5.1}$$

Here $\log(1/c)=\log\log T-\log\log X\in[\log 2,\log\log T-\log\log 2]$. For fixed c both the main term and the error are bounded; the theorem does not identify a limit at fixed c. It concerns the defect at the true ordinate and does not select a root of $F_X(t)=k-\frac12$.

Corollary 5.2 (fixed cutoff)**.** *For fixed $X\ge 2$, $\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}u_k(X)^2=\frac{1}{2\pi^2}\log\log T+O_X(\sqrt{\log\log T})$, and $\#\{k\in\mathcal{I}_T:|u_k(X)|\le y_0\}=o(\#\mathcal{I}_T)$ for every fixed $y_0>0$.*

Proof. The first statement is (5.1). Since $|u_k(X)-s_k|\le\sigma_X$, $|u_k(X)|\le y_0$ implies $|s_k|\le y_0+\sigma_X$; apply Remark 3.15(a). □

Corollary 5.3 (power cutoff)**.** *For fixed $0<c\le\frac12$ and $X=T^c$,*

$$\limsup_{T\to\infty}\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}u_k(T^c)^2\le\frac{1}{2\pi^2}\log\frac1c+C\Big(1+\sqrt{\log(1/c)}\Big), \tag{5.2}$$

and $\#\{k\in\mathcal{I}_T:|u_k(T^c)|>w_T\}=o(\#\mathcal{I}_T)$ whenever $w_T\to\infty$. The same holds for $X=N(T)^c$ with fixed $0<c<\frac12$, since then $X=T^{c'}$ with $c'=c+O(\log\log T/\log T)$.

Proof. (5.1) and Chebyshev's inequality. □

Corollary 5.4 (dichotomy)**.** *Let $X=X(T)$ with $2\le X\le T^{1/2}$. The mean square of $u_k(X)$ over $\mathcal{I}_T$ is bounded as $T\to\infty$ if and only if $\liminf_{T\to\infty}\log X/\log T>0$. For fixed $X_0\ge 2$ and $c\in(0,\frac12]$,*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(u_k(X_0)^2-u_k(T^c)^2\Big)=\frac{1}{2\pi^2}\log\log T+O\Big(\sqrt{\log\log T}\Big). \tag{5.3}$$

Proof. Both directions follow from (5.1), because $y \mapsto \frac{y}{2\pi^2} \pm C(1+\sqrt{y})$ is bounded on bounded sets and tends to infinity; (5.3) is the difference of the two cases of (5.1). □

Proposition 5.5 (Rounding the corrected count misassigns a positive proportion)**.** *There are absolute constants $c_* > 0$, $\kappa_0 > 0$ and T_0 such that for $T \ge T_0$ and $2 \le X \le T^{c_*}$,*

$$\frac{1}{T}\operatorname{meas}\Big\{t \in [T, 2T] : \Big\lfloor F_X(t) + \tfrac{1}{2}\Big\rfloor \ne N(t)\Big\} \ge \kappa_0.$$

Proof. If $\lfloor F_X(t) + \frac{1}{2}\rfloor = N(t)$ then $|N(t) - F_X(t)| \le \frac{1}{2}$, i.e. $|E| \le \frac{1}{2}$ with $E = E_X + \varpi/\pi$. Let $\mathfrak{m}$ be the measure of $\{t \in [T, 2T] : |E| > \frac{1}{2}\}$. Then $\int_T^{2T} E^2 \le \frac{T}{4} + (\int_T^{2T} E^4)^{1/2}\mathfrak{m}^{1/2}$. By (5.10) and (5.11) (with $|\varpi| \ll 1/T$), $\int_T^{2T} E^2 \ge \frac{T}{2\pi^2}V - CT(1+\sqrt{V})$ and $\int_T^{2T} E^4 \le CT(1+V)^2$, so $\mathfrak{m}/T \ge (\frac{V}{2\pi^2} - C(1+\sqrt{V}) - \frac{1}{4})_+^2/(C(1+V)^2)$, which is $\ge \kappa_0 > 0$ once $V \ge V_0$, that is $c \le c_*$ by Lemma 5.8. □

Rounding the corrected count to the nearest integer therefore does not assign the correct index to the zeros, for small cutoffs, on a set of positive density: the obstruction is the size of $S - S_X$, not the sawtooth $N - N_0$. For c near $\frac{1}{2}$ the question depends on the constant in (5.1) (Question 10.1), and for $c > \frac{1}{2}$ nothing is proved.

Remark 5.6. Since $\log \#\{p \le X\} = \log X + O(\log\log X)$, the condition of Corollary 5.4 says that the logarithm of the number of primes must grow proportionally to $\log T$. A fixed number of primes leaves almost all zeros violating the corrected equation by more than any fixed amount; growth of the number of primes alone is necessary but not sufficient. The range $c > \frac{1}{2}$ is not covered. In Theorem 5.1 the cutoff is constant on each window $\mathcal{I}_T$; Theorem 5.12 below allows the cutoff $X_k = (\tau_k/2\pi)^c$, which varies at every ordinate.

Remark 5.7 (Localization of the ordinates). If $M_X \le \frac{1}{2}\mathcal{L}(T)$, then $F_X' \ge \mathcal{L} - M_X \ge \frac{1}{2}\mathcal{L}(T)$ on $[T, 2T]$, so F_X is strictly increasing there. For $k \in \mathcal{I}_T$ with $F_X(T) \le k - \frac{1}{2} \le F_X(2T)$ the equation $F_X(t) = k - \frac{1}{2}$ has exactly one root $t_k^X \in [T, 2T]$ (intermediate value theorem), and $u_k(X) = F_X(t_k^X) - F_X(\gamma_k)$ gives $|\gamma_k - t_k^X| \le 2|u_k(X)|/\mathcal{L}(T)$ (mean value theorem). As $M_X \asymp \sqrt{X}$, this applies only for $X \ll \log^2 T$, and in that range the localization gains nothing in order (Proposition 7.6). For $X = T^c$, F_X need not be monotone, and Theorem 5.1 controls the equation at the true ordinates without selecting a root; Theorem 7.3 replaces the root by one Newton-type step.

5.1. Lemmas on prime polynomials

Fix $T \ge T_0$ and $2 \le X \le T^{1/2}$, let $Y_0 = T^{1/160}$, and split $S_X = S_X^{(1)} + S_X^{(2)} + S_X^{(\ge 3)}$ according to $m = 1$, $m = 2$, $m \ge 3$ in (2.10); the three series converge absolutely and uniformly. Let $\mathcal{J}$ be the set of primes in (X, Y_0) if $X < Y_0$ and in $[Y_0, X]$ if $X \ge Y_0$, and put

$$\begin{aligned} \mathcal{W}_1(t) &= -\frac{1}{\pi}\sum_{p \in \mathcal{J}} \frac{\sin(t\log p)}{\sqrt{p}}, \qquad V = \sum_{p \in \mathcal{J}} \frac{1}{p}, \\ \mathcal{W} &= S_{Y_0}^{\mathrm{pr}} - S_X = \pm\mathcal{W}_1 - S_X^{(2)} - S_X^{(\ge 3)}, \end{aligned} \tag{5.4}$$

with $+$ if $X < Y_0$ and $-$ if $X \ge Y_0$. The last identity holds because $S_X^{(1)}$ is the sum of $-\frac{1}{\pi}p^{-1/2}\sin(t\log p)$ over the primes $p \le X$, while $S_{Y_0}^{\mathrm{pr}}$ is the same sum over $p < Y_0$.

Lemma 5.8. $V = \log(1/c) + O(1)$ *uniformly for* $\log 2/\log T \le c \le \frac{1}{2}$.

Proof. By Theorem 3.8, $V = |\log\log Y_0 - \log\log X| + O(1) = |\log \frac{1}{160c}| + O(1)$. For $c \le \frac{1}{160}$ this is $\log\frac{1}{c} - \log 160 + O(1)$; for $\frac{1}{160} < c \le \frac{1}{2}$ both V and $\log\frac{1}{c}$ are $O(1)$. □

Lemma 5.9. *For real* z_n, $2 \le n \le Y$,

$$\int_T^{2T} \Big(\sum_{n\le Y} z_n \sin(t\log n)\Big)^2 dt = \frac{T}{2}\sum_{n\le Y} z_n^2 + O\Big(\sum_{n\le Y} n z_n^2 + \Big(\sum_{n\le Y}|z_n|\Big)^2\Big), \tag{5.5}$$

and the same with cos *in place of* sin.

Proof. With the finite sum $\mathcal{D}(t) = \sum z_n n^{-it}$, $\sum z_n \sin(t\log n) = -\Im\mathcal{D}$ and $(\Im\mathcal{D})^2 = \frac{1}{2}(|\mathcal{D}|^2 - \Re\mathcal{D}^2)$. Theorem 3.7 evaluates $\int|\mathcal{D}|^2$, and $|\int_T^{2T}(n_1n_2)^{-it}dt| \le 2/\log(n_1n_2) \le 2/\log 4$ gives $|\int\mathcal{D}^2| \le (2/\log 4)(\sum|z_n|)^2$. For cos use $\Re\mathcal{D}$ and $(\Re\mathcal{D})^2 = \frac{1}{2}(|\mathcal{D}|^2 + \Re\mathcal{D}^2)$. □

Lemma 5.10. *With absolute implied constants,*

$$\int_T^{2T} \mathcal{W}^2\,dt = \frac{T}{2\pi^2}V + O\Big(T(1+\sqrt{V})\Big), \tag{5.6}$$

$$\int_T^{2T} \mathcal{W}^4\,dt \ll T(1+V)^2, \tag{5.7}$$

$$\int_T^{2T} S_X'(t)^2\,dt \ll T(1+\log X)^2. \tag{5.8}$$

Proof. All primes involved are $\le \max(X, Y_0) \le T^{1/2}$. Lemma 5.9 with $z_p = \frac{1}{\pi\sqrt{p}}$ ($p \in \mathcal{J}$) gives $\int\mathcal{W}_1^2 = \frac{T}{2\pi^2}V + O(T^{1/2} + (\sum_{p\le T^{1/2}} p^{-1/2})^2) = \frac{T}{2\pi^2}V + O(T^{1/2})$. For $S_X^{(2)}(t) = -\frac{1}{2\pi}\sum_{p\le X} p^{-1}\sin(t\log p^2)$ the same lemma gives $\int(S_X^{(2)})^2 \ll T\sum_p p^{-2} + \#\{p \le X\} + (\sum_{p\le X} p^{-1})^2 \ll T$. Finally $|S_X^{(\ge 3)}| \le \frac{1}{\pi}\sum_p\sum_{m\ge 3}\frac{1}{mp^{m/2}} \le \frac{1}{\pi}\sum_p \frac{p^{-3/2}}{1-p^{-1/2}} < \infty$, uniformly in X. Expanding $\mathcal{W}^2$ and bounding the cross terms by Cauchy–Schwarz gives (5.6).

For (5.7) use $(y_1+y_2+y_3)^4 \le 27(y_1^4+y_2^4+y_3^4)$. Let $\mathcal{D}_1(t) = \sum_{p\in\mathcal{J}} p^{-1/2}p^{-it}$, so that $|\mathcal{W}_1| \le |\mathcal{D}_1|/\pi$ and $\mathcal{D}_1^2 = \sum_n \eta_n n^{-it}$ with $\eta_{p^2} = p^{-1}$, $\eta_{p_1p_2} = 2(p_1p_2)^{-1/2}$ for $p_1 < p_2$, all $n \le T$. Theorem 3.7 gives $\int|\mathcal{D}_1|^4 = \sum|\eta_n|^2(T+O(n)) \ll T\sum|\eta_n|^2 \le T(1+2V^2)$. For $S_X^{(2)}$ let $\mathcal{D}_2(t) = \sum_{p\le X} p^{-1}(p^2)^{-it}$, so that $|S_X^{(2)}| \le |\mathcal{D}_2|/2\pi$ and $\mathcal{D}_2^2 = \sum_n \eta_n' n^{-it}$ over $n = (p_1p_2)^2$ with $|\eta_n'| \le 2(p_1p_2)^{-1}$; then $\sum|\eta_n'|^2 \ll 1$ and $\sum n|\eta_n'|^2 \le 4\#\{p\le X\}^2 \le 4X^2 \le 4T$. $S_X^{(\ge 3)}$ is bounded.

For (5.8), $S_X'(t) = \sum_{n\in A_X} a_n \log n\cos(t\log n)$ (termwise differentiation, Section 2.4). The terms $m = 1$ have $|z_p| = \frac{\log p}{\pi\sqrt{p}}$; Lemma 5.9 and Theorem 3.8 give $\ll T\log X\sum_{p\le X}\frac{\log p}{p} + \sum_{p\le X}\log^2 p + (\sum_{p\le X}\frac{\log p}{\sqrt{p}})^2 \ll T(1+\log X)^2 + X\log X$, and $X \le T^{1/2}$. The terms $m = 2$ have $|z_{p^2}| = \frac{\log p}{\pi p}$ and contribute $\ll T + X\log X + \log^2 X$. The terms $m \ge 3$ are bounded by $\frac{1}{\pi}\sum_p \log p\sum_{m\ge 3} p^{-m/2} < \infty$. Combine with $(y_1+y_2+y_3)^2 \le 3(y_1^2+y_2^2+y_3^2)$. □

5.2. Proof of Theorem 5.1

Proof. Integrability. On $[T, 2T]$, S is bounded (Theorem 3.1) and piecewise continuous, and S_X, S_X', ϖ are continuous; all integrals below exist.

Second and fourth moments of E_X. Let $E_0 = S - S^{\mathrm{pr}}_{Y_0}$ and $E_X = S - S_X = E_0 + \mathcal{W}$. Theorem 3.5 and Hölder's inequality give

$$\int_T^{2T} E_0^4\,dt \ll T, \qquad \int_T^{2T} E_0^2\,dt \le T^{1/2}\Big(\int_T^{2T} E_0^4\,dt\Big)^{1/2} \ll T. \tag{5.9}$$

Let $\mathcal{E}_2 = \int_T^{2T} E_X^2\,dt$. From (5.6), (5.9) and Cauchy–Schwarz,

$$\mathcal{E}_2 = \frac{T}{2\pi^2}V + O\Big(T(1+\sqrt{V})\Big), \tag{5.10}$$

and from (5.7), (5.9) and $(y_1+y_2)^4 \le 8(y_1^4+y_2^4)$,

$$\int_T^{2T} E_X^4\,dt \ll T(1+V)^2. \tag{5.11}$$

Transfer to the ordinates. Let k_1, k_2 be the smallest and largest indices in $\mathcal{I}_T$ ($\mathcal{I}_T \neq \emptyset$ for $T \ge T_0$) and apply (4.7). By Lemma 2.2 and (2.11), $|u_k(X)| \ll \log T + \sqrt{X} \ll T^{1/4}$, so the boundary terms and $u_{k_1}(X)^2 + u_{k_2}(X)^2$ are $O(T^{3/4})$. The intervals $[T, \gamma_{k_1}]$ and $[\gamma_{k_2}, 2T]$ have bounded length (Theorem 3.1), and on them $E_X^2(\mathcal{L} + |S'_X|) \ll T^{1/2}\cdot T^{1/4}$. Since $|\varpi| \ll 1/T$, replacing $(E_X + \varpi/\pi)^2$ by E_X^2 costs $\ll T^{-1}\int_T^{2T}(|E_X|+1)(\mathcal{L}+|S'_X|)\,dt \ll (1+V)^{1/2}\log T$ by Cauchy–Schwarz, (5.10) and (5.8). Hence

$$\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \int_T^{2T} E_X^2\mathcal{L}\,dt + \int_T^{2T} E_X^2 S'_X\,dt - \frac{\#\mathcal{I}_T}{12} + O(T^{3/4}). \tag{5.12}$$

Main term and derivative term. Since $0 \le \mathcal{L}(t) - \mathcal{L}(T) \le \frac{\log 2}{2\pi}$ on $[T, 2T]$, the first integral is $\mathcal{L}(T)\mathcal{E}_2 + O(\mathcal{E}_2)$. By Cauchy–Schwarz, (5.11) and (5.8),

$$\Big|\int_T^{2T} E_X^2 S'_X\,dt\Big| \le \Big(\int_T^{2T} E_X^4\Big)^{1/2}\Big(\int_T^{2T} S_X'^2\Big)^{1/2} \ll T(1+V)(1+\log X). \tag{5.13}$$

Dividing by $\#\mathcal{I}_T = T\mathcal{L}(T) + O(T)$ and using (5.10),

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{V}{2\pi^2} + O\Big(1 + \sqrt{V} + \frac{1+V}{\log T} + (1+V)\,c\Big), \tag{5.14}$$

because $\log X = c\log T$. By Lemma 5.8, $(1+V)c \ll c(1+\log\frac{1}{c}) \ll 1$ on $(0, \frac12]$; $V \ll \log\log T$ since $c \ge \log 2/\log T$, so $(1+V)/\log T = o(1)$. A second use of Lemma 5.8 replaces $\frac{V}{2\pi^2}$ by $\frac{1}{2\pi^2}\log\frac{1}{c} + O(1)$ and $\sqrt{V}$ by $\sqrt{\log(1/c)} + O(1)$, which proves (5.1). □

5.3. A cutoff varying with the index

Let $\mathcal{I}_T^\tau = \{k : T \le \tau_k < 2T\}$; by (2.5), $\#\mathcal{I}_T^\tau = N_0(2T) - N_0(T) + O(1) \asymp T\log T$.

Lemma 5.11. *Let $T \ge T_0$ and $2 \le X_- < X_+ \le (T/3)^{1/2}$ with $X_+ \le 4X_-$; let $V_\pm = \sum_{X_-<p\le X_+} p^{-1}$ and*

$$\mathcal{Q}(t) = \max_{X_-\le y\le X_+}\Big|\sum_{X_-<p\le y} p^{-1/2-it}\Big|.$$

Then $\mathcal{Q}$ is Lipschitz, $|\mathcal{Q}'(t)| \le \max_{X_-\le y\le X_+} |\sum_{X_-<p\le y} \log p\, p^{-1/2-it}|$ for almost every t, and

$$\int_{T-3}^{2T+3} \mathcal{Q}^4\,dt \ll TV_\pm^2, \qquad \int_{T-3}^{2T+3} |\mathcal{Q}'|^4\,dt \ll T(\log X_+)^4 V_\pm^2, \tag{5.15}$$

$$\langle \mathcal{Q}(\gamma_k)^2\rangle_{\mathcal{I}_T} \ll V_\pm\Big(1+\frac{\log X_+}{\log T}\sqrt{\log\log T}\Big)+\frac{X_+}{T}, \qquad \langle \mathcal{Q}(\tau_k)^2\rangle_{\mathcal{I}_T^\tau} \ll V_\pm. \tag{5.16}$$

Proof. If $(X_-, X_+]$ contains no prime, $\mathcal{Q}=0$. Otherwise let $p_1<\dots<p_J$ be its primes and $P_j(t)=\sum_{i\le j} p_i^{-1/2-it}$ ($P_0=0$), so that $\mathcal{Q}=\max_{0\le j\le J}|P_j|$, a maximum of finitely many Lipschitz functions; hence $\mathcal{Q}$ is Lipschitz and differentiable almost everywhere. At a point t where $\mathcal{Q}$ and all $|P_j|$ are differentiable and $\mathcal{Q}(t)=|P_j(t)|$, the function $\mathcal{Q}-|P_j|\ge 0$ has a minimum, so $\mathcal{Q}'(t)=|P_j|'(t)$, and $\big||P_j|'\big|\le |P_j'|$ wherever $|P_j|$ is differentiable. This gives the bound for $\mathcal{Q}'$.

Window. Fix $\mathsf{s}\in C_c^\infty(-\frac12,\frac12)$, real, even, nonnegative and not identically 0, let $\check{\mathsf{s}}(x)=\int \mathsf{s}(\omega)e^{i\omega x}d\omega$ and $\mathsf{W}=|\check{\mathsf{s}}|^2/\check{\mathsf{s}}(0)^2$. Then $\mathsf{W}\ge 0$ is a Schwartz function with $\mathsf{W}(0)=1$, and $\mathcal{F}W$ is supported in $[-1,1]$. Choose $0<\mathsf{y}\le\frac14$ with $\mathsf{W}\ge\frac12$ on $[-\mathsf{y},\mathsf{y}]$ and put $\mathfrak{w}(t)=\mathsf{W}(\mathsf{y}(t-\frac32 T)/(\frac{T}{2}+3))$. Then $\mathfrak{w}\ge\frac12$ on $[T-3,2T+3]$, $\|\mathfrak{w}\|_4^4\asymp T$, and $\mathcal{F}\mathfrak{w}$ and $\mathcal{F}(\mathfrak{w}^4)$ are supported in $|\omega|\le 2\mathsf{y}/T$ and $|\omega|\le 8\mathsf{y}/T$.

Proof of (5.15). Let $f_j=\mathfrak{w}P_j$. Then $\mathcal{F}f_j(\omega)=\sum_{i\le j}p_i^{-1/2}\mathcal{F}\mathfrak{w}(\omega+\log p_i)$, a sum of packets supported in $|\omega+\log p_i|\le 2\mathsf{y}/T$. Since $\log p_{i+1}-\log p_i\ge\log(1+1/X_+)\ge (3/T)^{1/2}/2>4\mathsf{y}/T$ for $T\ge T_0$, the packets are disjoint, and f_j is obtained from f_J by removing the packets of $p_{j+1},\dots,p_J$, which lie below a point ω_j separating them from the others: $f_j(t)=f_J(t)-\frac{1}{2\pi}\int_{-\infty}^{\omega_j}\mathcal{F}f_J(\omega)e^{i\omega t}d\omega$. Hence $\max_j|f_j|\le|f_J|+\mathcal{C}f_J$, and on $[T-3,2T+3]$, $\mathcal{Q}\le 2\max_j|f_j|$. By Theorem 3.11,

$$\int_{T-3}^{2T+3}\mathcal{Q}^4\le 16\int_{\mathbb{R}}\Big(|f_J|+\mathcal{C}f_J\Big)^4\ll\int_{\mathbb{R}}\mathfrak{w}^4|P_J|^4\,dt.$$

Write $P_J^2=\sum_n \eta_n n^{-it}$ with $\eta_{p^2}=p^{-1}$ and $\eta_{p_ip_{i'}}=2(p_ip_{i'})^{-1/2}$ for $i<i'$, all $n\le X_+^2\le T/3$ (unique factorization makes the representation unique). Expanding, $\int\mathfrak{w}^4|P_J^2|^2=\sum_{n,n'}\eta_n\eta_{n'}\mathcal{F}(\mathfrak{w}^4)(\log(n/n'))$, and $|\log(n/n')|\ge 1/\max(n,n')\ge 3/T>8\mathsf{y}/T$ for $n\ne n'$, so only the diagonal remains: $\int\mathfrak{w}^4|P_J|^4=\|\mathfrak{w}\|_4^4\sum_n\eta_n^2\ll T\big(\sum_p p^{-2}+4V_\pm^2\big)\ll TV_\pm^2$, because $\sum_{X_-<p\le X_+}p^{-2}\le V_\pm/X_-\le 4V_\pm/X_+\le 4V_\pm^2$. The same argument with the polynomials $\sum_{i\le j}\log p_i\, p_i^{-1/2-it}$, whose maximum dominates $|\mathcal{Q}'|$, gives the second bound, with η_n multiplied by $\log p_i\log p_{i'}\le(\log X_+)^2$.

At the ordinates. Let $T, 2T$ be non-ordinates (otherwise move them by less than the distance to the nearest ordinate, which changes neither side). As Lebesgue–Stieltjes measures on $(T,2T]$, $dN=\mathcal{L}\,dt+dG$, and $\mathcal{Q}^2$ is Lipschitz, so integration by parts [7, §3.5] gives

$$\sum_{k\in\mathcal{I}_T}\mathcal{Q}(\gamma_k)^2=\int_T^{2T}\mathcal{Q}^2\mathcal{L}\,dt+\Big[\mathcal{Q}^2G\Big]_T^{2T}-2\int_T^{2T}G\,\mathcal{Q}\mathcal{Q}'\,dt, \tag{5.17}$$

multiplicities included. By Cauchy–Schwarz and (5.15) the first term is $\ll\log T\cdot T^{1/2}(TV_\pm^2)^{1/2}=TV_\pm\log T$. The boundary term is $\ll X_+\log T$, since $\mathcal{Q}\le\sum_{p\le X_+}p^{-1/2}\ll X_+^{1/2}$ and $G\ll\log T$. By Hölder's inequality with exponents $2,4,4$, Theorem 3.3 ($\int_T^{2T}G^2\ll$

$T\log\log T$) and (5.15), the last term is $\ll (T\log\log T)^{1/2}(TV_\pm^2)^{1/4}(T(\log X_+)^4V_\pm^2)^{1/4} = T\sqrt{\log\log T}\,(\log X_+)V_\pm$. Divide by $\#\mathcal{I}_T \gg T\log T$.

At the clock points. For consecutive indices, $\tau_{k+1}-\tau_k = \int_k^{k+1} dx/\mathcal{L}(\tau_x) \geq 2\pi/\log(3T) = \mathfrak{b}_T$ while $\tau_{k+1} \leq 3T$, so the intervals $\mathcal{U}_k = [\tau_k, \tau_k+\mathfrak{b}_T]$, $k \in \mathcal{I}_T^\tau$, are disjoint and contained in $[T, 2T+1]$. For an absolutely continuous g on an interval $\mathcal{U}$ of length $\mathfrak{b}$ and $x \in \mathcal{U}$, $|g(x)| \leq \frac{1}{\mathfrak{b}}\int_{\mathcal{U}}|g| + \int_{\mathcal{U}}|g'|$, hence $g(x)^2 \leq \frac{2}{\mathfrak{b}}\int_{\mathcal{U}} g^2 + 2\mathfrak{b}\int_{\mathcal{U}} g'^2$. With $g = \mathcal{Q}$ and summing, $\sum_{k\in\mathcal{I}_T^\tau}\mathcal{Q}(\tau_k)^2 \leq \frac{2}{\mathfrak{b}_T}\int_T^{2T+1}\mathcal{Q}^2 + 2\mathfrak{b}_T\int_T^{2T+1}\mathcal{Q}'^2 \ll TV_\pm\log T + \frac{T(\log X_+)^2V_\pm}{\log T}$, by Cauchy–Schwarz and (5.15). Divide by $\#\mathcal{I}_T^\tau \gg T\log T$ and use $\log X_+ \leq \log T$. □

Theorem 5.12. *For $\frac{\log 2}{\log T} \leq c \leq \frac{1}{2}$ let $X_k = \max\{2, (\tau_k/2\pi)^c\}$ and $\tilde{u}_k = s_k - S_{X_k}(\gamma_k)$. Uniformly in c,*

$$\left|\langle \tilde{u}_k^2\rangle_{\mathcal{I}_T} - \frac{1}{2\pi^2}\log\frac{1}{c}\right| \ll 1 + \sqrt{\log(1/c)}. \tag{5.18}$$

For fixed c, $\langle(\tilde{u}_k - u_k(X_-))^2\rangle_{\mathcal{I}_T} \to 0$, with X_- as in the proof.

Proof. For $T \geq T_0$ and $k \in \mathcal{I}_T$, $|\delta_k| \leq 1$ by (3.4), so $\tau_k \in [T-1, 2T+1]$ and $X_- \leq X_k \leq X_+$ with $X_\pm$ the values of $\max\{2,(y/2\pi)^c\}$ at $y = T-1$ and $y = 2T+1$. Then $X_+ \leq ((2T+1)/2\pi)^{1/2} \leq (T/3)^{1/2}$ and $X_+/X_- \leq ((2T+1)/(T-1))^{1/2} \leq 2$. The prime powers p^m with $X_- < p \leq X_k$ are those in $S_{X_k} - S_{X_-}$, so by (2.10)

$$|S_{X_k}(\gamma_k) - S_{X_-}(\gamma_k)| \leq \frac{1}{\pi}\mathcal{Q}(\gamma_k) + \frac{1}{\pi}\sum_{X_-<p\leq X_+}\sum_{m\geq 2}\frac{1}{mp^{m/2}} \leq \frac{1}{\pi}\mathcal{Q}(\gamma_k) + 1.1\,V_\pm,$$

since the terms $m=1$ are $\frac{1}{\pi}\Im\sum_{X_-<p\leq X_k} p^{-1/2-i\gamma_k}$ and $\sum_{m\geq 2}p^{-m/2}/m \leq p^{-1}(1-2^{-1/2})^{-1}$. By Theorem 3.8, $V_\pm = \log\frac{\log X_+}{\log X_-} + O(\frac{1}{\log X_-}) \ll \frac{1}{\log X_-} \ll \frac{1}{1+c\log T}$ (and $V_\pm = 0$ if $X_- = X_+ = 2$). Lemma 5.11 gives

$$\langle(\tilde{u}_k - u_k(X_-))^2\rangle_{\mathcal{I}_T} \ll V_\pm + \frac{(1+c\log T)\sqrt{\log\log T}}{(1+c\log T)\log T} + T^{-1/2} + V_\pm^2 \ll 1,$$

which tends to 0 for fixed c. Theorem 5.1 with $X = X_-$, for which $\log(1/c_-) = \log(1/c) + O(1)$ with $c_- = \log X_-/\log T$, gives (5.1) for $u_k(X_-)$; (5.18) follows from $|\langle\tilde{u}^2\rangle - \langle u^2\rangle| \leq 2\langle u^2\rangle^{1/2}\langle(\tilde{u}-u)^2\rangle^{1/2} + \langle(\tilde{u}-u)^2\rangle$. □

6. Theorem C: exact accounting for the one-step clock

For a finite nonempty set I of indices and a prime power n, define the response statistics at the zeros and at the clock points,

$$\mathcal{R}_n^\gamma(I) = \frac{2}{a_n}\langle s_k\sin(\gamma_k\log n)\rangle_I, \qquad \mathcal{R}_n^\tau(I) = \frac{2}{a_n}\langle s_k\sin(\tau_k\log n)\rangle_I, \tag{6.1}$$

and, for a set A of prime powers with $\sum_{n\in A}|a_n| < \infty$, the cross term

$$\mathrm{Cr}_A(I) = \langle S_A(\tau_k)^2\rangle_I - v_A. \tag{6.2}$$

For a prime p, $2/a_p = -2\pi\sqrt{p}$. These are finite-window statistics of the zeros of the Riemann zeta function, defined without any hypothesis; Cr_A contains the finite-window diagonal discrepancies and all off-diagonal products and has no fixed sign.

Theorem 6.1 (Finite-window accounting)**.** *Let I be a finite nonempty set of indices and A a set of prime powers with $\sum_{n\in A}|a_n|<\infty$.*

(a) For every k and $X\ge 2$,

$$\begin{aligned} &r_k(X)=u_k(X)+\Big(S_X(\gamma_k)-S_X(\tau_k)\Big)+\chi_k,\\ &\chi_k=\frac{\gamma_k}{2\pi}\Big(\frac{\tau_k}{\gamma_k}-1-\log\frac{\tau_k}{\gamma_k}\Big),\qquad \frac{\delta_k^2}{4\pi\max(\gamma_k,\tau_k)}\le\chi_k\le\frac{\delta_k^2}{4\pi\min(\gamma_k,\tau_k)},\\ &|S_X(\gamma_k)-S_X(\tau_k)|\le M_X|\delta_k|. \end{aligned} \tag{6.3}$$

(b) For every real λ,

$$\Big\langle s_k^2-(s_k-\lambda S_A(\tau_k))^2\Big\rangle_I=\sum_{n\in A}a_n^2\Big(\lambda\,\mathcal{R}_n^\tau(I)-\frac{\lambda^2}{2}\Big)-\lambda^2\,\mathrm{Cr}_A(I). \tag{6.4}$$

In particular, for the full local factors,

$$\begin{aligned} &\langle s_k^2-(s_k-S_X(\tau_k))^2\rangle_I=D_X(I)-\mathrm{Cr}_X(I),\\ &D_X(I)=\sum_{n\in A_X}a_n^2\Big(\mathcal{R}_n^\tau(I)-\frac12\Big)=2\langle s_kS_X(\tau_k)\rangle_I-v_X, \end{aligned} \tag{6.5}$$

with $\mathrm{Cr}_X=\mathrm{Cr}_{A_X}$.

(c) In units of zeros, the gain of the one-step clock over the bare clock is

$$\begin{aligned} &\Big\langle(\mathcal{L}(\tau_k)\delta_k)^2-r_k(X)^2\Big\rangle_I=D_X(I)-\mathrm{Cr}_X(I)+2\langle\chi_kS_X(\tau_k)\rangle_I,\\ &|\langle\chi_kS_X(\tau_k)\rangle_I|\le\sigma_X\max_{k\in I}\chi_k. \end{aligned} \tag{6.6}$$

Proof. (a) By (2.12), $r_k(X)=\mathcal{L}(\tau_k)\delta_k-S_X(\tau_k)$. Put $\chi_k=\mathcal{L}(\tau_k)\delta_k-s_k$; since $s_k=N_0(\tau_k)-N_0(\gamma_k)=\int_{\gamma_k}^{\tau_k}\mathcal{L}$,

$$\chi_k=\int_{\gamma_k}^{\tau_k}\Big(\mathcal{L}(\tau_k)-\mathcal{L}(y)\Big)dy=\frac{1}{2\pi}\int_{\gamma_k}^{\tau_k}\log\frac{\tau_k}{y}\,dy=\frac{\gamma_k}{2\pi}\Big(\frac{\tau_k}{\gamma_k}-1-\log\frac{\tau_k}{\gamma_k}\Big),$$

the last step by $\int\log y\,dy=y\log y-y$. If $\gamma_k<\tau_k$, then $\frac{\tau_k-y}{\tau_k}\le\log\frac{\tau_k}{y}\le\frac{\tau_k-y}{y}\le\frac{\tau_k-y}{\gamma_k}$ for $y\in[\gamma_k,\tau_k]$, and integration gives $\frac{\delta_k^2}{4\pi\tau_k}\le\chi_k\le\frac{\delta_k^2}{4\pi\gamma_k}$; if $\tau_k<\gamma_k$, write $\chi_k=\frac{1}{2\pi}\int_{\tau_k}^{\gamma_k}\log\frac{y}{\tau_k}\,dy$ and use $\frac{y-\tau_k}{\gamma_k}\le\frac{y-\tau_k}{y}\le\log\frac{y}{\tau_k}\le\frac{y-\tau_k}{\tau_k}$; if $\tau_k=\gamma_k$, $\chi_k=0$. Insert $s_k=u_k(X)+S_X(\gamma_k)$. The last bound is the mean value theorem with (2.11).

(b) The series $S_A(\tau_k)=\sum_n a_n\sin(\tau_k\log n)$ converges absolutely, and so does the double series for $S_A(\tau_k)^2$; since I is finite, the mean over I may be exchanged with these sums. Expanding, the left side is $2\lambda\langle s_kS_A(\tau_k)\rangle_I-\lambda^2\langle S_A(\tau_k)^2\rangle_I$, with $\langle s_kS_A(\tau_k)\rangle_I=\sum_n a_n\langle s_k\sin(\tau_k\log n)\rangle_I=\frac12\sum_n a_n^2\mathcal{R}_n^\tau(I)$ and $\langle S_A(\tau_k)^2\rangle_I=v_A+\mathrm{Cr}_A(I)$. Taking $\lambda=1$ and $A=A_X$ gives (6.5).

(c) By (a), $\mathcal{L}(\tau_k)\delta_k=s_k+\chi_k$ and $r_k(X)=s_k+\chi_k-S_X(\tau_k)$, so $(\mathcal{L}(\tau_k)\delta_k)^2-r_k(X)^2=2s_kS_X(\tau_k)-S_X(\tau_k)^2+2\chi_kS_X(\tau_k)$; average and use (6.5) and $|S_X|\le\sigma_X$. □

Remark 6.2. (6.5) is an exact sign criterion for the gain of the full Euler phase at the clock points: the gain is positive exactly when the diagonal responses outweigh the cross term. No independence of the phases $\tau_k \log p$ is assumed. A threshold $\mathcal{R}_p^\tau = \frac{1}{2}$ describes a single diagonal prime contribution and does not determine the optimal cutoff when Cr_X and the prime powers matter.

Remark 6.3 (Optimal trigonometric correction). For a finite set A of frequencies let $\varsigma_n(k) = \sin(\tau_k \log n)$, $\mathbf{Q} = (\langle \varsigma_{n_1}\varsigma_{n_2}\rangle_I)_{n_1,n_2\in A}$, $\mathbf{b} = (\langle s_k \varsigma_n(k)\rangle_I)_{n\in A}$ and $\Upsilon_{\boldsymbol{\lambda}} = \sum_n \lambda_n \varsigma_n$. Ordinary least squares gives $\langle (s_k - \Upsilon_{\boldsymbol{\lambda}}(k))^2\rangle_I = \langle s_k^2\rangle_I - 2\boldsymbol{\lambda}^{\mathsf{T}}\mathbf{b} + \boldsymbol{\lambda}^{\mathsf{T}}\mathbf{Q}\boldsymbol{\lambda}$, minimized by $\boldsymbol{\lambda} = \mathbf{Q}^\dagger\mathbf{b}$ with gain $\mathbf{b}^{\mathsf{T}}\mathbf{Q}^\dagger\mathbf{b}$; for prime powers $b_n = \frac{1}{2}a_n\mathcal{R}_n^\tau(I)$. The Euler phase is the choice $\lambda_n = a_n$, and (6.4) measures what it loses through $\mathcal{R}_n^\tau \neq 1$ and $\mathrm{Cr}_A \neq 0$.

Remark 6.4 (Clock point against zero). For a prime power n, $\sin(\gamma_k\ell_n) - \sin(\tau_k\ell_n) = \int_0^1 \frac{d}{dy}\sin((\tau_k - y\delta_k)\ell_n)\,dy$, whence, exchanging the finite mean and the integral,

$$\mathcal{R}_n^\gamma(I) - \mathcal{R}_n^\tau(I) = -\frac{2\ell_n}{a_n}\int_0^1 \Big\langle s_k\delta_k \cos\Big((\tau_k - y\delta_k)\ell_n\Big)\Big\rangle_I\,dy. \tag{6.7}$$

For fixed n and $I = \mathcal{I}_T$ the right side is $O(\langle|s_k\delta_k|\rangle_{\mathcal{I}_T}) = O(\log\log T/\log T)$ by (2.6) and Remark 3.15; for $\log n = \mathsf{q}\log T$ with fixed $0 < \mathsf{q} < 1$ the phase change $\delta_k\ell_n$ has mean square $\mathsf{q}^2\log^2 T\,\langle\delta_k^2\rangle_{\mathcal{I}_T} \sim 2\mathsf{q}^2\log\log T$, so it is not small, and the transport integral need not be small; no asymptotic statement about this difference is proved.

Remark 6.5 (Fixed prime sets). For fixed X, $|\delta_k^X - \delta_k| \le \sigma_X/\mathcal{L}(\tau_k) \to 0$ and $|u_k(X) - s_k| \le \sigma_X$. A fixed set of primes changes neither $\limsup_k |\delta_k|$ nor the unboundedness of the defect.

7. Theorem D: a computable clock that approaches the zeros

The one-step clock $\tau_k^X = \tau_k - y_k$, $y_k = S_X(\tau_k)/\mathcal{L}(\tau_k)$, uses only τ_k and the primes $p \le X$. By Theorem 6.1(a),

$$r_k(X) = u_k(X) + \mathfrak{t}_k(X) + \chi_k, \qquad \mathfrak{t}_k(X) = S_X(\gamma_k) - S_X(\tau_k), \tag{7.1}$$

where Theorem 5.1 controls $u_k(X)$ and $0 \le \chi_k \ll 1/T$ on $\mathcal{I}_T$. This section controls the transport term $\mathfrak{t}_k(X)$ in mean square, with no assumption on the joint behaviour of δ_k and the phases $\tau_k \log p$.

Lemma 7.1. *For $T \ge T_0$ and $2 \le X \le T^{1/2}$,*

$$\int_{T-3}^{2T+3} S_X'(t)^4\,dt \ll T(1+\log X)^4, \qquad \int_{T-3}^{2T+3} S_X''(t)^4\,dt \ll T(1+\log X)^8.$$

Proof. By Section 2.4, the series

$$S_X'(t) = \sum_{n\in A_X} a_n \log n \cos(t\log n), \qquad S_X''(t) = -\sum_{n\in A_X} a_n \log^2 n \sin(t\log n)$$

converge absolutely and uniformly. Split according to $m = 1$, $m = 2$, $m \ge 3$ in $n = p^m$ and use $(y_1+y_2+y_3)^4 \le 27(y_1^4+y_2^4+y_3^4)$. For $\nu \in \{1,2\}$ the terms $m \ge 3$ of the ν-th derivative are bounded by $\frac{1}{\pi}\sum_p \sum_{m\ge 3} \frac{(m\log p)^\nu}{mp^{m/2}} < \infty$, uniformly in X and t. The terms $m = 1$ are bounded

by $\frac{1}{\pi}|\mathcal{D}_\nu(t)|$ with $\mathcal{D}_\nu(t)=\sum_{p\le X}\log^\nu p\; p^{-1/2-it}$; $\mathcal{D}_\nu^2=\sum_n \eta_n n^{-it}$ over $n=pp'\le X^2\le T$ with $\sum_n \eta_n^2\le 2(\sum_{p\le X}\log^{2\nu}p/p)^2\ll(1+\log X)^{4\nu}$ (Theorem 3.8 and partial summation), so Theorem 3.7 on an interval of length $T+6$ gives $\int|\mathcal{D}_\nu|^4=\sum\eta_n^2(T+6+O(n))\ll T(1+\log X)^{4\nu}$. The terms $m=2$ are bounded by $\frac{1}{2\pi}|\mathcal{D}'_\nu(t)|$ with $\mathcal{D}'_\nu(t)=\sum_{p\le X}(2\log p)^\nu p^{-1-2it}$; here $\mathcal{D}'^2_\nu=\sum_n\eta'_n n^{-it}$ over $n=(pp')^2\le X^4$, with $\sum_n\eta'^2_n\ll 1$ and $\sum_n n\eta'^2_n\ll(\sum_{p\le X}\log^{2\nu}p)^2\ll X^2(1+\log X)^{4\nu}\le T(1+\log X)^{4\nu}$, so again $\int|\mathcal{D}'_\nu|^4\ll T(1+\log X)^{4\nu}$. □

Lemma 7.2 (Transport)**.** *For $T\ge T_0$ and $2\le X\le T^{1/2}$,*

$$\Big\langle \mathfrak{t}_k(X)^2\Big\rangle_{\mathcal{I}_T}\ll\frac{\log\log T\,(1+\log X)^2}{\log^2 T}. \tag{7.2}$$

Proof. Fourth moment of δ_k. For $T\ge T_0$ and $k\in\mathcal{I}_T$, $|\delta_k|\le 1$ by (3.4); so the point ξ_k of (2.6) is $\ge T-1$ and $\mathcal{L}(\xi_k)\gg\log T$. By (3.3) with $j=2$,

$$\langle\delta_k^4\rangle_{\mathcal{I}_T}=\langle s_k^4\mathcal{L}(\xi_k)^{-4}\rangle_{\mathcal{I}_T}\ll\frac{(\log\log T)^2}{\log^4 T}. \tag{7.3}$$

Maximal function. Let $\varphi\in C^\infty(\mathbb{R})$ with $0\le\varphi\le 1$, $\varphi=1$ on $[T-2,2T+2]$, $\varphi=0$ outside $[T-3,2T+3]$ and $|\varphi'|\le 2$, and let $\mathfrak{h}=\varphi S'_X$. For $k\in\mathcal{I}_T$ the segment between γ_k and τ_k lies in $[T-1,2T+1]$, where $\mathfrak{h}=S'_X$, so

$$|\mathfrak{t}_k(X)|=\Big|\int_{\tau_k}^{\gamma_k}\mathfrak{h}(y)\,dy\Big|\le\int_{\tau_k-|\delta_k|}^{\tau_k+|\delta_k|}|\mathfrak{h}|\le 2|\delta_k|\,\mathcal{M}\mathfrak{h}(\tau_k) \tag{7.4}$$

(trivially if $\delta_k=0$). The function $\mathfrak{h}$ is C^∞ with compact support. By Theorem 3.10, $g=\mathcal{M}\mathfrak{h}$ is locally absolutely continuous with $\|g\|_4\ll\|\mathfrak{h}\|_4$ and $\|g'\|_4\le\|\mathcal{M}(\mathfrak{h}')\|_4\ll\|\mathfrak{h}'\|_4$; by Lemma 7.1, $\|\mathfrak{h}\|_4^4\ll T(1+\log X)^4$ and, since $\mathfrak{h}'=\varphi' S'_X+\varphi S''_X$, $\|\mathfrak{h}'\|_4^4\ll T(1+\log X)^8$.

Sampling at the clock points. As in the proof of Lemma 5.11, the intervals $\mathcal{U}_k=[\tau_k,\tau_k+\mathfrak{b}_T]$, $k\in\mathcal{I}_T$, are disjoint and contained in $[T-1,2T+2]$. For $x\in\mathcal{U}$, an interval of length $\mathfrak{b}$, $|g(x)|\le\frac{1}{\mathfrak{b}}\int_{\mathcal{U}}|g|+\int_{\mathcal{U}}|g'|$; by Hölder's inequality and $(y_1+y_2)^4\le 8(y_1^4+y_2^4)$, $g(x)^4\le\frac{8}{\mathfrak{b}}\int_{\mathcal{U}}g^4+8\mathfrak{b}^3\int_{\mathcal{U}}|g'|^4$. Summing over $k\in\mathcal{I}_T$ and dividing by $\#\mathcal{I}_T\gg T\log T$,

$$\langle g(\tau_k)^4\rangle_{\mathcal{I}_T}\ll\frac{1}{T}\int_{\mathbb{R}}g^4+\frac{1}{T\log^4 T}\int_{\mathbb{R}}|g'|^4\ll(1+\log X)^4+\frac{(1+\log X)^8}{\log^4 T}\ll(1+\log X)^4.$$

Conclusion. By (7.4) and Cauchy–Schwarz, $\langle\mathfrak{t}_k^2\rangle\le 4\langle\delta_k^2 g(\tau_k)^2\rangle\le 4\langle\delta_k^4\rangle^{1/2}\langle g(\tau_k)^4\rangle^{1/2}$; insert (7.3). □

Theorem 7.3 (Theorem D)**.** *There are absolute constants T_0 and C such that for $T\ge T_0$ and $2\le X\le T^{1/2}$, with $c=\log X/\log T$,*

$$\Big|\Big\langle r_k(X)^2\Big\rangle_{\mathcal{I}_T}-\frac{1}{2\pi^2}\log\frac{1}{c}\Big|\le C\Big(1+\sqrt{\log(1/c)}\Big)\Big(1+c^2\log\log T\Big). \tag{7.5}$$

Proof. By Theorem 6.1(a), $0\le\chi_k\le\delta_k^2/(4\pi(T-1))\le 1/T$ on $\mathcal{I}_T$. Since $1+\log X\le(1+\frac{1}{\log 2})c\log T$, Lemma 7.2 gives $\langle\mathfrak{t}_k^2\rangle\ll c^2\log\log T$, so $d_k=r_k(X)-u_k(X)=\mathfrak{t}_k+\chi_k$ has $\langle d_k^2\rangle\ll c^2\log\log T+T^{-2}$. Theorem 5.1 gives $\langle u_k(X)^2\rangle\ll 1+\log(1/c)$. Then $|\langle r_k^2\rangle-\langle u_k^2\rangle|\le 2\langle u_k^2\rangle^{1/2}\langle d_k^2\rangle^{1/2}+\langle d_k^2\rangle\ll(1+\sqrt{\log(1/c)})(c\sqrt{\log\log T}+T^{-1})+c^2\log\log T$, which is bounded by the right side of (7.5) because $y\le 1+y^2$. Combine with (5.1). □

Corollary 7.4. *Let $X = T^{1/\sqrt{\log\log T}}$. Then*

$$\langle r_k(X)^2\rangle_{\mathcal{I}_T} = \frac{1}{4\pi^2}\log\log\log T + O\Big(\sqrt{\log\log\log T}\Big), \tag{7.6}$$

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(\tau_k^X - \gamma_k\Big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}, \tag{7.7}$$

while $\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}(\tau_k-\gamma_k)^2 = (2+o(1))\log\log T/\log^2 T$; the ratio is asymptotic to $\log\log\log T/(2\log\log T)$.

Proof. Here $c = (\log\log T)^{-1/2}$, $2 \le X \le T^{1/2}$ for $T \ge T_0$, $\log(1/c) = \frac12\log\log\log T$ and $c^2\log\log T = 1$, so (7.5) gives (7.6). On $\mathcal{I}_T$, $\tau_k \in [T-1, 2T+1]$ and $\mathcal{L}(\tau_k) = \frac{\log T}{2\pi}(1+O(\frac{1}{\log T}))$; divide r_k^2 by $\mathcal{L}(\tau_k)^2$. For the bare clock, $\delta_k = s_k/\mathcal{L}(\xi_k)$ with $\mathcal{L}(\xi_k) = \frac{\log T}{2\pi}(1+O(\frac{1}{\log T}))$, and $\langle s_k^2\rangle \sim \frac{1}{2\pi^2}\log\log T$ by (3.3). □

Corollary 7.5 (A cutoff read off the clock)**.** *For $\tau_k > e^e$ let $X_k^{\mathrm{loc}} = \tau_k^{1/\sqrt{\log\log\tau_k}}$ and $\tau_k^{\mathrm{loc}} = \tau_k - S_{X_k^{\mathrm{loc}}}(\tau_k)/\mathcal{L}(\tau_k)$. Then*

$$\frac{1}{\#\mathcal{I}_T^\tau}\sum_{k\in\mathcal{I}_T^\tau}\Big(\tau_k^{\mathrm{loc}} - \gamma_k\Big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}.$$

Proof. Let $X_0 = T^{1/\sqrt{\log\log T}}$ and $X_0' = (2T)^{1/\sqrt{\log\log 2T}}$. The function $y \mapsto y/\sqrt{\log y}$ is increasing for $y > e^{1/2}$, so $X_0 \le X_k^{\mathrm{loc}} \le X_0'$ for $k \in \mathcal{I}_T^\tau$, and $\log X_0' - \log X_0 \le \log 2/\sqrt{\log\log T} = o(1)$; hence $X_0' \le 2X_0$ and $X_0' \le (T/3)^{1/2}$ for $T \ge T_0$. As in the proof of Theorem 5.12, $|S_{X_k^{\mathrm{loc}}}(\tau_k) - S_{X_0}(\tau_k)| \le \frac1\pi\mathcal{Q}(\tau_k) + 1.1V_\pm$ with $X_\pm$ replaced by X_0, X_0', and Lemma 5.11 gives a mean square over $\mathcal{I}_T^\tau$ that is $\ll V_\pm \ll 1/\log X_0 = \sqrt{\log\log T}/\log T$. The sets $\mathcal{I}_T$ and $\mathcal{I}_T^\tau$ differ by $O(\log T)$ indices, those with γ_k or τ_k within distance 1 of T or $2T$, and for them $|r_k(X_0)| \le |s_k| + \chi_k + \sigma_{X_0} \ll \log T + X_0^{1/2}$ (Lemma 2.2); so the means of $r_k(X_0)^2$ over the two sets differ by $O(T^{-1/2})$, and (7.6) holds over $\mathcal{I}_T^\tau$. Finally $\mathcal{L}(\tau_k)(\tau_k^{\mathrm{loc}} - \gamma_k) = r_k(X_0) - (S_{X_k^{\mathrm{loc}}}(\tau_k) - S_{X_0}(\tau_k))$; conclude with Cauchy–Schwarz and $\mathcal{L}(\tau_k) = \frac{\log T}{2\pi}(1+O(\frac{1}{\log T}))$. □

Proposition 7.6 (The root of the corrected equation in the monotone regime)**.** *Let $X = X(T) \ge 2$ with $M_X \le \frac12\mathcal{L}(T)$, and let $\mathcal{I}_T'$ be the set of $k \in \mathcal{I}_T$ with $F_X(T) \le k - \frac12 \le F_X(2T)$, so that the root $t_k^X \in [T, 2T]$ of Remark 5.7 exists. Then $X \ll \log^2 T$, $\#(\mathcal{I}_T \setminus \mathcal{I}_T') \ll \log T$, and*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T'}\Big(t_k^X - \gamma_k\Big)^2 \ge \frac{4}{9\mathcal{L}(2T)^2}\Big(\frac{1}{2\pi^2}\log\log T - O\Big(\sqrt{\log\log T}\Big)\Big) \gg \frac{\log\log T}{\log^2 T}.$$

Proof. $M_X \ge \frac1\pi\sum_{p\le X}\log p/\sqrt{p} \gg \sqrt{X}$ by Chebyshev's estimate, so $X \ll \mathcal{L}(T)^2 \ll \log^2 T$ and $c = \log X/\log T \le (2\log\log T + O(1))/\log T$. On $[T, 2T]$, $0 < F_X' \le \mathcal{L} + M_X \le \frac32\mathcal{L}(2T)$, so $|u_k(X)| = |F_X(t_k^X) - F_X(\gamma_k)| \le \frac32\mathcal{L}(2T)|t_k^X - \gamma_k|$ for $k \in \mathcal{I}_T'$. An index $k \in \mathcal{I}_T$ is missing from $\mathcal{I}_T'$ only if $k - \frac12 < F_X(T)$ or $k - \frac12 > F_X(2T)$; since $N(T) < k \le N(2T)$ and $|F_X - N| \le |G| + \sigma_X \ll \log T$ at T and $2T$, there are $O(\log T)$ such k, each with $|u_k(X)| \ll \log T$ (Lemma 2.2). Hence $\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T'}u_k(X)^2 = \langle u_k(X)^2\rangle_{\mathcal{I}_T} + O(\frac{\log^3 T}{T\log T})$, and Theorem 5.1 with $\log(1/c) \ge \log\log T - \log(2\log\log T + O(1))$ bounds it from below. □

Thus, in the regime $M_X \le \frac{1}{2}\mathcal{L}(T)$ where the corrected equation is monotone on the whole window, its root gains nothing in order: the error in height of t_k^X has mean square of the same order $\log\log T/\log^2 T$ as that of τ_k. The proposition says nothing about roots at larger cutoffs selected by a local criterion. Theorem 7.3 avoids roots and takes one Newton-type step instead.

Remark 7.7 (The Newton term and the restriction on c). Let ξ_k^X be a point between γ_k and τ_k with $\mathfrak{t}_k(X) = -S_X'(\xi_k^X)\delta_k$ (mean value theorem). Since $\delta_k = (r_k(X) + S_X(\tau_k))/\mathcal{L}(\tau_k)$ by (2.12), (7.1) becomes the exact relation

$$r_k(X)\Big(1 + \frac{S_X'(\xi_k^X)}{\mathcal{L}(\tau_k)}\Big) = u_k(X) + \chi_k - y_k\, S_X'(\xi_k^X). \tag{7.8}$$

The term $y_k S_X'(\xi_k^X)$ is the second-order term of Newton's method. Heuristically, $\langle S_X^2\rangle \approx v_X \approx \frac{1}{2\pi^2}\log\log X$ and $\langle S_X'^2\rangle/\mathcal{L}(T)^2 \approx \sum_{p\le X}\frac{\log^2 p}{2\pi^2 p}/\mathcal{L}(T)^2 \approx c^2$, so this term has mean square of order $c^2\log\log T$ if the two factors decorrelate: the order of Lemma 7.2 is the order of the Newton term, and the lemma cannot be improved by bounds that ignore the correlation between $u_k(X)$ and $\mathfrak{t}_k(X)$. Whether the transport cancels against $u_k(X)$ is open; a proof of such a cancellation would allow a fixed c in Corollary 7.4 (Question 10.2). The denominator in (7.8) can vanish only when $S_X'(\xi_k^X) = -\mathcal{L}(\tau_k)$; its sign cannot be decided from $F_X'(\tau_k)$ alone, because the derivative here is evaluated at the intermediate point ξ_k^X. The construction (2.12) does not divide by this denominator or select a root of F_X.

8. Iterating the Newton-type step

Theorem 7.3 takes one Newton-type step from τ_k. Its restriction $c \le (\log\log T)^{-1/2}$ comes from the transport over the interval between τ_k and γ_k, whose length is that of δ_k. After one step the error is smaller, so a second step can use a larger cutoff. This section proves the second step and then every further step, with constants uniform in the number of steps (Section 8.5).

8.1. The exact recursion

For a sequence of cutoffs $X_1, X_2, \dots \ge 2$ define $x_k^{(0)} = \tau_k$ and

$$x_k^{(j)} = x_k^{(j-1)} - \frac{N_0(x_k^{(j-1)}) + S_{X_j}(x_k^{(j-1)}) - (k-\frac{1}{2})}{\mathcal{L}(x_k^{(j-1)})}, \qquad e_{j,k} = x_k^{(j)} - \gamma_k, \tag{8.1}$$

so that $x_k^{(1)} = \tau_k^{X_1}$ and $e_{0,k} = \delta_k$. For a finite index set I and $r \ge 1$ let $\|a\|_{r,I} = \langle |a_k|^r\rangle_I^{1/r}$.

Proposition 8.1 (Exact recursion). *Let $j \ge 1$ and suppose $x_k^{(j-1)} > 2\pi$. Then*

$$\mathcal{L}(x_k^{(j-1)})\, e_{j,k} = u_k(X_j) + \Big(S_{X_j}(\gamma_k) - S_{X_j}(x_k^{(j-1)})\Big) + \chi_{j,k}, \qquad 0 \le \chi_{j,k} \le \frac{e_{j-1,k}^2}{4\pi\min(\gamma_k, x_k^{(j-1)})}. \tag{8.2}$$

Let $\varphi_j \in C^\infty(\mathbb{R})$, $0 \le \varphi_j \le 1$, be 1 on an interval containing all segments between γ_k and $x_k^{(j-1)}$, $k \in I$, and $\mathsf{g}_j = \mathcal{M}(\varphi_j S'_{X_j})$. Then for every $r \ge 1$,

$$\begin{aligned} \Big\|\mathcal{L}(x^{(j-1)})e_j\Big\|_{2r,I} &\le \|u(X_j)\|_{2r,I} + 2\,\|e_{j-1}\|_{4r,I}\,\|\mathsf{g}_j(x^{(j-1)})\|_{4r,I} + \|\chi_j\|_{2r,I}, \\ \|\chi_j\|_{2r,I} &\le \frac{\|e_{j-1}\|_{4r,I}^2}{4\pi \min_{k\in I}\min(\gamma_k, x_k^{(j-1)})}. \end{aligned} \tag{8.3}$$

Proof. Write $x = x_k^{(j-1)}$. By (8.1), $\mathcal{L}(x)e_{j,k} = \mathcal{L}(x)(x-\gamma_k) - N_0(x) - S_{X_j}(x) + k - \frac12$. As in the proof of Theorem 6.1(a), $N_0(x) - N_0(\gamma_k) = \int_{\gamma_k}^x \mathcal{L} = \mathcal{L}(x)(x-\gamma_k) - \chi_{j,k}$ with $\chi_{j,k} = \int_{\gamma_k}^x (\mathcal{L}(x) - \mathcal{L}(y))\,dy$, which lies between $(x-\gamma_k)^2/(4\pi\max(\gamma_k,x))$ and $(x-\gamma_k)^2/(4\pi\min(\gamma_k,x))$ by the same estimates of log. Hence $\mathcal{L}(x)e_{j,k} = k - \frac12 - N_0(\gamma_k) - S_{X_j}(x) + \chi_{j,k} = u_k(X_j) + S_{X_j}(\gamma_k) - S_{X_j}(x) + \chi_{j,k}$. For (8.3), $|S_{X_j}(\gamma_k) - S_{X_j}(x)| \le \int_{x-|e_{j-1,k}|}^{x+|e_{j-1,k}|} |\varphi_j S'_{X_j}| \le 2|e_{j-1,k}|\,\mathsf{g}_j(x)$, and Minkowski's and Hölder's inequalities in $\ell^{2r}(I)$. □

The recursion (8.3) is exact and uses no independence between the errors and the prime phases. It reduces the error of each step to three quantities: the moments of the defect $u_k(X_j)$ at the true ordinates (Theorem 5.1 and its higher-moment versions), the moments of the previous error, and the moments of the maximal function g_j sampled at the computable points $x_k^{(j-1)}$. The last is the only new difficulty: the points $x_k^{(j-1)}$ are not separated like the τ_k.

8.2. Moments of order $2r$ of the defect

Lemma 8.2. *There is an absolute constant C such that for every integer $r \ge 1$, every $T \ge T_0$ with $T^{1/(160r)} \ge 2$ and every $2 \le X \le T^{1/(160r)}$, with V as in Lemma 5.8 computed with $Y_{2r} = T^{1/(160r)}$ in place of Y_0,*

$$\int_T^{2T} E_X^{4r}\,dt \le \Big(C(r^2 + \sqrt{r(1+V)})\Big)^{4r} T, \qquad \langle u_k(X)^{2r}\rangle_{\mathcal{I}_T} \le \Big(C(r^2 + \sqrt{r(1+V)})\Big)^{2r},$$

and for every $r \ge 1$ and $2 \le X \le T^{1/(2r)}$,

$$\int_{T-3}^{2T+3} |S_X^{(\nu)}(t)|^{4r} dt \le (Cr)^{2r}\, T(1+\log X)^{4r\nu} \qquad (\nu = 1, 2).$$

Proof. The first bound: $E_X = E_0 + \mathcal{W}$ with $E_0 = S - S^{\mathrm{pr}}_{Y_{2r}}$, $\|E_0\|_{L^{4r}(T,2T)} \le C(2r)^2 T^{1/4r}$ by Theorem 3.5 with $k = 2r$, and $\mathcal{W}$ as in Lemma 5.10: $|\mathcal{W}_1| \le |\mathcal{D}_1|/\pi$, $\mathcal{D}_1^{2r} = \sum_n \eta_n n^{-it}$ over products of $2r$ primes $\le \max(X, Y_{2r})$, so $n \le T^{1/80}$, with $\sum|\eta_n|^2 \le (2r)!\,(V+1)^{2r}$ (each n has at most $(2r)!$ ordered factorizations, so $\sum_n |\eta_n|^2 \le (2r)! \sum_{p_1,\dots,p_{2r}} \prod p_i^{-1}$); Theorem 3.7 gives $\int |\mathcal{D}_1|^{4r} \le 2T\sum|\eta_n|^2$, whence $\|\mathcal{W}_1\|_{L^{4r}} \le C\sqrt{r}\sqrt{1+V}\,T^{1/4r}$ by $((2r)!)^{1/4r} \le \sqrt{2r}$. The terms $m = 2$ and $m \ge 3$ are $O(T^{1/4r})$ in the same norm (products of $2r$ squares of primes, $\sum|\eta_n'|^2 \le (2r)!\,(\sum_p p^{-2})^{2r} \le (2r)!$, and a bounded function). Minkowski's inequality in L^{4r} gives the first bound. The second: Theorem 4.1 with $j = r$ and $F = F_X$, $\Psi_r(u) \ge u^{2r}$, gives $\sum_{k\in\mathcal{I}_T} u_k^{2r} \le \int_T^{2T} (|E_X| + T^{-1})^{2r}(\mathcal{L} + |S'_X|)\,dt + C_{\mathrm{b}}\Big(4^r T^{1/100} + (2\log T)^{2r+1}\Big)$ with an absolute C_{b}, the last term bounding the boundary terms and the end pieces, since $|u_k(X)| \le \log T + \sqrt{X}$ (Lemma 2.2 and (2.11)) and $X^{r+1/2} \le T^{1/100}$. Since $r \le \log T/(160\log 2)$,

$4^r T^{1/100} \le T^{1/80+1/100} \le T^{1/2}$. For $(2\log T)^{2r+1}$ there are two cases: if $r \ge \sqrt{2\log T/C}$ then $2\log T \le Cr^2$ and $(2\log T)^{2r+1} \le (Cr^2)^{2r}\, 2\log T \le (Cr^2)^{2r}\, T\log T$; if $r < \sqrt{2\log T/C}$ then $(2\log T)^{2r+1} \le \exp\bigl(3\sqrt{2\log T/C}\,\log(2\log T)\bigr) \le T^{1/2}$ for $T \ge T_0$. In both cases the boundary term is at most $C_{\mathrm{b}}(Cr^2)^{2r}(T^{1/2}+T\log T) \le 9\pi C_{\mathrm{b}}(Cr^2)^{2r}\,\#\mathcal{I}_T$ for $T \ge T_0$. With $\mathcal{E}_{4r} = \int_T^{2T} E_X^{4r}$, Hölder's inequality gives $\int_T^{2T} E_X^{2r} \le T^{1/2}\mathcal{E}_{4r}^{1/2}$ and, by (5.8), $\int_T^{2T} E_X^{2r}|S_X'| \le \mathcal{E}_{4r}^{1/2} C_{\mathrm{d}} T^{1/2}(1+\log X)$, C_{d} the constant of (5.8); so the integral is at most $(\mathcal{L}(2T) + C_{\mathrm{d}}(1+\log X))T^{1/2}(\mathcal{E}_{4r}^{1/2} + T^{-2r+1/2})$. Dividing by $\#\mathcal{I}_T \ge \frac{1}{2}T\mathcal{L}(T)$ and using $\mathcal{L}(2T) \le 2\mathcal{L}(T)$ and $1+\log X \le 2\pi\mathcal{L}(T)$, $\langle u_k(X)^{2r}\rangle_{\mathcal{I}_T} \le (4+4\pi C_{\mathrm{d}})T^{-1/2}\mathcal{E}_{4r}^{1/2} + (9\pi C_{\mathrm{b}}+1)(Cr^2)^{2r}$, and the first bound gives the second with $(9\pi C_{\mathrm{b}}+2)C$ in place of C, since $(9\pi C_{\mathrm{b}}+2)^{2r} \ge 9\pi C_{\mathrm{b}}+2$. The third: Lemma 7.1 with products of $2r$ primes, $n \le X^{2r} \le T$, and $\sum \eta_n^2 \le (2r)!\,(\sum_{p\le X}\log^{2\nu} p/p)^{2r} \le (Cr)^{2r}(1+\log X)^{4r\nu}$, by Theorem 3.8 and partial summation; the terms $m=2$ and $m \ge 3$ as in Lemma 7.1. □

The dependence on r matters in Theorem 8.11, where r grows with the number of Newton-type steps; for fixed r the three bounds read $\ll_r T(1+V)^{2r}$, $\ll_r (1+V)^r$ and $\ll_r T(1+\log X)^{4r\nu}$.

The moments of all orders give a tail bound for the defect that no second-moment argument can give: the proportion of zeros whose corrected index is off by more than λ positions decays exponentially in $\sqrt{\lambda}$, provided the cutoff is chosen in a range that depends on λ.

Theorem 8.3 (Exponential tails of the ordinal defect)**.** *There are absolute constants $C_8 \ge 1$, λ_0 and T_0 such that, for $T \ge T_0$, every $\lambda \ge \lambda_0$ and every $X = T^c \ge 2$ with*

$$\exp\bigl(-(\lambda/C_8)^{3/2}\bigr) \le c \le \frac{1}{C_8\sqrt{\lambda}}, \tag{8.4}$$

$$\#\{k \in \mathcal{I}_T : |u_k(X)| > \lambda\} \le 2^{\,2-\sqrt{\lambda/C_8}}\,\#\mathcal{I}_T. \tag{8.5}$$

Proof. Let C be the constant of Lemma 8.2 and $r = \lfloor\sqrt{\lambda/(4C)}\rfloor$, so $r \ge 1$ for $\lambda \ge 4C$ and $Cr^2 \le \lambda/4$. The lemma applies: $X \le T^{1/(160r)}$ because $c \le 1/(C_8\sqrt{\lambda}) \le 1/(160r)$ once $C_8 \ge 80$, since $160r \le 160\sqrt{\lambda/(4C)} \le 80\sqrt{\lambda}$; and $T^{1/(160r)} \ge X \ge 2$. With $Y_{2r} = T^{1/(160r)} \ge X$, Lemma 5.8 gives $V = \log(1/(160rc)) + O(1) \le \log(1/c) + C_9 \le (\lambda/C_8)^{3/2} + C_9$ by (8.4), so $1+V \le 2(\lambda/C_8)^{3/2}$ for $\lambda \ge C_8(C_9+1)^{2/3}$. Then $r(1+V) \le \sqrt{\lambda/(4C)}\cdot 2\lambda^{3/2}C_8^{-3/2} = \lambda^2/(\sqrt{C}\,C_8^{3/2})$ and $C\sqrt{r(1+V)} \le C^{3/4}C_8^{-3/4}\lambda \le \lambda/4$ once $C_8 \ge 7C$. Hence $C(r^2+\sqrt{r(1+V)}) \le \lambda/2$, and Chebyshev's inequality with the second bound of the lemma gives

$$\#\{k \in \mathcal{I}_T : |u_k(X)| > \lambda\} \le \lambda^{-2r}\sum_{k\in\mathcal{I}_T} u_k(X)^{2r} \le 2^{-2r}\,\#\mathcal{I}_T \le 2^{\,2-\sqrt{\lambda/C}}\,\#\mathcal{I}_T,$$

since $2r \ge 2\sqrt{\lambda/(4C)} - 2$. Take $C_8 = \max(80, 7C)$ and $\lambda_0 = C_8(C_9+1)^{2/3}$; then $\lambda \ge \lambda_0 \ge 4C$, and $2^{-\sqrt{\lambda/C}} \le 2^{-\sqrt{\lambda/C_8}}$. □

For the cutoff $X = T^{1/(C_8\sqrt{\lambda})}$, Theorem 5.1 gives $\langle u_k(X)^2\rangle_{\mathcal{I}_T} = \frac{1}{4\pi^2}\log\lambda + O(\sqrt{\log\lambda})$, so Chebyshev's inequality alone bounds the proportion in (8.5) by $\lambda^{-2}\log\lambda$; the moments of order $2r \asymp \sqrt{\lambda}$ replace this by $2^{2-\sqrt{\lambda/C_8}}$. The bound (8.5) is for cutoffs that depend on λ; for a fixed cutoff exponent c it applies only to $\lambda \le (C_8 c)^{-2}$, and no exponential tail is claimed beyond that. The range (8.4) is the price of the moment order: Lemma 8.2 needs $X \le T^{1/(160r)}$, and Fujii's approximation 3.5 with $2r$ in place of k needs the prime polynomial of length $T^{1/(160r)}$.

8.3. Multiplicity of the one-step clock points

Let $X_1 = T^{c_1}$ with $c_1 = (\log\log T)^{-1/2}$, $x_k^{(1)} = \tau_k^{X_1} = \tau_k - y_k$ with $y_k = S_{X_1}(\tau_k)/\mathcal{L}(\tau_k)$, $\mathfrak{b}_T = 2\pi/\log(3T)$ as in Lemma 5.11, and

$$\tilde{n}(u) = \#\{k \in \mathcal{I}_T :\ u - \mathfrak{b}_T \le x_k^{(1)} \le u\} \qquad (u \in \mathbb{R}).$$

Note that $|y_k| \le \sigma_{X_1}/\mathcal{L}(\tau_k) \ll T^{c_1/2}$, so $x_k^{(1)} \in [T - T^{1/2}, 2T + T^{1/2}]$ for $T \ge T_0$.

Lemma 8.4 (Multiplicity)**.** $\displaystyle\int_{\mathbb{R}} \tilde{n}(u)^2\,du = \sum_{j,k\in\mathcal{I}_T} \Big(\mathfrak{b}_T - |x_j^{(1)} - x_k^{(1)}|\Big)_+ \ll \mathfrak{b}_T\,\#\mathcal{I}_T.$

Proof. The identity: $\tilde{n} = \sum_k \mathbf{1}_{[x_k^{(1)}, x_k^{(1)}+\mathfrak{b}_T]}$, and the measure of the intersection of two such intervals is $(\mathfrak{b}_T - |x_j^{(1)} - x_k^{(1)}|)_+$. Write $b = \mathfrak{b}_T$ and $\mathsf{M} = \#\mathcal{I}_T$. The pairs with $|j-k| \le 1$ contribute at most $3b\mathsf{M}$. Let $j = k+m$ with $m \ge 2$ and $|x_{k+m}^{(1)} - x_k^{(1)}| \le b$. Since $\tau_{k+m} - \tau_k \ge mb$ (proof of Lemma 5.11), this forces

$$|y_{k+m} - y_k| \ge (m-1)b. \tag{8.6}$$

Two consequences of (8.6) are counted separately; a close pair at distance m satisfies both.

Variation. $y = S_{X_1}/\mathcal{L}$ is C^∞ with $y' = S'_{X_1}/\mathcal{L} - S_{X_1}\mathcal{L}'/\mathcal{L}^2$, and on $J_{k,m} = [\tau_k, \tau_{k+m}]$, of length $|J_{k,m}| \le mb(1+o(1))$,

$$|y_{k+m} - y_k| \le \frac{1+o(1)}{\mathcal{L}(T)}\int_{J_{k,m}} |S'_{X_1}| + \frac{\sigma_{X_1}|J_{k,m}|}{2\pi(T-1)\mathcal{L}(T)^2},$$

and the last term is $\ll T^{c_1/2-1}mb/\log^2 T = o((m-1)b)$ uniformly in $m \ge 2$. As $b\mathcal{L}(T) = 1 - o(1)$, (8.6) gives $\frac{1}{|J_{k,m}|}\int_{J_{k,m}} |S'_{X_1}| \ge \frac{1}{3}\mathcal{L}(T)$ for $T \ge T_0$. Each t lies in at most m of the intervals $J_{k,m}$ (k varying), and $|J_{k,m}| \ge mb$; by Chebyshev's inequality and Jensen's inequality,

$$\begin{aligned}\#\{k : \text{variation}\} &\le \frac{9}{\mathcal{L}(T)^2}\sum_k \frac{1}{|J_{k,m}|}\int_{J_{k,m}} |S'_{X_1}|^2 \\ &\le \frac{9}{\mathcal{L}(T)^2\,b}\int_{T-1}^{2T+1} |S'_{X_1}|^2 \ll T\log T\,c_1^2 \asymp \mathsf{M}\,c_1^2.\end{aligned}$$

by (5.8), uniformly in m.

Size. (8.6) gives $\max(|y_k|, |y_{k+m}|) \ge \frac{m-1}{2}b$, hence $\max(|S_{X_1}(\tau_k)|, |S_{X_1}(\tau_{k+m})|) \ge \frac{m-1}{3}$ for $T \ge T_0$. Sampling at the clock points as in the proof of Lemma 5.11 (Sobolev on the disjoint intervals $[\tau_k, \tau_k + b]$), $\langle S_{X_1}(\tau_k)^4\rangle_{\mathcal{I}_T} \ll \frac{1}{T}\int S_{X_1}^4 + \frac{1}{T\log^4 T}\int |S'_{X_1}|^4 \ll (1+\log\log T)^2$, the first

integral by Theorem 3.7 with products of two primes ($\sum_{p\le X_1} p^{-1} \ll \log\log T$) and the second by Lemma 7.1. By Chebyshev's inequality of order 4, $\#\{k : \text{size}\} \ll \mathsf{M}(\log\log T)^2/(m-1)^4$.

Sum. With $H_0 = \log\log T$ and $c_1^2 = H_0^{-1}$, the number of close ordered pairs at distance $m \ge 2$ is $\ll \mathsf{M}\min(H_0^{-1}, H_0^2(m-1)^{-4})$, and

$$\sum_{m\ge 2} \min\Big(H_0^{-1}, \frac{H_0^2}{(m-1)^4}\Big) \le \sum_{m-1\le H_0^{3/4}} H_0^{-1} + \sum_{m-1> H_0^{3/4}} \frac{H_0^2}{(m-1)^4} \ll H_0^{-1/4}.$$

Hence the number of pairs (j,k) with $|x_j^{(1)} - x_k^{(1)}| \le b$ is $\ll \mathsf{M}$, and each contributes at most b. □

Lemma 8.5 (Sampling at points of bounded multiplicity)**.** *Let x_k, $k \in \mathcal{I}_T$, be real points of $[T - 2T^{1/2}, 2T + 2T^{1/2}]$ with multiplicity function $\tilde n(u) = \#\{k \in \mathcal{I}_T : u - \mathfrak{b}_T \le \mathsf{x}_k \le u\}$ satisfying $\int_{\mathbb{R}} \tilde n^2 \le \mathfrak{k}\,\mathfrak{b}_T \#\mathcal{I}_T$ for some $\mathfrak{k} \ge 1$. Let $r \ge 1$ be an integer, $2 \le X \le T^{1/(4r)}$, φ a cutoff equal to 1 on $[T - 2T^{1/2}, 2T + 2T^{1/2}]$ as in Lemma 7.2, and $g = \mathcal{M}(\varphi S_X')$. Then, with an absolute constant C,*

$$\|g(\mathsf{x}_k)\|_{4r,\mathcal{I}_T} \le C\,\mathfrak{k}^{1/(8r)}\sqrt{r}\,(1+\log X).$$

In particular $\langle g(x_k^{(1)})^4\rangle_{\mathcal{I}_T} \ll (1+\log X)^4$ at the one-step clock points $x_k^{(1)}$, for $2 \le X \le T^{1/4}$.

Proof. Let $\mathsf{F} = g^{4r}$, locally absolutely continuous with $\mathsf{F}' = 4rg^{4r-1}g'$ (Theorem 3.10). For $x \in [u-b, u]$, $\mathsf{F}(x) \le \frac{1}{b}\int_{u-b}^u F + \int_{u-b}^u |\mathsf{F}'|$. Averaging over $u \in [\mathsf{x}_k, \mathsf{x}_k + b]$ and summing over k,

$$\sum_{k\in\mathcal{I}_T} \mathsf{F}(\mathsf{x}_k) \le \frac{1}{b}\int_{\mathbb{R}} \tilde n(u)\Big(\frac{1}{b}\int_{u-b}^u F + \int_{u-b}^u |\mathsf{F}'|\Big)du \le \frac{1}{b}\|\tilde n\|_2\big(\|\mathsf{F}\|_2 + b\|\mathsf{F}'\|_2\big),$$

by Cauchy–Schwarz and because the averaging operator $h \mapsto \frac{1}{b}\int_{u-b}^u h$ has norm ≤ 1 on $L^2(\mathbb{R})$. Here $\|\mathsf{F}\|_2 = \|g\|_{8r}^{4r} \le (\mathcal{A}_{8r}\|\varphi S_X'\|_{8r})^{4r} \le (C\sqrt{r}(1+\log X))^{4r} T^{1/2}$ and $\|\mathsf{F}'\|_2 \le 4r\|g\|_{8r}^{4r-1}\|g'\|_{8r} \le 4r(1+\log X)\,(C\sqrt{r}(1+\log X))^{4r}T^{1/2}$, by Theorem 3.10 ($p = 8r$, $\mathcal{A}_p$ bounded) and Lemma 8.2 (third bound with $2r$ in place of r, which needs $X^{4r} \le T$). Since $b \cdot 4r(1+\log X) \le 8\pi r\,c\,(1+o(1)) \le 7$ for $c = \log X/\log T \le 1/(4r)$, we get $\sum_k F(\mathsf{x}_k) \le 8\,b^{-1}(\mathfrak{k}b\mathsf{M})^{1/2}T^{1/2}(C\sqrt{r}(1+\log X))^{4r} \le C'\sqrt{\mathfrak{k}}\,\mathsf{M}\,(C\sqrt{r}(1+\log X))^{4r}$, since $b^{-1}T \asymp \mathsf{M} = \#\mathcal{I}_T$; take the $4r$-th root. For the $x_k^{(1)}$, $\mathfrak{k} \ll 1$ by Lemma 8.4. □

8.4. The second step

Theorem 8.6 (Second Newton-type step)**.** *Let $H_0 = \log\log T$, $H_1 = \log H_0$, $H_2 = \log H_1$, $X_1 = T^{H_0^{-1/2}}$, $X_2 = T^{H_1^{-1/2}}$, and let $x_k^{(2)}$ be given by* (8.1)*. Then*

$$\begin{aligned}
&\Big\langle \mathcal{L}(x_k^{(1)})^2 e_{2,k}^2\Big\rangle_{\mathcal{I}_T} = \frac{1}{4\pi^2}H_2 + O\Big(1+\sqrt{H_2}\Big),\\
&\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(x_k^{(2)} - \gamma_k\Big)^2 = \frac{\log\log\log\log T + O\Big(\sqrt{\log\log\log\log T}\Big)}{\log^2 T}.
\end{aligned} \tag{8.7}$$

Proof. Fourth moment of e_1. By (8.2) with $j=1$ (that is (7.1)), $\mathcal{L}(\tau_k)e_{1,k} = u_k(X_1)+\mathfrak{t}_k(X_1)+\chi_k$. Lemma 8.2 with $r=2$ gives $\langle u_k(X_1)^4\rangle \ll (1+\log(1/c_1))^2 \ll H_1^2$. For the transport, $|\mathfrak{t}_k| \le 2|\delta_k|\mathfrak{g}_1(\tau_k)$ with $\mathfrak{g}_1 = \mathcal{M}(\varphi S'_{X_1})$, so $\langle \mathfrak{t}_k^4\rangle \le 16\langle \delta_k^8\rangle^{1/2}\langle \mathfrak{g}_1(\tau_k)^8\rangle^{1/2}$; here $\langle\delta_k^8\rangle \ll H_0^4/\log^8 T$ by (3.3) with $j=4$, and $\langle \mathfrak{g}_1(\tau_k)^8\rangle \ll (c_1\log T)^8$ by the sampling argument of Lemma 7.2 in L^8 (Sobolev on the intervals $\mathcal{U}_k$ with g^8, Theorem 3.10 with $p=8$, Lemma 8.2 with $r=2$). Hence $\langle \mathfrak{t}_k^4\rangle \ll (c_1^2H_0)^2 = 1$, and $\chi_k \ll 1/T$. Dividing by $\mathcal{L}(\tau_k)^4 \asymp \log^4 T$,

$$\langle e_{1,k}^4\rangle_{\mathcal{I}_T} \ll \frac{H_1^2}{\log^4 T}. \tag{8.8}$$

Transport of the second step. The interval centred at $x_k^{(1)}$ of radius $|e_{1,k}|$ contains γ_k and lies in $[T-2T^{1/2}, 2T+2T^{1/2}]$, so $|S_{X_2}(\gamma_k) - S_{X_2}(x_k^{(1)})| \le 2|e_{1,k}|\mathfrak{g}_2(x_k^{(1)})$ with $\mathfrak{g}_2 = \mathcal{M}(\varphi S'_{X_2})$, and by Cauchy–Schwarz, (8.8) and Lemma 8.5 (applicable since $X_2 \le T^{1/8}$),

$$\Big\langle (S_{X_2}(\gamma_k) - S_{X_2}(x_k^{(1)}))^2\Big\rangle \le 4\langle e_{1,k}^4\rangle^{1/2}\langle \mathfrak{g}_2(x_k^{(1)})^4\rangle^{1/2} \ll \frac{H_1}{\log^2 T}(c_2\log T)^2 = c_2^2 H_1 = 1.$$

Conclusion. By (8.2) with $j=2$, $0 \le \chi_{2,k} \le e_{1,k}^2/(2\pi T)$, and $\langle \chi_{2,k}^2\rangle \ll T^{-2}$. Theorem 5.1 with $X = X_2$ (allowed, $2 \le X_2 \le T^{1/2}$) gives $\langle u_k(X_2)^2\rangle = \frac{1}{2\pi^2}\log\frac{1}{c_2} + O(1+\sqrt{\log(1/c_2)})$ with $\log\frac{1}{c_2} = \frac12 H_2$. Combining as in the proof of Theorem 7.3 gives the first statement of (8.7), and $\mathcal{L}(x_k^{(1)}) = \frac{\log T}{2\pi}(1+O(T^{-1/2}))$ on $\mathcal{I}_T$ gives the second. □

8.5. All further steps: the exact identity, the local deformation, and the multiplicity

This subsection proves the Newton-type steps $m \ge 3$ with constants uniform in m, up to $m = \log^* T + O(1)$ steps, where the error becomes $O(1)$ in units of zeros.

Lemma 8.7 (The second-level identity)**.** *For $j \ge 1$, with $y_k^{(j)} = \tau_k - x_k^{(j)}$ and $\chi''_{j,k} = \int_{x_k^{(j-1)}}^{\tau_k}(\mathcal{L}(\tau_k)-\mathcal{L}(y))\,dy$,*

$$\begin{aligned} y_k^{(j)} &= \Big(1 - \frac{\mathcal{L}(\tau_k)}{\mathcal{L}(x_k^{(j-1)})}\Big)y_k^{(j-1)} + \frac{S_{X_j}(x_k^{(j-1)}) + \chi''_{j,k}}{\mathcal{L}(x_k^{(j-1)})}, \\ \textit{hence}\qquad \mathcal{L}(\tau_k)y_k^{(j)} &= S_{X_j}(x_k^{(j-1)}) + \mathfrak{r}_{j,k}, \end{aligned} \tag{8.9}$$

with $|\mathfrak{r}_{j,k}| \ll T^{c_j-1}$ if $|y_k^{(j-1)}| \le T^{c_{j-1}/2}$ and $c_{j-1} \le c_j$ (true by (2.11) and induction): the three contributions are $\mathcal{L}(\tau_k)(1-\mathcal{L}(\tau_k)/\mathcal{L}(x))y_k^{(j-1)} \ll (y_k^{(j-1)})^2/T$, $(\mathcal{L}(\tau_k)/\mathcal{L}(x)-1)S_{X_j}(x) \ll |y_k^{(j-1)}|\sigma_{X_j}/(T\log T)$ and $\chi''_{j,k} \le (y_k^{(j-1)})^2/(4\pi(T-T^{1/2}))$. Newton's step does not accumulate the corrections of the previous levels: it replaces the previous one by the phase of the new cutoff evaluated at the previous point.

Proof. $N_0(x_k^{(j-1)}) - (k-\frac12) = N_0(x_k^{(j-1)}) - N_0(\tau_k) = -\mathcal{L}(\tau_k)y_k^{(j-1)} + \chi''_{j,k}$, with $0 \le \chi''_{j,k} \le (y_k^{(j-1)})^2/(4\pi(T-T^{1/2}))$; insert in (8.1) and use $1-\mathcal{L}(\tau_k)/\mathcal{L}(x) = O(|y_k^{(j-1)}|/(T\log T))$ and $|S_{X_j}| \le \sigma_{X_j}$. □

Proposition 8.8 (Local deformation of the second clock)**.** *Let $X_1 = T^{c_1}$, $X_2 = T^{c_2}$ with $c_1 = H_0^{-1/2}$, $c_2 = H_1^{-1/2}$, and $d_{j,k} = x_{k+1}^{(j)} - x_k^{(j)}$ ($d_{0,k} = \tau_{k+1} - \tau_k$). Then, for $T \ge T_0$,*

$$\Big\langle \Big| \frac{d_{2,k} - d_{0,k}}{d_{0,k}} \Big|^4 \Big\rangle_{\mathcal{I}_T} \ll c_2^4 = \frac{1}{H_1^2}, \qquad \frac{1}{\#\mathcal{I}_T} \#\Big\{ k \in \mathcal{I}_T : \Big| \frac{d_{2,k}}{d_{0,k}} - 1 \Big| > \alpha \Big\} \ll \frac{1}{\alpha^4 H_1^2} \quad (0 < \alpha \le 1), \tag{8.10}$$

in particular $\#\{k \in \mathcal{I}_T : d_{2,k} \le 0\} \ll \#\mathcal{I}_T/(\log\log\log T)^2$: the second clock preserves the order and a spacing comparable to that of the RvM clock outside a proportion $O(H_1^{-2})$ of the indices.

Proof. By (8.9) with $j = 2$, $d_{2,k} - d_{0,k} = -\Delta_k\big(S_{X_2}(x^{(1)})/\mathcal{L}(\tau)\big) - \Delta_k(\mathfrak{r}_2/\mathcal{L}(\tau))$, where $\mathfrak{r}_2$ is the sequence $\mathfrak{r}_{2,k}$ of Lemma 8.7, $\Delta_k a = a_{k+1} - a_k$; the last term is $\ll T^{c_1 - 1}$, negligible against $d_{0,k} \asymp 1/\log T$, and $\Delta_k(1/\mathcal{L}(\tau)) \ll 1/(T\log^3 T)$ is negligible too, so it suffices to bound $\Delta_k S_{X_2}(x^{(1)})/(d_{0,k}\mathcal{L}(\tau_k))$ in $\ell^4(\mathcal{I}_T)$. Let $\mathsf{g}_j = \mathcal{M}(\varphi S'_{X_j})$ with $\varphi = 1$ on $[T - 2T^{1/2}, 2T + 2T^{1/2}]$. Two applications of $|S(a) - S(b)| \le 2|a - b|\,\mathcal{M}(S')(a)$:

$$|x_{k+1}^{(1)} - x_k^{(1)}| \le d_{0,k} + |y_{k+1} - y_k| \le d_{0,k}\Big(1 + \frac{2\mathsf{g}_1(\tau_k)}{\mathcal{L}(\tau_k)}\Big) + O\Big(T^{c_1/2 - 1}\Big),$$

$$|\Delta_k S_{X_2}(x^{(1)})| \le 2|x_{k+1}^{(1)} - x_k^{(1)}|\,\mathsf{g}_2(x_k^{(1)}).$$

Hence, with $\mathcal{L}(\tau_k) \asymp \log T$,

$$\Big\| \frac{\Delta S_{X_2}(x^{(1)})}{d_0\,\mathcal{L}(\tau)} \Big\|_{4,\mathcal{I}_T} \ll \Big\| \frac{g_2(x^{(1)})}{\log T} \Big\|_{4,\mathcal{I}_T} + \Big\| \frac{g_2(x^{(1)})}{\log T} \Big\|_{8,\mathcal{I}_T} \Big\| \frac{g_1(\tau)}{\log T} \Big\|_{8,\mathcal{I}_T} + o(1).$$

Sampling at the τ_k in L^8 (Sobolev on the disjoint intervals $\mathcal{U}_k$ as in Lemma 7.2, with Lemma 8.2 for $\int |S'_{X_1}|^8$, $\int |S''_{X_1}|^8$ and Theorem 3.10 with $p = 8$) gives $\|\mathsf{g}_1(\tau)/\log T\|_8 \ll c_1$. Sampling at the $x_k^{(1)} = x_k^{(1)}$ in L^8 is Lemma 8.5 with $\mathsf{F} = \mathsf{g}_2^8$: $\|\mathsf{F}\|_2 = \|\mathsf{g}_2\|_{16}^8$ and $\|\mathsf{F}'\|_2 \le 8\|\mathsf{g}_2\|_{16}^7 \|\mathsf{g}_2'\|_{16}$, which need Theorem 3.7 with products of 8 primes ($X_2^{16} \le T$, true for $T \ge T_0$ since $c_2 \to 0$) and Theorem 3.10 with $p = 16$; the result is $\|\mathsf{g}_2(x^{(1)})/\log T\|_8 \ll c_2$. The first inequality of (8.10) follows, and the second is Markov's inequality; the last statement is $\alpha = 1$. □

Proposition 8.8 controls consecutive differences; the multiplicity of the points $x_k^{(j)}$ at scale $\mathfrak{b}_T$, which Lemma 8.5 needs, is obtained for every j by comparing $x_k^{(j)}$ with γ_k through two unconditional counts of near pairs and near triples of ordinates. The ordinates enter the proof of the bound, as in Theorem 5.1, and not the construction of the points.

Lemma 8.9 (Near pairs and triples of ordinates)**.** *Let $b = \mathfrak{b}_T$, $\mathsf{M} = \#\mathcal{I}_T$ and, for $\mathsf{R} \ge 1$ and real u, $n_{\mathsf{R}}(u) = \#\{l \in \mathcal{I}_T : |\gamma_l - u| \le \mathsf{R}b\}$. There are absolute constants C_1, C_2 such that for $T \ge T_0$ and every $\mathsf{R} \ge 1$,*

$$\#\{(k,l) \in \mathcal{I}_T^2 : |\gamma_k - \gamma_l| \le \mathsf{R}b\} \le C_1 \mathsf{R}\mathsf{M}, \qquad \sum_{k \in \mathcal{I}_T} n_{\mathsf{R}}(\gamma_k)^2 \le C_2 \mathsf{R}^2 \mathsf{M}. \tag{8.11}$$

Proof. If $4\mathsf{R}b > 1$, then $N(u+h) - N(u) \le Ch\log T$ for $1 \le h \le T$ and $u \in [T-1, 2T]$, by (2.3) and Theorem 3.1. Since only $l \in \mathcal{I}_T$ are counted, $n_{\mathsf{R}}(\gamma_k)$ is at most the number of ordinates in an interval of length $h = \min(2\mathsf{R}b, T)$ contained in $[T-1, 2T]$, which is $\le Ch\log T \le C'\mathsf{R}$ (for

$h = T$ use $T\log T \le 2\mathsf{R}b\log T \le 4\pi\mathsf{R}$); both bounds follow. Let $4\mathsf{R}b \le 1$ and $h \in \{2\mathsf{R}b, 4\mathsf{R}b\}$. For $u \in [T-1, 2T]$, $N(u+h) - N(u) = h\mathcal{L}(u) + S(u+h) - S(u) + O(1/T)$ with $h\mathcal{L}(u) \le 5\mathsf{R}$, so Theorem 3.6 with $k = 1, 2$ and $\log(2 + h\log T) \le \log(30\mathsf{R}) \le 30\mathsf{R}$ give

$$\int_{T-1}^{2T} \Big(N(u+h) - N(u)\Big)^{2k} du \le C\mathsf{R}^{2k}T \qquad (k = 1, 2).$$

A pair (k, l) with $|\gamma_k - \gamma_l| \le \mathsf{R}b$ lies in $[u, u+2\mathsf{R}b]$ for every $u \in [\min(\gamma_k, \gamma_l) - \mathsf{R}b, \min(\gamma_k, \gamma_l)]$, a set of measure $\mathsf{R}b$ contained in $[T-1, 2T]$; hence the number of pairs is at most $\frac{1}{\mathsf{R}b}\int_{T-1}^{2T}(N(u+2\mathsf{R}b) - N(u))^2 du \le C\mathsf{R}T/b \le C_1\mathsf{R}\mathsf{M}$, as $\mathsf{M} \asymp T/b$. A triple (k, l, l') with $|\gamma_l - \gamma_k|, |\gamma_{l'} - \gamma_k| \le \mathsf{R}b$ lies in $[u, u+4\mathsf{R}b]$ for u in a set of measure $2\mathsf{R}b$; by Hölder's inequality the number of triples is at most $\frac{1}{2\mathsf{R}b}\int_{T-1}^{2T}(N(u+4\mathsf{R}b) - N(u))^3 du \le \frac{T^{1/4}(C\mathsf{R}^4T)^{3/4}}{2\mathsf{R}b} \le C_2\mathsf{R}^2\mathsf{M}$; the triples with $l, l' \in \mathcal{I}_T$ are among them. □

Theorem 8.10 (Multiplicity of the corrected clock points)**.** *Let $j \ge 0$ and let $x_k^{(j)}$ be given by* (8.1) *with cutoffs $X_i = T^{c_i}$, $c_i \le \frac{1}{2}$, so that $x_k^{(j)} \in [T - 2T^{1/2}, 2T + 2T^{1/2}]$ for $k \in \mathcal{I}_T$ and $T \ge T_0$ (Lemma 8.7). Let $\tilde{n}_j(u) = \#\{k \in \mathcal{I}_T : u - \mathfrak{b}_T \le x_k^{(j)} \le u\}$, and let $\mathsf{A} \ge 1$, $\mathsf{H} \ge 1$ be such that $\|\mathcal{L}(\tau_k)e_{j,k}\|_{4,\mathcal{I}_T} \le \mathsf{A}\sqrt{\mathsf{H}}$. Then, with an absolute constant C_3,*

$$\int_{\mathbb{R}} \tilde{n}_j(u)^2\, du \le C_3\, \mathsf{A}\sqrt{\mathsf{H}}\; \mathfrak{b}_T\, \#\mathcal{I}_T. \tag{8.12}$$

Proof. Write $b = \mathfrak{b}_T$, $\mathsf{M} = \#\mathcal{I}_T$, $x_k = x_k^{(j)}$, $e_k = e_{j,k}$. As in Lemma 8.4, $\int \tilde{n}_j^2 = \sum_{k,l\in\mathcal{I}_T}(b - |x_k - x_l|)_+ \le b\,\mathsf{P}$ with $\mathsf{P} = \#\{(k,l) \in \mathcal{I}_T^2 : |x_k - x_l| < b\}$. Let $\mathsf{R}_0 \ge 2$. The pairs with $|\gamma_k - \gamma_l| \le \mathsf{R}_0 b$ number at most $C_1\mathsf{R}_0\mathsf{M}$ by (8.11). For the others, let $\mathsf{R} = 2^i\mathsf{R}_0$, $i \ge 0$, and consider the pairs with $\mathsf{R}b < |\gamma_k - \gamma_l| \le 2\mathsf{R}b$ and $|x_k - x_l| < b$. For them $|e_k - e_l| \ge (\mathsf{R}-1)b \ge \mathsf{R}b/2$, and $b\mathcal{L}(\tau_k) \ge \frac{1}{2}$ for $T \ge T_0$, so $\mathcal{L}(\tau_k)|e_k - e_l| \ge \mathsf{R}/4$ and

$$1 \le \Big(\frac{4\mathcal{L}(\tau_k)|e_k - e_l|}{\mathsf{R}}\Big)^2 \le \frac{32}{\mathsf{R}^2}\Big(\mathcal{L}(\tau_k)^2 e_k^2 + \mathcal{L}(\tau_l)^2 e_l^2\Big)\Big(1 + O(1/\log T)\Big).$$

Summing over the ordered pairs of the block and symmetrizing, their number P_R satisfies

$$\mathsf{P}_\mathsf{R} \le \frac{65}{\mathsf{R}^2}\sum_{k\in\mathcal{I}_T}\mathcal{L}(\tau_k)^2 e_k^2\, n_{2\mathsf{R}}(\gamma_k) \le \frac{65}{\mathsf{R}^2}\Big(\sum_k \mathcal{L}(\tau_k)^4 e_k^4\Big)^{1/2}\Big(\sum_k n_{2\mathsf{R}}(\gamma_k)^2\Big)^{1/2} \le \frac{130\sqrt{C_2}\,\mathsf{A}^2\mathsf{H}}{\mathsf{R}}\,\mathsf{M},$$

by Cauchy–Schwarz, the hypothesis and (8.11) (both k and l lie in $\mathcal{I}_T$, so $n_{2\mathsf{R}}$ counts every l of the block). Summing over $i \ge 0$ gives at most $260\sqrt{C_2}\,\mathsf{A}^2\mathsf{H}\mathsf{M}/\mathsf{R}_0$. With $\mathsf{R}_0 = \mathsf{A}\sqrt{\mathsf{H}}\,(260\sqrt{C_2}/C_1)^{1/2}$ (which is ≥ 2 for C_2 large enough) the two parts are equal and $\mathsf{P} \le 2\sqrt{260C_1\sqrt{C_2}}\,\mathsf{A}\sqrt{\mathsf{H}}\,\mathsf{M}$. □

Only the fourth moment of the error of the j-th clock enters. For $j = 2$, with A bounded (shown below), the multiplicity of the second clock is $\ll \sqrt{H_2}\,\mathfrak{b}_T\#\mathcal{I}_T$. The factor $\sqrt{\mathsf{H}}$ is not a loss in the iteration: it is absorbed by the cutoff of the next step.

All further steps

Let $H_0 = \log\log T$, $c_1 = H_0^{-1/2}$, $c_2 = H_1^{-1/2}$ with $H_1 = \log H_0$, and define recursively, with an absolute constant $0 < C_\star \le 1$ fixed in the proof of Theorem 8.11,

$$H_j = 2\log\frac{1}{c_j} \quad (j \ge 1), \qquad c_j = C_\star\, H_{j-1}^{-9/16} \quad (j \ge 3), \tag{8.13}$$

so that $H_2 = \log H_1$ as in Theorem 8.6, $H_j = \frac{9}{8}\log H_{j-1} + 2\log(1/C_\star)$ for $j \ge 3$, and $\log H_{j-1} \le H_j$ for all j. All cutoffs depend only on T. Put

$$A_j(r) = \frac{\|\mathcal{L}(\tau_k)e_{j,k}\|_{2r,\mathcal{I}_T}}{\sqrt{H_j}} \quad (j \ge 1), \qquad A_0(r) = \frac{\|\mathcal{L}(\tau_k)\delta_k\|_{2r,\mathcal{I}_T}}{\sqrt{H_0}},$$

and let $m^* = m^*(T)$ be the first index m with $H_m < H^*$, where H^* is the absolute constant of the next theorem. Since $\log H_{j-1} \le H_j \le \frac{9}{8}\log H_{j-1} + 2\log(1/C_\star)$, the sequence H_j decreases from $\log\log T$ to below H^* in $m^* = \log^* T + O(1)$ steps, where $\log^*$ is the iterated logarithm.

Theorem 8.11 (All Newton-type steps, uniformly)**.** *There are absolute constants $C_\star$, H^*, C and T_0 such that, with the cutoffs* (8.13)*, for every $T \ge T_0$ and every $1 \le m \le m^*$,*

$$\Big\langle \mathcal{L}(\tau_k)^2 e_{m,k}^2 \Big\rangle_{\mathcal{I}_T} = \frac{H_m}{4\pi^2} + \vartheta_m\, C\Big(1 + \sqrt{H_m}\Big), \qquad |\vartheta_m| \le 1, \tag{8.14}$$

and $\|\mathcal{L}(\tau_k)e_{j,k}\|_{4,\mathcal{I}_T} \le C\sqrt{H_j}$ for $1 \le j < m^*$. In particular, at $m = m^* = \log^* T + O(1)$,

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(x_k^{(m^*)} - \gamma_k\Big)^2 \le \frac{C'}{\log^2 T}\,: \tag{8.15}$$

the clock corrected $\log^* T + O(1)$ times with the Euler phases of the primes up to T^{c_j} locates the ordinates with a mean-square error of $O(1)$ mean spacings, with an absolute constant, using no ordinate in its construction.

Proof. Multiplicities. Let $\mathfrak{k}_j$ be the least $\mathfrak{k} \ge 1$ with $\int \tilde{n}_j^2 \le \mathfrak{k}\,\mathfrak{b}_T \#\mathcal{I}_T$. Then $\mathfrak{k}_0 = 1$ (the τ_k are separated by at least $\mathfrak{b}_T$, so only the diagonal contributes), $\mathfrak{k}_1 \le C$ (Lemma 8.4) and $\mathfrak{k}_j \le C_3 \max(A_j(2), 1)\sqrt{H_j}$ for $j \ge 2$ (Theorem 8.10 with $\mathsf{A} = \max(A_j(2), 1)$ and $\mathsf{H} = H_j$).

Base. By Lemma 8.2 with $X = 2$, $|S_2| \le \sigma_2 < 1$ and $0 \le \chi_k \le 1/T$, $\|\mathcal{L}(\tau_k)\delta_k\|_{2r} \le \|u_k(2)\|_{2r} + 2 \le C_0(r^2 + \sqrt{rH_0})$, so $A_0(r) \le C_0 r^2$ for $r \ge 1$.

One step. Let $j \ge 1$ and $r \ge 1$ with $X_j \le T^{1/(160r)}$. By (8.3), $\mathcal{L}(x_k^{(j-1)}) = \mathcal{L}(\tau_k)(1 + O(T^{-1/2}))$, Lemma 8.2 (with $V \le \log\frac{1}{c_j} + O(1) \le H_j$), Lemma 8.5 at the points $x_k^{(j-1)}$ with multiplicity constant $\mathfrak{k}_{j-1}$, and $\|\chi_j\|_{2r} \ll T^{c_{j-1}-1}$,

$$\sqrt{H_j}\,A_j(r) \le C\Big(r^2 + \sqrt{rH_j}\Big) + 2\sqrt{H_{j-1}}\,A_{j-1}(2r)\cdot\frac{C\mathfrak{k}_{j-1}^{1/(8r)}\sqrt{r}\,(1+\log X_j)}{\mathcal{L}(\tau_k)}\Big(1 + o(1)\Big) + o(1),$$

and $(1 + \log X_j)/\mathcal{L}(\tau_k) \le 2\pi c_j(1 + o(1))$, $\mathfrak{k}_{j-1}^{1/(8r)} \le \mathfrak{k}_{j-1}^{1/8}$. For $j \le 2$, $\sqrt{H_{j-1}}c_j = 1$ and $\mathfrak{k}_{j-1} \le C$; for $j \ge 3$, $\mathfrak{k}_{j-1}^{1/8}\sqrt{H_{j-1}}c_j \le (C_3\max(A_{j-1}(2),1))^{1/8} H_{j-1}^{1/16+1/2-9/16} C_\star =$

$C_\star(C_3\max(A_{j-1}(2),1))^{1/8}$, and $\max(A_{j-1}(2),1)\le\max(A_{j-1}(2r),1)$ (monotonicity of the discrete norms), so that $A_{j-1}(2r)\max(A_{j-1}(2r),1)^{1/8}=\max(A_{j-1}(2r),A_{j-1}(2r)^{9/8})$. Fix $C_\star=\min(1,(4\pi C\,C_3^{1/8})^{-1})$. Then, for all $j\ge1$ and $r\ge1$,

$$A_j(r)\le C\Big(\sqrt{r}+\frac{r^2}{\sqrt{H_j}}\Big)+\frac{C_5\sqrt{r}}{\sqrt{H_j}}\max\Big(A_{j-1}(2r),A_{j-1}(2r)^{9/8}\Big),\tag{8.16}$$

with absolute C,C_5, provided $X_j\le T^{1/(160r)}$.

Induction. Let $B_\star=\max(2C,C_0,1)$ and $C_6=\Big(4C/B_\star+4C_5(4B_\star)^{9/8}/B_\star\Big)^2$. We claim that for every $m\le m^*$,

$$A_j(r)\le B_\star r^2\qquad(0\le j\le m-1,\ 1\le r\le 2^{m-j}).\tag{8.17}$$

For $j=0$ this is the base. Suppose it holds at $j-1$ for $r\le2^{m-j+1}$, let $1\le r\le2^{m-j}$, and assume $H_j\ge C_6 4^{m-j}$ and $X_j\le T^{1/(160r)}$. Since $A_{j-1}(2r)\le4B_\star r^2$ and $4B_\star r^2\ge1$, (8.16) gives

$$A_j(r)\le C\sqrt{r}+\frac{Cr^2+C_5(4B_\star)^{9/8}r^{11/4}}{\sqrt{H_j}}\le\frac{B_\star}{2}r^2+\frac{\big(C+C_5(4B_\star)^{9/8}\big)r^3}{\sqrt{H_j}}\le B_\star r^2,$$

because $r^3/\sqrt{H_j}\le r^2\cdot2^{m-j}/\sqrt{H_j}\le r^2/\sqrt{C_6}$ and $\sqrt{C_6}\ge2(C+C_5(4B_\star)^{9/8})/B_\star$.

The two conditions. Let H^* be an absolute constant with $H^*\ge\max(4C_6,160^2\cdot4)$ and so large that $e^{\frac89(H-2\log(1/C_\star))}\ge4H$ and $e^H\ge4H$ for $H\ge H^*$. If $m\le m^*$ then $H_{m-1}\ge H^*$ (by definition of m^*), and from $H_{j-1}\ge e^{H_j}$ ($j\le2$) or $H_{j-1}=e^{\frac89(H_j-2\log(1/C_\star))}$ ($j\ge3$) we get $H_{j-1}\ge4H_j$ whenever $H_j\ge H^*$; by downward induction, $H_{m-1-i}\ge4^iH^*$ for $0\le i\le m-1$. Hence $H_j\ge4^{m-1-j}H^*\ge C_64^{m-j}$ for $1\le j\le m-1$, and $H_{j-1}\ge4^{m-j}H^*\ge160^2\,4^{m-j}$ for $1\le j\le m$, which gives $c_j\le H_{j-1}^{-1/2}\le2^{-(m-j)}/160$, i.e. $X_j\le T^{1/(160r)}$ for $r\le2^{m-j}$, and $X_j\ge2$ because $c_j\ge C_\star(\log\log T)^{-9/16}$. So (8.17) holds for every $m\le m^*$, which is the statement $\|\mathcal{L}e_j\|_{4,\mathcal{I}_T}\le4B_\star\sqrt{H_j}$ for $j<m^*$.

The last step. For $m\le m^*$ and $r=1$, (8.17) at $j=m-1$ gives $A_{m-1}(2)\le4B_\star$, so the transport term of the m-th step is $\|\mathcal{L}(\tau_k)\mathfrak{t}_k^{(m)}\|_{2,\mathcal{I}_T}\le C_5(4B_\star)^{9/8}$, with $\mathfrak{t}_k^{(m)}=S_{X_m}(\gamma_k)-S_{X_m}(x_k^{(m-1)})$. Theorem 5.1 at $X=X_m$ ($2\le X_m\le T^{1/160}$) gives $\langle u_k(X_m)^2\rangle=\frac{1}{2\pi^2}\log\frac{1}{c_m}+O(1+\sqrt{H_m})=\frac{H_m}{4\pi^2}+O(1+\sqrt{H_m})$, the cross term is at most $2\|u(X_m)\|_2\,C_5(4B_\star)^{9/8}\ll1+\sqrt{H_m}$, and $\chi_{m,k}\ll T^{-1/2}$; (8.2) gives (8.14). At $m=m^*$, $H_{m^*}<H^*$ and $\mathcal{L}(\tau_k)=\frac{\log T}{2\pi}(1+O(1/\log T))$ give (8.15). □

Corollary 8.12 (Third step)**.** *With $c_3=C_\star H_2^{-9/16}$ and $H_3=\frac98\log H_2+2\log(1/C_\star)$,*

$$\Big\langle\mathcal{L}(\tau_k)^2e_{3,k}^2\Big\rangle_{\mathcal{I}_T}=\frac{H_3}{4\pi^2}+O\Big(1+\sqrt{H_3}\Big),\qquad\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(x_k^{(3)}-\gamma_k\Big)^2=\frac{H_3+O(\sqrt{H_3})}{\log^2T},$$

and the multiplicity of the second clock satisfies $\int\tilde n_2^2\ll\sqrt{H_2}\,\mathfrak{b}_T\#\mathcal{I}_T$.

Proof. Theorem 8.11 with $m=3$ (for $T\ge T_0$, $H_2\ge H^*$), and Theorem 8.10 with $\mathsf{A}=A_2(2)\le4B_\star$ and $\mathsf{H}=H_2$. □

The induction (8.17) bounds the moments of every order $2r \le 2^{m^*-j+1}$ of the j-th error, not only its mean square, and the number of available moments grows with T. This gives tails.

Corollary 8.13 (Tails of the corrected clocks)**.** *With the cutoffs* (8.13)*, for $T \ge T_0$, every $1 \le j < m^*$, every integer $1 \le q \le 2^{m^*-j}$ and every $\lambda > 0$,*

$$\#\Big\{k \in \mathcal{I}_T : \mathcal{L}(\tau_k)|e_{j,k}| > \lambda\sqrt{H_j}\Big\} \le \Big(\frac{B_\star q^2}{\lambda}\Big)^{2q} \#\mathcal{I}_T. \tag{8.18}$$

For fixed j and q the right side is $\ll_q \lambda^{-2q}\#\mathcal{I}_T$ for $T \ge T_0(j,q)$, since $2^{m^*-j} \to \infty$: for every fixed q, and T large in terms of q, the j-th corrected clock misses its ordinate by more than $\lambda\sqrt{H_j}$ positions on a proportion $\ll_q \lambda^{-2q}$ of the indices. The orders available at a given T are $q \le 2^{m^*-j}$; nothing is claimed uniformly in q.

Proof. (8.17) with $m = m^*$ gives $\|\mathcal{L}(\tau_k)e_{j,k}\|_{2q,\mathcal{I}_T} \le B_\star q^2\sqrt{H_j}$ for $j \le m^*-1$ and $q \le 2^{m^*-j}$; apply Chebyshev's inequality to the $2q$-th moment. □

The Newton points of a level need not be in increasing order (Proposition 8.8 bounds the proportion of inversions of the second clock). Sorting them costs nothing.

Lemma 8.14 (Sorted Newton-type steps)**.** *Let $\gamma_1 \le \cdots \le \gamma_{\mathsf{M}}$ and let $z_1, \dots, z_{\mathsf{M}}$ be arbitrary reals, with $z^\uparrow$ the same values in increasing order. Then $\sum_k |z_k^\uparrow - \gamma_k|^q \le \sum_k |z_k - \gamma_k|^q$ for every $q \ge 1$, and $\{z_k^\uparrow\} = \{z_k\}$ as multisets. Consequently, if at some level j the points $x_k^{(j)}$, $k \in I$, are replaced by their sorted version and the iteration* (8.1) *is continued from the sorted points, then $\|e_j\|_{r,I}$ does not increase for any r, the multiplicity function $\tilde{n}_j$ is unchanged, and the bounds of Proposition 8.1, Lemma 8.5, Theorems 8.6, 8.10, 8.11 and Corollary 8.13 hold for the continued iteration with the same constants; the inversions $d_{j,k} \le 0$ of Proposition 8.8 are removed.*

Proof. For $a \le b$, $x \le y$ and φ convex, $\varphi(a-x)+\varphi(b-y) \le \varphi(a-y)+\varphi(b-x)$ (the points $a-y \le a-x, b-y \le b-x$ have the same sum in both pairings, and the outer pair majorizes the inner one). With $\varphi(t) = |t|^q$, each transposition of an inverted pair of z's does not increase the sum, and finitely many transpositions sort z. For the second statement: Proposition 8.1 holds for the Newton step from any points $x_k^{(j)} > 2\pi$, and its bound (8.3) for the next level involves the level j only through $\|e_j\|_{4r,I}$, which does not increase, and through $\|\mathsf{g}_{j+1}(x^{(j)})\|_{4r,I}$, a sum over the point set $\{x_k^{(j)}\}$, which is unchanged; Lemma 8.5 depends on the point set only through $\tilde{n}_j$; the induction in the proof of Theorem 8.11 uses nothing else. □

Corollary 8.15 (An increasing computable clock)**.** *Let $\mathcal{I}_T^\tau = \{k : T \le \tau_k < 2T\}$, a set of consecutive indices defined by the clock alone, and let $x_k^\uparrow$, $k \in \mathcal{I}_T^\tau$, be the values $x_k^{(m^*)}$ of Theorem 8.11 in increasing order. Then $x_k^\uparrow \le x_{k+1}^\uparrow$, every $x_k^\uparrow$ is computed from the clock and the primes alone, and for $T \ge T_0$*

$$\frac{1}{\#\mathcal{I}_T^\tau}\sum_{k\in\mathcal{I}_T^\tau}\Big(x_k^\uparrow - \gamma_k\Big)^2 \le \frac{C'+1}{\log^2 T}.$$

Proof. By Lemma 8.14 with $q = 2$ on the index set $\mathcal{I}_T^\tau$, the sum is at most $\sum_{k\in\mathcal{I}_T^\tau}(x_k^{(m^*)}-\gamma_k)^2$. Since $|\delta_k| \le C_{10}$ for all k (Theorem 3.1), the symmetric difference $\mathcal{I}_T^\tau \triangle \mathcal{I}_T$ consists of indices with $|\gamma_k - T| \le C_{10}$ or $|\gamma_k - 2T| \le C_{10}$, and has $O(\log T)$ elements by (2.3). For every $k \in \mathcal{I}_T^\tau$ the points $x_k^{(j)}$ lie in $[T - 2T^{1/2}, 2T + 2T^{1/2}]$ and Lemma 8.7 gives $|x_k^{(j)} - \tau_k| = |y_k^{(j)}| \le (\sigma_{X_j} + 1)/\mathcal{L}(\tau_k) \ll T^{1/320}/\log T$, because $X_j \le T^{1/160}$ for every $j \le m^*$ (proof of Theorem 8.11, *the two conditions*, with $r = 1$) and $\sigma_X \ll \sqrt{X}/\log X$. Hence $|x_k^{(m^*)} - \gamma_k| \le C_{10} + O(T^{1/320}/\log T)$ for $k \in \mathcal{I}_T^\tau \triangle \mathcal{I}_T$, these indices contribute $O(\log T \cdot T^{1/160}/\log^2 T) = O(T^{1/160})$ to the sum, and (8.15) gives

$$\sum_{k\in\mathcal{I}_T^\tau} \bigl(x_k^{\uparrow} - \gamma_k\bigr)^2 \le \frac{C'\,\#\mathcal{I}_T}{\log^2 T} + O\bigl(T^{1/160}\bigr) \le \frac{(C'+1)\,\#\mathcal{I}_T^\tau}{\log^2 T}$$

for $T \ge T_0$, since $\#\mathcal{I}_T^\tau = \#\mathcal{I}_T + O(\log T)$ and $\#\mathcal{I}_T \asymp T\log T$. □

The cutoff of the third step, T^{c_3} with $c_3 \asymp H_2^{-9/16}$ instead of $H_2^{-1/2}$, is what the multiplicity costs: the exponent $\frac{9}{16} = \frac{1}{2} + \frac{1}{16}$ comes from the factor $\sqrt{H_2}$ of (8.12) through the exponent $\frac{1}{8r} \le \frac{1}{8}$ of Lemma 8.5. The constants of Fujii's and Montgomery–Vaughan's bounds, which grow with the moment order 2^{m-j} used at level j, are absorbed by the normalization $H_j^{-1/2}$, because H_j grows as a tower in $m - j$ while the orders grow geometrically.

A bounded error in units of zeros with a *fixed* final cutoff T^{c_0} is not obtained by this method: the Newton term has mean square of order $c^2 \log\log T$ unless transport and defect cancel (Remark 7.7), and no estimate here proves such a cancellation.

9. Theorem E: the pointwise envelope

Let $\mathfrak{a} = \limsup_{t\to\infty} |S(t)|/\log t$. By Theorem 3.2, $\mathfrak{a} \le 0.10076$, and $\mathfrak{a} \le 0.1013$ also follows from [28]; whether $\mathfrak{a} = 0$ is not known.

Proposition 9.1 (The envelope of the clock is the envelope of S)**.** *(a)* $\limsup_{k\to\infty} |\tau_k - \gamma_k| = 2\pi\mathfrak{a}$. *(b) If* $X_k \ge 2$ *and* $\sigma_{X_k} = o(\log \tau_k)$, *in particular if* $X_k \le (\log\tau_k)^{2-\varepsilon}$, *then* $\limsup_{k\to\infty} |\tau_k^{X_k} - \gamma_k| = 2\pi\mathfrak{a}$.

Proof. (a) The values $S(\gamma_k^{\pm})$ are limits of values of S at non-ordinates. For $\varepsilon > 0$, (2.8) gives $|s_k| \le (\mathfrak{a} + \varepsilon)\log\gamma_k + O_\varepsilon(1)$, and $|\delta_k| = |s_k|/\mathcal{L}(\xi_k)$ with $\mathcal{L}(\xi_k) = \frac{\log\gamma_k}{2\pi}(1 + o(1))$ since $\delta_k = O(1)$; so $\limsup|\delta_k| \le 2\pi\mathfrak{a}$. Conversely, let $t > \gamma_1$ be a non-ordinate, let $\gamma < t < \gamma'$ be the nearest ordinates and $k = N(t)$, so that $\gamma_k = \gamma$ and $\gamma_{k+1} = \gamma'$. On (γ, γ') the function G is continuous and decreasing ($G' = -\mathcal{L}$), so $G(\gamma'^-) \le G(t) \le G(\gamma^+)$, while $s_k = k - \frac12 - N_0(\gamma) = G(\gamma^+) - \frac12$ and $s_{k+1} = k + \frac12 - N_0(\gamma') = G(\gamma'^-) + \frac12$. Hence $s_k \ge G(t) - \frac12$ and $s_{k+1} \le G(t) + \frac12$, so $\max(|s_k|, |s_{k+1}|) \ge |G(t)| - \frac12$, and $|G(t) - S(t)| = |\varpi(t)|/\pi \to 0$. Both ordinates are within bounded distance of t (Theorem 3.1). Choosing non-ordinates $t \to \infty$ with $|S(t)| \ge (\mathfrak{a} - \varepsilon)\log t$ gives indices $k \to \infty$ with $|\delta_k| = |s_k|/\mathcal{L}(\xi_k) \ge 2\pi(\mathfrak{a} - \varepsilon) + o(1)$.

(b) $|\tau_k^{X_k} - \tau_k| = |S_{X_k}(\tau_k)|/\mathcal{L}(\tau_k) \le \sigma_{X_k}/\mathcal{L}(\tau_k) \to 0$, and $\sigma_X \ll \sqrt{X}/\log X$. □

So the corrected clocks with slowly growing cutoffs have exactly the pointwise envelope of the bare clock, and for every cutoff the envelope of τ_k is that of S. A smaller coefficient for the clock is a smaller coefficient for $S(t)$.

Theorem 9.2 (Theorem E)**.** *Let $\mathfrak{e}$ be continuous on $[\frac12,1]$ with $\mathfrak{e}(1)=0$, such that for every $\varepsilon>0$, $\zeta(\sigma+it)\ll_\varepsilon |t|^{\mathfrak{e}(\sigma)+\varepsilon}$ uniformly for $\frac12\le\sigma\le1$ and $|t|\ge2$, and put $\mathfrak{e}(\sigma)=\frac12-\sigma+\mathfrak{e}(1-\sigma)$ for $0\le\sigma<\frac12$. Then*

$$\mathfrak{a}\le\frac{1}{2\pi\log2}\int_0^1\frac{\mathfrak{e}(\sigma)}{\sqrt{\sigma(2-\sigma)}}\,d\sigma. \tag{9.1}$$

For $J\ge0$ let $(\mathsf{k}_0,\mathsf{l}_0)=(\frac{13}{84},\frac{55}{84})$, $(\mathsf{k}_j,\mathsf{l}_j)=\mathscr{A}^j(\mathsf{k}_0,\mathsf{l}_0)$ with $\mathscr{A}(\mathsf{k},\mathsf{l})=\big(\frac{\mathsf{k}}{2\mathsf{k}+2},\frac12+\frac{\mathsf{l}}{2\mathsf{k}+2}\big)$, $\sigma_j=\mathsf{l}_j-\mathsf{k}_j$, and let $\mathfrak{e}_J$ be the piecewise linear function through the points (σ_j,k_j), $0\le j\le J$, and $(1,0)$; the first vertices are $(\frac12,\frac{13}{84})$, $(\frac{139}{194},\frac{13}{194})$, $(\frac{173}{207},\frac{13}{414})$, $(\frac{773}{854},\frac{13}{854})$. Let C_J be the right side of (9.1) for $\mathfrak{e}=\mathfrak{e}_J$. Then $\mathfrak{a}\le C_J$ for every J; the sequence C_J decreases to a limit C_∞ with $0\le C_J-C_\infty\le 3\mathsf{k}_J\sqrt{1-\sigma_J}/(2\pi\log2)$, and

$$C_\infty=0.0968151142790\ldots,\qquad \limsup_{k\to\infty}|\tau_k-\gamma_k|=2\pi\mathfrak{a}\le2\pi C_\infty=0.608307303\ldots; \tag{9.2}$$

for comparison $C_0=0.1010228\ldots$ (the exponent $\frac{13}{84}$ alone) and $C_7=0.0968160\ldots$.

Proof. Uniform bounds. Extend $\mathfrak{e}$ by 0 on $[1,\infty)$ and by $\frac12-\sigma$ on $(-\infty,0)$; the extension $\mathfrak{e}^*$ is continuous ($\mathfrak{e}(0)=\frac12+\mathfrak{e}(1)=\frac12$). For $\sigma\ge1$, $\zeta(\sigma+it)\ll\log|t|$ uniformly [27, Thm. 3.5]. For $-\frac{1}{50}\le\sigma<\frac12$, $\zeta(\sigma+it)=\chi(\sigma+it)\zeta(1-\sigma-it)$ with $|\chi(\sigma+it)|=(|t|/2\pi)^{1/2-\sigma}(1+O(1/|t|))$ uniformly [27, (4.12.3)], and $|\zeta(1-\sigma-it)|=|\zeta(1-\sigma+it)|$. Hence $\zeta(\sigma+it)\ll_\varepsilon|t|^{\mathfrak{e}^*(\sigma)+\varepsilon}$ uniformly for $-\frac{1}{50}\le\sigma\le4$, $|t|\ge2$.

Backlund's method. Let $0<\varepsilon_\circ<\frac{1}{100}$, $\sigma_\circ=1+\varepsilon_\circ$, $\varrho=1+2\varepsilon_\circ$. Then $1<\sigma_\circ<\varrho$, $\sigma^\sharp=\frac54+\frac32\varepsilon_\circ$, $\delta^\sharp=\frac14+\frac{\varepsilon_\circ}{2}\in(0,\frac12)$, and $\varrho/(\sigma_\circ-\frac12)=2$. For $s=\sigma_\circ+\varrho e^{i\theta}$, $\Re s\in[-\varepsilon_\circ,2+3\varepsilon_\circ]$ and $|\Im s|\le2$, so for T large

$$\begin{aligned}\frac{1}{\varkappa}\log|\mathfrak{f}_\varkappa(s)|&\le\log\max_{\pm}|(s\pm iT-1)\zeta(s\pm iT)|\\&\le\log(T+4)+(\mathfrak{e}^*(\Re s)+\varepsilon)\log(T+2)+C_\varepsilon=:F_\circ(\theta).\end{aligned}$$

an even continuous function. By Theorem 3.14(a), along $\varkappa_m\to\infty$,

$$\limsup_m\mathcal{Z}_{\varkappa_m}(\sigma_\circ,\varrho)\le\log T\cdot\frac{1}{\pi}\int_0^\pi\Big(\mathfrak{e}^*(\sigma_\circ+\varrho\cos\theta)+\varepsilon\Big)d\theta+O_{\varepsilon,\varepsilon_\circ}(1),$$

the terms $\log(T+4)$ and $-\log\sqrt{\varepsilon_\circ^2+T^2}$ cancelling up to $O(1/T)$. The argument defining $S(T)$ runs along $2\to2+iT\to\sigma^\sharp+iT\to\frac12+iT$, and on the first two segments $\Re s>1$, where $\log\zeta$ is holomorphic; so $\pi S(T)=\arg\zeta(\sigma^\sharp+iT)+\Delta^{1/2}_{\sigma=\sigma^\sharp}\arg((\sigma+iT-1)\zeta(\sigma+iT))-\Delta^{1/2}_{\sigma=\sigma^\sharp}\arg(\sigma+iT-1)$, with $|\arg\zeta(\sigma^\sharp+iT)|\le\log\zeta(\sigma^\sharp)$ and the last term bounded by π. Theorem 3.14(b) with $\varkappa=\varkappa_m\to\infty$ gives, for T large,

$$|S(T)|\le\frac{\log T}{2\log2}\cdot\frac{1}{\pi}\int_0^\pi\Big(\mathfrak{e}^*(1+\varepsilon_\circ+(1+2\varepsilon_\circ)\cos\theta)+\varepsilon\Big)d\theta+O_{\varepsilon,\varepsilon_\circ}(1).$$

Limit. Divide by $\log T$, let $T\to\infty$, then $\varepsilon\to0$ and $\varepsilon_\circ\to0$; $\mathfrak{e}^*$ is bounded and continuous, so by dominated convergence $\mathfrak{a}\le\frac{1}{2\pi\log2}\int_0^\pi\mathfrak{e}^*(1+\cos\theta)\,d\theta$. The integrand vanishes for $0\le\theta\le\frac\pi2$; on $[\frac\pi2,\pi]$ substitute $\sigma=1+\cos\theta$, $d\sigma=-\sin\theta\,d\theta$, $\sin\theta=\sqrt{\sigma(2-\sigma)}$, which gives (9.1).

The profile. Let $\varepsilon' > 0$, $(\mathsf{k}_0, \mathsf{l}_0) = (\frac{13}{84} + \varepsilon', \frac{55}{84} + \varepsilon')$ and $(\mathsf{k}_j, \mathsf{l}_j) = \mathscr{A}^j(\mathsf{k}_0, \mathsf{l}_0)$ for $1 \le j \le J$, exponent pairs by Theorem 3.12(a),(c), with $\mathsf{l}_j - \mathsf{k}_j \ge \frac12$; by (b), $|\zeta(\mathsf{l}_j - \mathsf{k}_j + it)| \ll |t|^{\mathsf{k}_j} \log|t|$, and $|\zeta(1+it)| \ll \log|t|$. The points $(\mathsf{l}_j - \mathsf{k}_j, \mathsf{k}_j)$ and $(1,0)$ have decreasing ordinates, and Theorem 3.13 between consecutive points gives the hypothesis of the theorem for the piecewise linear function through them. For $\varepsilon' = 0$ these are the vertices of $\mathfrak{e}_J$; the right side of (9.1) depends continuously on ε', so $\mathfrak{a} \le C_J$.

The limit. From $\mathsf{k}_{j+1} = \mathsf{k}_j/(2\mathsf{k}_j+2)$ and $\mathsf{l}_{j+1} - \mathsf{k}_{j+1} = \frac12 + (\mathsf{l}_j - \mathsf{k}_j)/(2\mathsf{k}_j+2)$ one gets $\mathsf{k}_j \le 2^{-j}\mathsf{k}_0$, $\sigma_j < \sigma_{j+1} < 1$ and $1 - \sigma_{j+1} = (1 - \sigma_j + \mathsf{k}_j)/(2\mathsf{k}_j+2) \le \frac12(1-\sigma_j) + \mathsf{k}_j$, so $\sigma_j \to 1$. Each new vertex $(\sigma_{J+1}, \mathsf{k}_{J+1})$ lies below the chord from (σ_J, k_J) to $(1,0)$: this is $\mathsf{k}_{J+1}(1-\sigma_J) \le \mathsf{k}_J(1-\sigma_{J+1})$, i.e. $\frac{1-\sigma_J}{2\mathsf{k}_J+2} \le \frac{1-\sigma_J+\mathsf{k}_J}{2\mathsf{k}_J+2}$, which holds. Hence $\mathfrak{e}_{J+1} \le \mathfrak{e}_J$ on $[\frac12, 1]$ (they agree on $[\frac12, \sigma_J]$), the reflected parts are ordered in the same way, and $C_{J+1} \le C_J$; the limit C_∞ exists. Since $0 \le \mathfrak{e}_J - \mathfrak{e}_{J'} \le \mathsf{k}_J$ for $J' \ge J$, supported on $[\sigma_J, 1]$ and, after reflection, on $[0, 1-\sigma_J]$, and $\sigma(2-\sigma) \ge \sigma$ on $[0,1]$,

$$\begin{aligned} 0 \le C_J - C_{J'} &\le \frac{\mathsf{k}_J}{2\pi\log 2}\Big(\int_{\sigma_J}^1 \frac{d\sigma}{\sqrt{\sigma(2-\sigma)}} + \int_0^{1-\sigma_J} \frac{d\sigma}{\sqrt{\sigma}}\Big) \\ &\le \frac{\mathsf{k}_J\Big((1-\sigma_J)\sqrt{2} + 2\sqrt{1-\sigma_J}\Big)}{2\pi\log 2} \le \frac{3\mathsf{k}_J\sqrt{1-\sigma_J}}{2\pi\log 2}. \end{aligned}$$

The values. On each linear piece $\mathfrak{e}(\sigma) = \alpha\sigma + \beta$ of $[0,1]$ a primitive of $\mathfrak{e}(\sigma)/\sqrt{\sigma(2-\sigma)}$ is $-\alpha\sqrt{\sigma(2-\sigma)} + (\alpha+\beta)\arcsin(\sigma-1)$; the vertices are rational, and the program `opc_envelope_iterations_p1.py` (Data availability) evaluates C_J from this primitive at 50 digits for $0 \le J \le 30$. It gives $C_{30} = 0.09681511427905377\ldots$ and $3\mathsf{k}_{30}\sqrt{1-\sigma_{30}}/(2\pi\log 2) < 5\cdot 10^{-15}$, hence $C_\infty \in [C_{30} - 5\cdot 10^{-15}, C_{30}]$ and (9.2); C_7 agrees with the 40-digit quadrature of `opc_envelope_coefficient_v14.py`. Multiply by 2π and use Proposition 9.1(a). □

Remark 9.3. (a) Trudgian [28, §5 of the preprint] obtained $\mathfrak{a} \le 0.1013$ from Huxley's exponent $\frac{32}{205}$ at $\sigma = \frac12$ by the same method; Theorem 9.2 replaces the exponents and lets the parameters of the disc tend to their limits. With the explicit exponents used by Bellotti and Wong ($\frac{27}{164}$ at $\sigma = \frac12$, Yang's lines $\sigma_j = 1 - j/(2^j-2)$ with exponents $1/(2^j-2)$, $4 \le j \le 9$ [1]) and linear interpolation between $\sigma = 0$ and $\sigma = \frac12$, as in [1, §3.1], the right side of (9.1) is 0.10074, the limit of their $C_1 = 0.10076$ as their parameters tend to $(1,1)$; the same right-hand profile reflected by the functional equation gives 0.09783. (b) Profiles of the growth exponent of $\zeta(\sigma+it)$ below $\mathfrak{e}_J$ are known: Trudgian and Yang [30] close a convex hull of known exponent pairs, Bourgain's among them, under van der Corput's $\mathscr{A}$- and $\mathscr{B}$-processes and the $\mathscr{C}$-process and deduce piecewise bounds for that exponent on $[\frac12, 1]$. Any such profile can be inserted in (9.1); we do not evaluate it here. (c) For the Lindelöf profile ($\mathfrak{e} = 0$ on $[\frac12, 1]$) the right side is 0.07863: Backlund's method with a disc cannot give less, whatever the exponents. (d) Theorem 9.2 is asymptotic: the explicit constants of the bounds for ζ do not enter, which is why non-explicit exponents may be used, and it does not replace the explicit bounds of [1], valid for all $T \ge e$.

10. Conclusions

10.1. What was known

Pointwise, $\delta_k = \tau_k - \gamma_k$ is bounded, with $\limsup|\delta_k| \le 0.6331$ from the explicit bound of Bellotti and Wong [1], and $\limsup|\delta_k| \le 0.6365$ from Trudgian's asymptotic bound [28]. On average, s_k is asymptotically Gaussian with variance $\frac{1}{2\pi^2}\log\log T$, by Selberg's formulae for the Gram displacement proved by Korolev [18, 14], so $\langle\delta_k^2\rangle \sim 2\log\log T/\log^2 T$: the clock is close to the zeros in height but misplaces almost every zero by an unbounded number of positions. Selberg's approximation [23, 8] replaces $S(t)$ by short prime sums in t-mean.

10.2. What this paper adds

1. *An exact dictionary between clock-type models and the zeros* (Theorems 4.1 and 4.2). For any C^1 model F and any continuous $\mathfrak{g}$, sums of $\mathfrak{g}$-increments of $k - \frac{1}{2} - F(\gamma_k)$ over the ordinates are exactly dF-integrals of $\mathfrak{g}(N - F)$ up to two boundary terms; for powers this gives the even moments with the constant $\frac{1}{12}$ per zero, for exponentials it transfers the signed law of $S(t)$ to s_k (Corollary 4.4).

2. *The prime information needed by the clock* (Theorems 5.1 and 5.12). The corrected clock equation $N_0(t) + S_X(t) = k - \frac{1}{2}$ holds at the true ordinates with mean-square ordinal defect $\frac{1}{2\pi^2}\log\frac{\log T}{\log X} + O(1 + \sqrt{\log(\log T/\log X)})$, also with a cutoff varying at every index; the defect is bounded in mean square if and only if the primes reach a fixed power of the height.

3. *An exact finite-height account* (Theorem 6.1). The gain of the one-step clock is a diagonal prime response minus a cross term, and its residual differs from the defect by a transport term and an explicit nonnegative second-order term.

4. *A computable clock that approaches the zeros* (Theorems 7.3, 8.6 and 8.11). With $X = T^{1/\sqrt{\log\log T}}$ the one-step clock has mean-square error $(\log\log\log T + O(\sqrt{\log\log\log T}))/\log^2 T$ in height, against $(2 + o(1))\log\log T/\log^2 T$ for the bare clock; the m-th Newton-type step has error $\frac{H_m}{4\pi^2} + O(1 + \sqrt{H_m})$ in units of zeros with absolute constants, and after $\log^* T + O(1)$ steps the error is $O(1)$ mean spacings, using no ordinate. The proof needs an exact moment recursion, a multiplicity bound for the corrected points of every level (Theorem 8.10) and a cutoff law that absorbs the multiplicity.

5. *Tails and order* (Theorem 8.3, Corollaries 8.13 and 8.15). The defect at the ordinates exceeds λ on at most $2^{2-\sqrt{\lambda/C}}$ of the zeros when the primes reach T^c with c in a range fixed by λ; the errors of the corrected clocks have all moments bounded in units of $\sqrt{H_j}$; and the sorted final clock is increasing with the same $O(1)$ error.

6. *The pointwise envelope* (Proposition 9.1, Theorem 9.2). $\limsup|\tau_k - \gamma_k| = 2\pi\limsup|S(t)|/\log t$ exactly, no slowly growing prime correction changes it, and Backlund's method with the profiles generated from Bourgain's exponent pair gives $\limsup|\tau_k - \gamma_k| \le 2\pi C_\infty = 0.6083073\ldots$.

10.3. Open problems

Question 10.1. For fixed $c \in (0, \frac{1}{2}]$, do the bounded means of $u_k(T^c)^2$ converge, and to $(2\pi^2)^{-1}\log(1/c)$ plus which constant?

Question 10.2. Does $\langle r_k(T^c)^2\rangle_{\mathcal{I}_T}$ stay bounded for fixed c, that is, does the transport cancel against the defect? Is the error of the corrected clock after $\log^* T + O(1)$ steps bounded in units of zeros for *almost every* index, and not only in mean square? Can Theorems 5.1 and 7.3 be extended to $c > \frac{1}{2}$?

Question 10.3. Can the absolute constants $C_\star$, H^* and $B_\star$ of Theorem 8.11, and those of Fujii's and Montgomery–Vaughan's bounds on which they rest, be made explicit, so that the cutoff law (8.13) becomes a formula that can be evaluated at a given height?

Theorem 4.1 applies to any counting staircase and any C^1 model, in particular to the zeros of a primitive Dirichlet L-function; an analogue of Theorem 5.1 would additionally need Selberg's approximation for $L(s,\chi)$ [24, 8] and uniformity in the conductor.

Declaration of competing interest

The author declares no competing interests.

Data availability

No zero dataset was used to choose a construction or parameter. Two programs, `opc_envelope_coefficient_v14.py` (40-digit quadrature) and `opc_envelope_iterations_p1.py` (closed form at 50 digits, $0 \le J \le 30$), evaluate the integral (9.1) for piecewise linear exponent profiles with rational vertices; they and their outputs are supplied in the accompanying compilation archive, in `code/` and `outputs/`. A third program, `verify_envelope_intervals_v5.py`, encloses C_{30} and the analytic tail by interval arithmetic at 80 digits; its output `envelope_intervals_v5.json` confirms the upper bound $2\pi C_\infty < 0.6083074$. The numerical manifest identifies each program, its precision and parameters, and the output supporting each displayed decimal. Python 3 and mpmath 1.3.0 were used. The programs evaluate consequences of the analytic formulae; no observed zero ordinate chooses the construction or a parameter.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used AI assistants (Claude, Anthropic, and ChatGPT, OpenAI) to edit the text, check proofs and write programs. The author reviewed and edited the content and takes full responsibility for it.

References

[1] C. Bellotti, P.-J. Wong, Improved estimates for the argument and zero-counting function of the Riemann zeta-function, Math. Comp. (2025) (statements numbered as in arXiv:2412.15470v2), https://doi.org/10.1090/mcom/4133. (Cited on pp. 3, 5, 6, 7, 12, 13, 14, 15, 41, and 42.)

[2] P. Billingsley, Probability and Measure, third ed., Wiley, New York, 1995. (Cited on pp. 15 and 17.)

[3] J. Bourgain, Decoupling, exponential sums and the Riemann zeta function, J. Amer. Math. Soc. 30 (1) (2017) 205–224 (theorems numbered as in arXiv:1408.5794v2), https://doi.org/10.1090/jams/860. (Cited on pp. 6 and 14.)

[4] L. Carleson, On convergence and growth of partial sums of Fourier series, Acta Math. 116 (1966) 135–157, https://doi.org/10.1007/BF02392815. (Cited on p. 14.)

[5] R.M. Corless, G.H. Gonnet, D.E.G. Hare, D.J. Jeffrey, D.E. Knuth, On the Lambert W function, Adv. Comput. Math. 5 (1996) 329–359, https://doi.org/10.1007/BF02124750. (Cited on pp. 3 and 11.)

[6] A. Fazzari, M. Gerspach, P. Minelli, Selberg's central limit theorem weighted by linear statistics of zeta zeros, preprint, arXiv:2507.04150, 2025, https://doi.org/10.48550/arXiv.2507.04150. (Cited on pp. 4, 5, and 14.)

[7] G.B. Folland, Real Analysis: Modern Techniques and Their Applications, second ed., Wiley, New York, 1999. (Cited on pp. 18 and 22.)

[8] A. Fujii, On the zeros of Dirichlet L-functions. I, Trans. Amer. Math. Soc. 196 (1974) 225–235, https://doi.org/10.1090/S0002-9947-1974-0349603-2. (Cited on pp. 4, 13, 42, and 43.)

[9] A. Fujii, On the zeros of Dirichlet L-functions. II (with corrections to "On the zeros of Dirichlet L-functions. I" and the subsequent papers), Trans. Amer. Math. Soc. 267 (1) (1981) 33–40. (Cited on p. 13.)

[10] E. Hasanalizade, Q. Shen, P.-J. Wong, Counting zeros of the Riemann zeta function, J. Number Theory 235 (2022) 219–241, https://doi.org/10.1016/j.jnt.2021.06.032. (Cited on pp. 3, 13, and 14.)

[11] R.A. Hunt, On the convergence of Fourier series, in: Orthogonal Expansions and their Continuous Analogues (Proc. Conf., Edwardsville, Ill., 1967), Southern Illinois Univ. Press, Carbondale, 1968, pp. 235–255. (Cited on p. 14.)

[12] A. Ivić, The Riemann Zeta-Function, Wiley, New York, 1985 (reprinted by Dover, Mineola, NY, 2003). (Cited on pp. 6 and 14.)

[13] J. Kinnunen, The Hardy–Littlewood maximal function of a Sobolev function, Israel J. Math. 100 (1997) 117–124, https://doi.org/10.1007/BF02773636. (Cited on p. 14.)

[14] M. Korolev, Gram's law and the argument of the Riemann zeta function, Publ. Inst. Math. (Beograd) (N.S.) 92(106) (2012) 53–78, https://doi.org/10.2298/PIM1206053K. (Cited on pp. 4, 13, and 42.)

[15] M.A. Korolev, Gram's law in the theory of the Riemann zeta-function. Part 1, Sovrem. Probl. Mat. 20 (2015) 3–161; English transl. Proc. Steklov Inst. Math. 292, suppl. 2 (2016) S1–S146, https://doi.org/10.4213/spm53. (Cited on p. 13.)

[16] A. Ivić, M.A. Korolev, On the distribution of values of the argument of the Riemann zeta-function, preprint, arXiv:1808.10768 (2018), https://doi.org/10.48550/arXiv.1808.10768. (Cited on p. 13.)

[17] M. Radziwiłł, Large deviations in Selberg's central limit theorem, preprint, arXiv:1108.5092 (2011), https://doi.org/10.48550/arXiv.1108.5092. (Cited on p. 13.)

[18] M.A. Korolev, On Gram's law in the theory of the Riemann zeta function, Izv. Math. 76 (2) (2012) 275–309, https://doi.org/10.1070/IM2012v076n02ABEH002583. (Cited on pp. 4, 13, and 42.)

[19] A.K. Lerner, On weighted norm inequalities for the Carleson operator, preprint, arXiv:1310.3352, 2013, https://doi.org/10.48550/arXiv.1310.3352. (Cited on p. 14.)

[20] J.E. Littlewood, On the zeros of the Riemann zeta-function, Proc. Cambridge Philos. Soc. 22 (1924) 295–318, https://doi.org/10.1017/S0305004100014225. (Cited on p. 12.)

[21] F. Mertens, Ein Beitrag zur analytischen Zahlentheorie, J. Reine Angew. Math. 78 (1874) 46–62, https://doi.org/10.1515/crll.1874.78.46. (Cited on p. 14.)

[22] H.L. Montgomery, R.C. Vaughan, Hilbert's inequality, J. Lond. Math. Soc. (2) 8 (1974) 73–82, https://doi.org/10.1112/jlms/s2-8.1.73. (Cited on p. 14.)

[23] A. Selberg, Contributions to the theory of the Riemann zeta-function, Arch. Math. Naturvid. 48 (5) (1946) 89–155. (Cited on pp. 4, 13, 14, and 42.)

[24] A. Selberg, Contributions to the theory of Dirichlet's L-functions, Skr. Norske Vid.-Akad. Oslo I 1946 (3) (1946) 2–62. (Cited on pp. 4 and 43.)

[25] E.M. Stein, Singular Integrals and Differentiability Properties of Functions, Princeton University Press, Princeton, NJ, 1970. (Cited on p. 14.)

[26] G. Tenenbaum, Introduction to Analytic and Probabilistic Number Theory, third ed., Grad. Stud. Math., vol. 163, Amer. Math. Soc., Providence, RI, 2015, https://doi.org/10.1090/gsm/163. (Cited on p. 14.)

[27] E.C. Titchmarsh, The Theory of the Riemann Zeta-Function, second ed., revised by D.R. Heath-Brown, Clarendon Press, Oxford, 1986. (Cited on pp. 3, 5, 10, 12, and 40.)

[28] T.S. Trudgian, An improved upper bound for the argument of the Riemann zeta-function on the critical line, Math. Comp. 81 (278) (2012) 1053–1061, https://doi.org/10.1090/S0025-5718-2011-02537-8; preprint: A modest improvement on the function $S(T)$, arXiv:1010.4596 (2010). (Cited on pp. 3, 6, 7, 13, 39, 41, and 42.)

[29] T.S. Trudgian, An improved upper bound for the argument of the Riemann zeta-function on the critical line II, J. Number Theory 134 (2014) 280–292, https://doi.org/10.1016/j.jnt.2013.07.017. (Cited on pp. 3, 6, 13, and 14.)

[30] T.S. Trudgian, A. Yang, Toward optimal exponent pairs, preprint, arXiv:2306.05599 (2023), https://doi.org/10.48550/arXiv.2306.05599. (Cited on p. 41.)

[31] K.-M. Tsang, The distribution of the values of the Riemann zeta-function, Ph.D. thesis, Princeton University, 1984. (Cited on pp. 4, 13, and 14.)